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AN 985: Nios® V Processor Tutorial

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Contents

1. AN 985: Nios® V Processor Tutorial

1.1. Overview

This application note is a fundamental guide for you to get started with building a Nios® V processor system and running a simple Hello World program. The target boards are the Altera FPGA development kits. The build software includes both Quartus® Prime software and Ashling* RiscFree* IDE for Altera® FPGAs.

The document provides the following information:

- Introduces you to the basic design flow for the Nios V processor.
- Provides instructions on how to generate a Nios V example design, create a software program, and run the program on an Altera FPGA device.

Related Information

- Nios V Processor Reference Manual
- Nios V Processor IP Release Notes
- Nios V Processor Software Developer Handbook
- Nios V Embedded Processor Design Handbook

2. Hello World on Agilex™ 7 FPGA F-Series Transceiver-SoC Development Kit

2.1. Hardware and Software Requirements

Hardware Requirements

- Any Altera FPGA Development Kits

  Note: This tutorial uses Agilex™ 7 FPGA F-Series Transceiver-SoC Development Kit (DK-SI-AGF014EA).
- Power adapter
- USB Blaster II

Software Requirements

- Quartus Prime Pro Edition Software
- Ashling RiscFree IDE for Altera FPGAs
- Questa* Intel® FPGA Edition

Note: You need to acquire the license for the Nios V processor to compile the design in Quartus Prime software.

Related Information

- Agilex 7 FPGA F-Series Transceiver-SoC Development Kit (P-Tile and E-Tile)
- Quartus Prime Pro Edition User Guide: Getting Started
- Quartus Prime Standard Edition User Guide: Getting Started
- Ashling* RiscFree Integrated Development Environment (IDE) for Altera FPGAs User Guide
- Questa*-Altera FPGA Edition Quick-Start: Quartus Prime Pro Edition
- Questa*-Altera FPGA Edition Quick-Start: Quartus Prime Standard Edition

2.2. Generating the System

The implementation of Altera FPGA devices requires a hardware system developed using the Quartus Prime software. The Nios V processor requires an additional software system that complements the processor hardware system. You can develop a Nios V processor software system that is compatible to your Nios V processor hardware system using Ashling RiscFree IDE for Altera FPGAs.

2.2.1. Building Hardware Design in Platform Designer Overview

Begin designing the system hardware by instantiating the Nios V processor and its peripherals into Platform Designer. After configuring the system assignments and constraints, complete the hardware design by performing a successful compilation.

Table 1. Component Description

Components	Description
Nios V/m Processor IP	Runs application by executing instructions.
JTAG UART IP	Enables serial character communication between Nios V/m processor and host computer
On-Chip Memory II IP	Stores data and instructions.
Reset Release IP	Recommended reset output in SDM-based devices.

You can design the system hardware by using one of the following methods:

- Manual instantiation
- Board-aware flow
- Configurable example design

2.2.2. Building Hardware Design in Platform Designer — Manual Instantiation

2.2.2.1. Creating a New Project

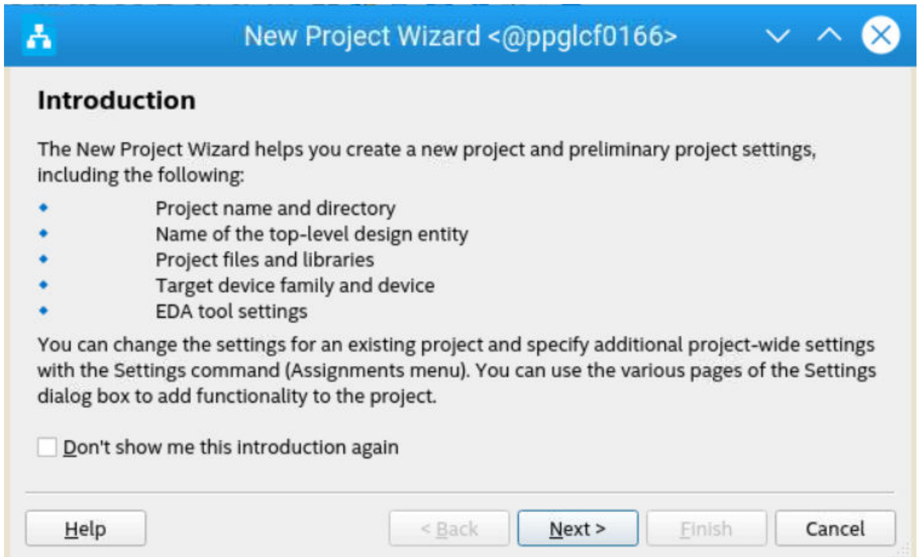
Start developing the Nios V processor system by creating a new Quartus Prime software project.

1. Launch **Quartus Prime Pro Edition** software.
2. Click **New Project Wizard** to create a new project.

Figure 1. New Project Wizard

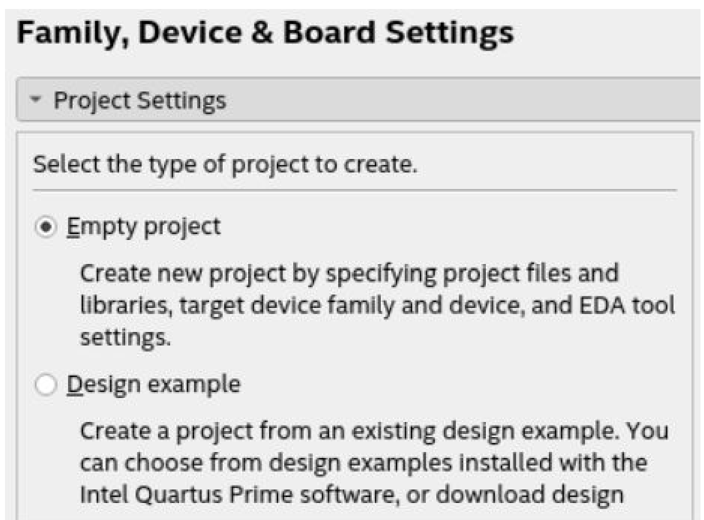
3. Read the **Introduction** and click **Next** to proceed.

Figure 2. New Project Wizard — Introduction



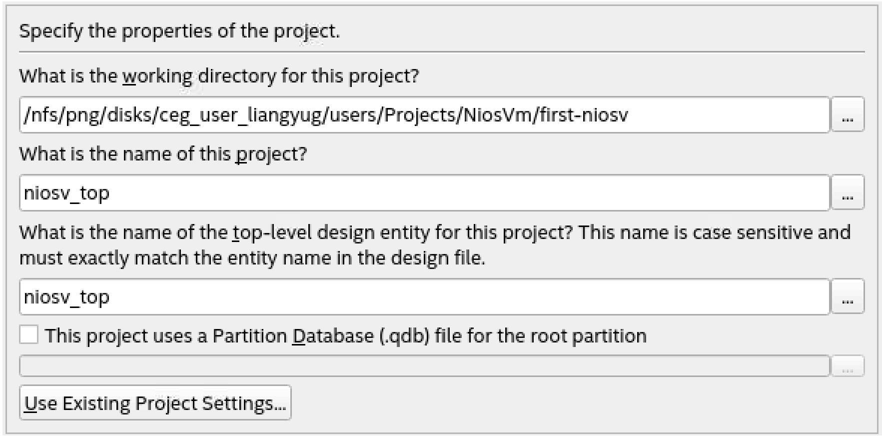
4. In **Family, Device & Board Settings** window,
    a. Select **Empty project**.

Figure 3. Selecting Project Settings



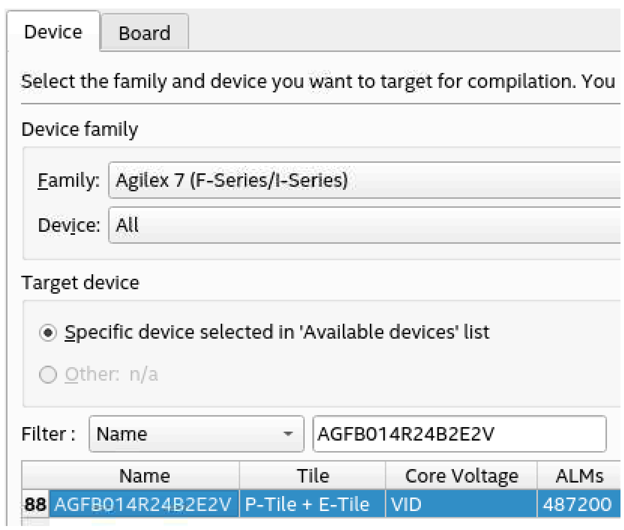
    b. Enter your working directory, define the name of the project, and enter the top-level design entity as `niosv_top`.

Figure 4. Specifying the Properties of the Project



c. Filter **Device Name** as **AGFB014R24B2E2V** which is for Agilex 7 FPGA F-Series Transceiver-SoC Development Kit.

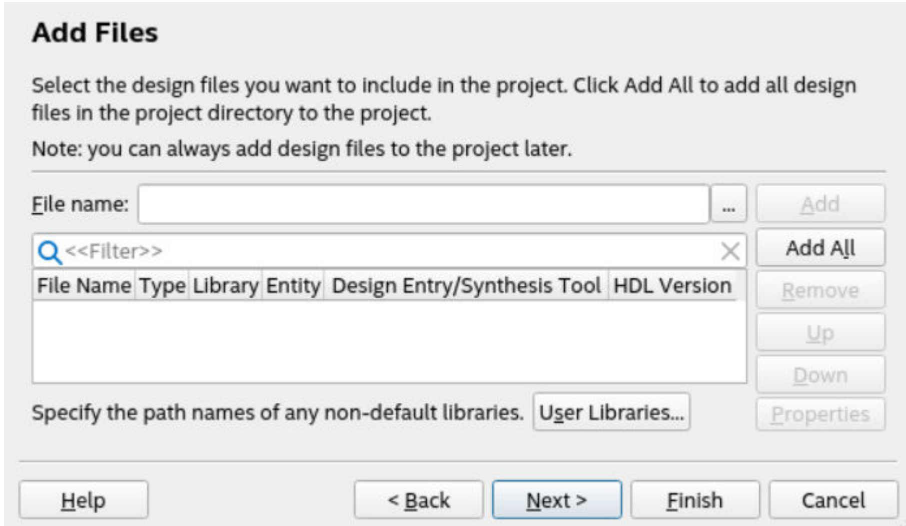
Figure 5. Device Tab



d. Click **Next** to proceed.

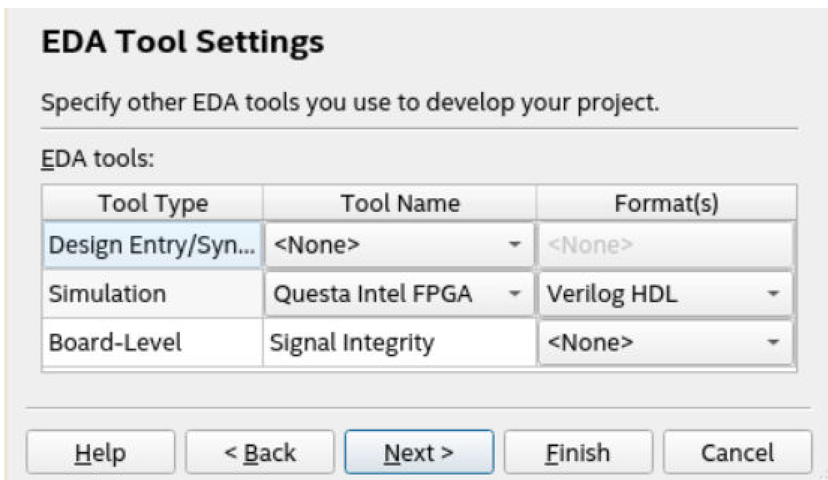
5. In **Add Files** window, leave it empty and click **Next** to proceed.

Figure 6. Add Files Window



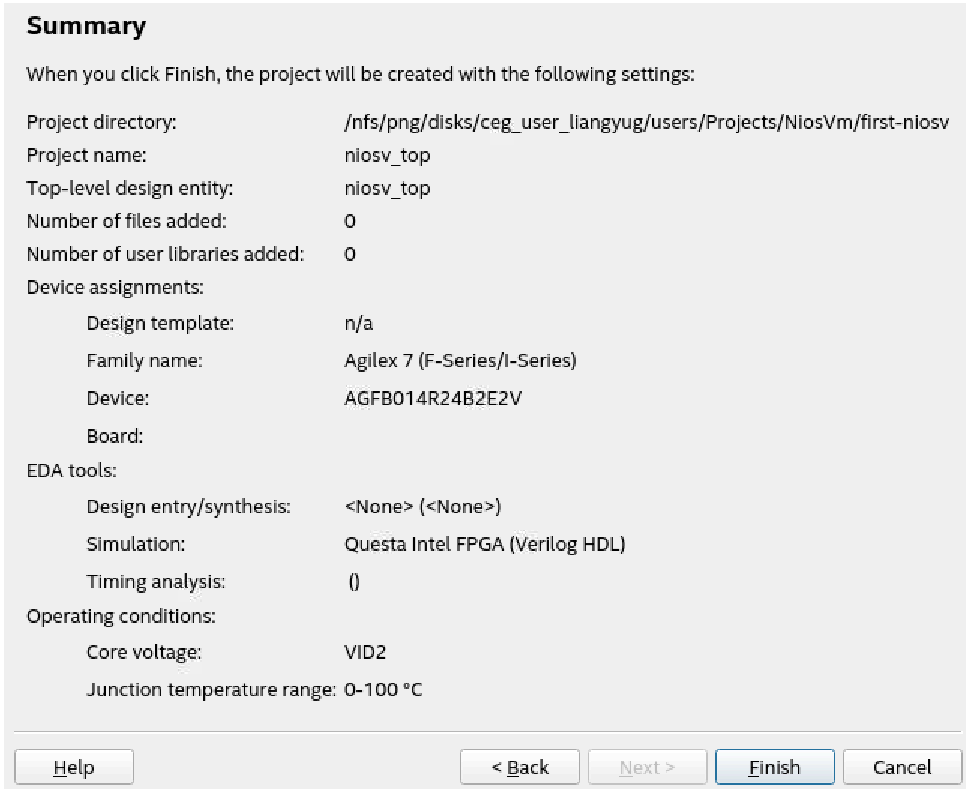
6. In **EDA Tool Settings** windows, select **Questa Intel FPGA**. (This is for simulation in later chapter.)

Figure 7. EDA Tool Settings



7. Click **Next** to review the summary of the new project. Then click **Finish**.

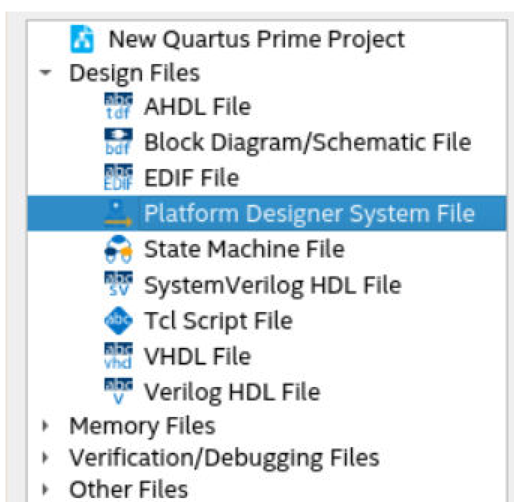
Figure 8. Summary



2.2.2.2. Creating a Platform Designer System

1. Click **New**, select **Platform Designer System File**, and click **OK**.

Figure 9. New Window



2. In the **Open System** window, check the project information.

a. **Quartus project**: `<Working directory>/niosv_top.qpf`

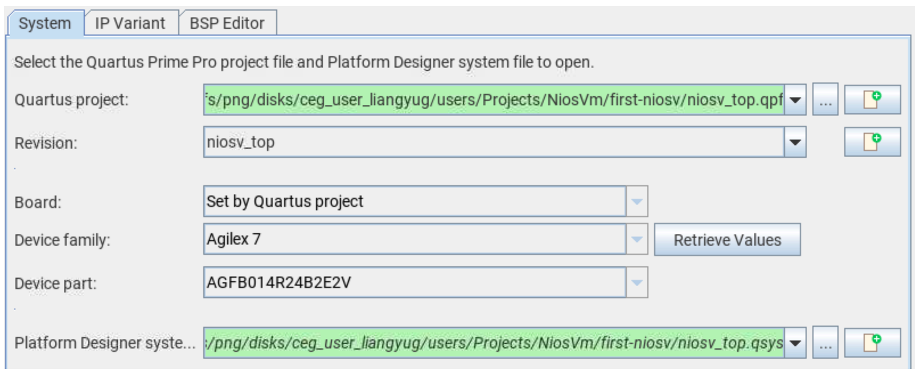
b. **Revision**: `niosv_top`

c. **Device family**: Agilex 7

d. **Device part**: AGFB014R24B2E2V

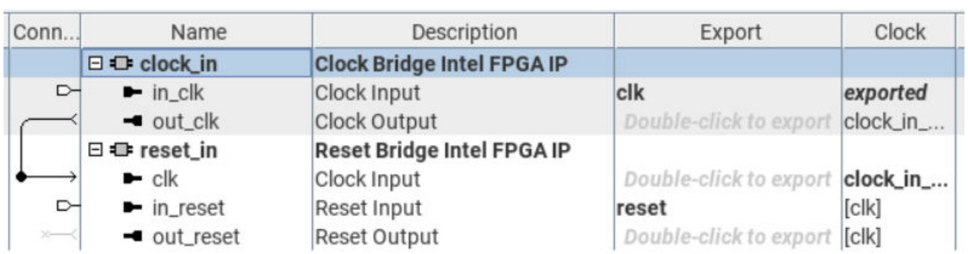
e. **Platform Designer system**: Click **Create new Platform Designer System** icon, and name the QSYS file as `niosv_top`.

Figure 10. Open System Window



3. Click **Create**.
4. In the Platform Designer system, the software instantiates the Clock Bridge and Reset Bridge IP by default.
5. In Clock Bridge IP, configure **Explicit clock rate** as `100000000` Hz (100 MHz).
6. In Reset Bridge IP, enable it as an **Active low reset**.

Figure 11. Default Platform Designer System

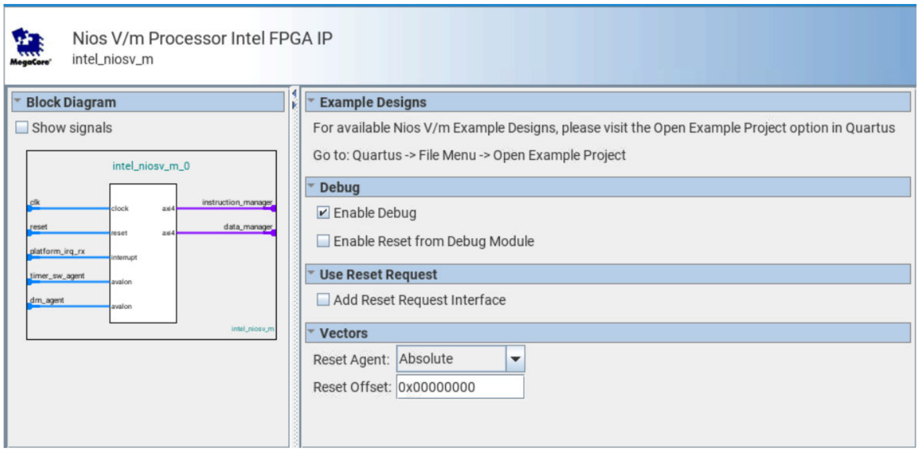


Conn...	Name	Description	Export	Clock
	clock_in	Clock Bridge Intel FPGA IP		
	in_clk	Clock Input	clk	exported
	out_clk	Clock Output	Double-click to export	clock_in_...
	reset_in	Reset Bridge Intel FPGA IP		
	clk	Clock Input	Double-click to export	clock_in_...
	in_reset	Reset Input	reset	[clk]
	out_reset	Reset Output	Double-click to export	[clk]

2.2.2.2.1. Adding Nios V/m Processor IP

1. Search for **Nios V/m Processor** in the **IP Catalog**.
2. Add the **Nios V/m Processor IP** under **Processors and Peripherals ➤ Embedded Processors** section. The New IP Variation window appears.
3. Click **Finish** to instantiate the processor. Leave it at the default settings.

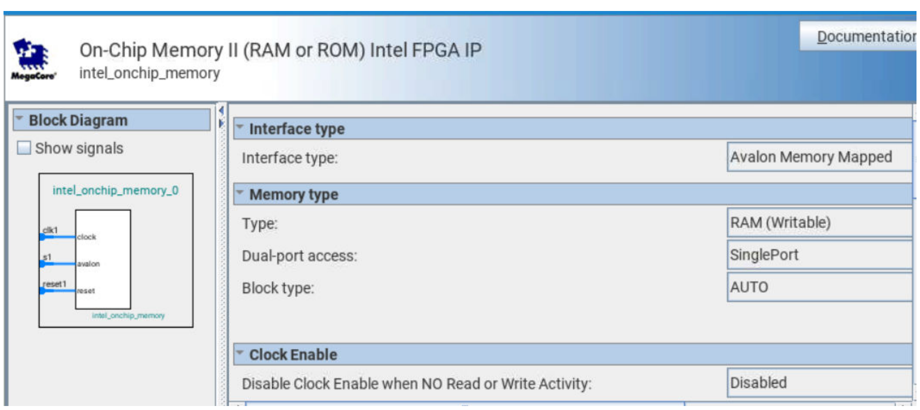
Figure 12. Nios V/m Processor IP Parameter Editor



2.2.2.2.2. Adding On-Chip Memory II (RAM or ROM) IP

1. Search for **On-Chip Memory** in the **IP Catalog**.
2. Add the **On-Chip Memory II (RAM or ROM) IP** under **Basic Functions ➤ On Chip Memory**. The New IP Variation window appears.
3. Configure the **Total memory size** as **262144**.
4. Turn on the option **Enable non-default initialization file** and provide the filename `hello.hex`.
5. Leave other settings at default.
6. Click **Finish** to instantiate the peripheral.

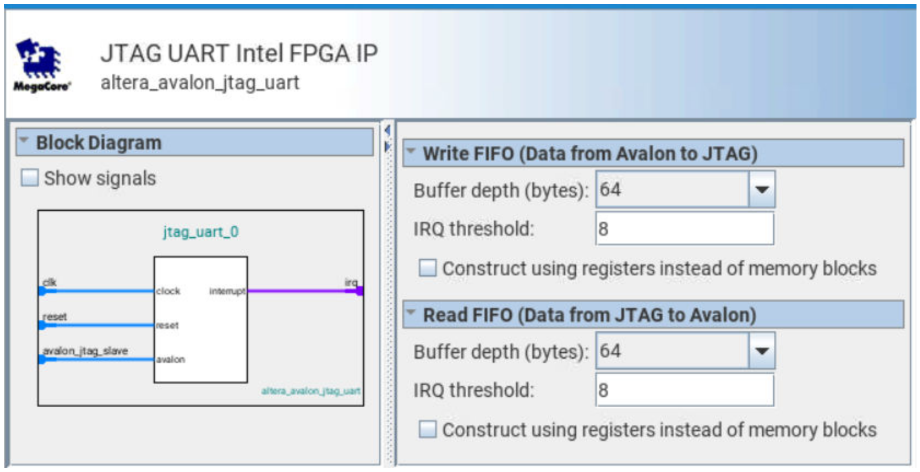
Figure 13. On-Chip Memory II (RAM or ROM) IP Parameter Editor



2.2.2.2.3. Adding JTAG UART IP

1. Search for **JTAG UART** in the **IP Catalog**.
2. Add the **JTAG UART IP** under **Interface Protocols ➤ Serial** section. The New IP Variation window appears.
3. Click **Finish** to instantiate the peripheral. Leave it at the default settings.

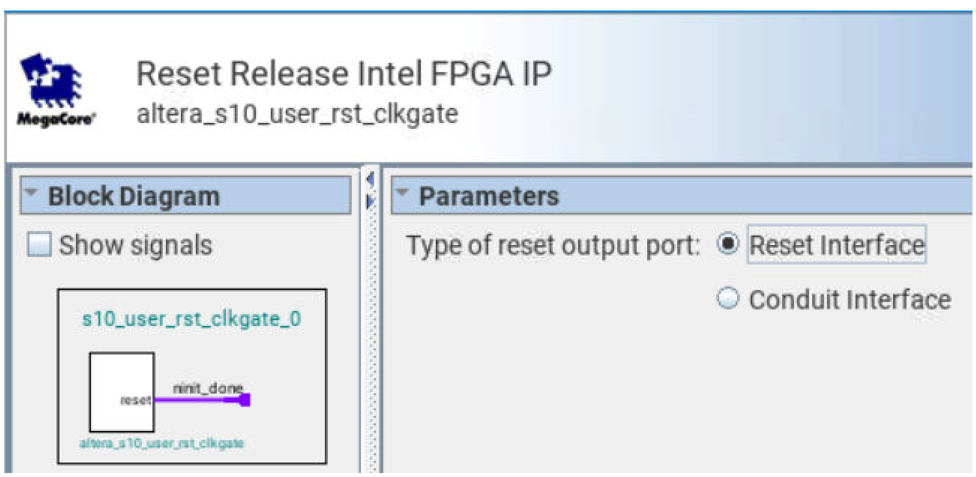
Figure 14. JTAG UART IP Parameter Editor



2.2.2.2.4. Adding Reset Release IP

1. Search for **Reset Release** in the **IP Catalog**.
2. Add the **Reset Release IP** under **Basic Functions ➤ Configuration and Programming** section. The New IP Variation window appears.
3. Select the type of reset output port as **Reset Interface**.
4. Click **Finish** to instantiate the peripheral.

Figure 15. Reset Release IP Parameter Editor



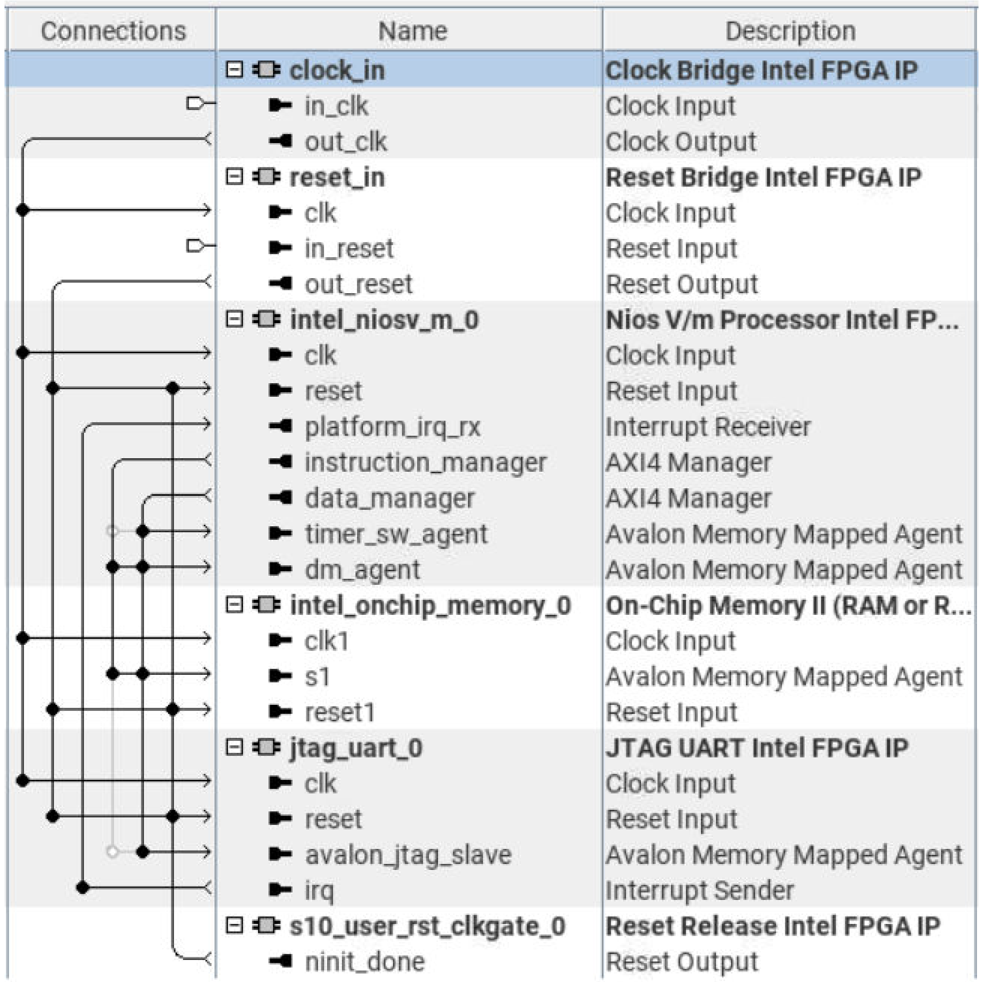
2.2.2.2.5. Connect Interfaces and Signals

Connect the clock bridge, reset bridge, reset release IP, and Nios V processor to the peripherals.

Table 2. Connection between Host and Agent

IP	Host	Peripheral
Clock Bridge IP	out_clk	intel_niosv_m_0.clk
		intel_onchip_memory_0.clk1
		jtag_uart_0.clk
Reset Bridge IP	out_reset	intel_niosv_m_0.reset
		intel_onchip_memory_0.reset1
		jtag_uart_0.reset
Reset Release IP	ninit_done	intel_niosv_m_0.reset
		intel_onchip_memory_0.reset1
		jtag_uart_0.reset
Nios V/m Processor IP	platform_irq_rx	jtag_uart_0.irq
	instruction_manager	intel_onchip_memory_0.s1
	data_manager	intel_onchip_memory_0.s1
		jtag_uart_0.avalon_jtag_slave

Figure 16. Full System Connection



2.2.2.2.6. Clear System Warnings and Errors

1. Navigate to the **System** menu bar.
2. Click **Assign Base Address**.

Figure 17. Example of System Connectivity Issues

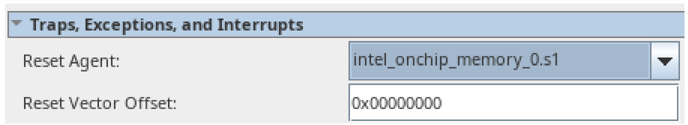
top.intel_niosv_m_0.instruction_manager	**intel_onchip_memory_0.s1** (0x0..0xffff) overlaps **intel_niosv_m_0.dm_agent** (0x0..0xffff)
top.intel_niosv_m_0.data_manager	**intel_onchip_memory_0.s1** (0x0..0xffff) overlaps **intel_niosv_m_0.dm_agent** (0x0..0xffff)
top.intel_niosv_m_0.data_manager	**jtag_uart_0.avalon_jtag_slave** (0x0..0x7) overlaps **intel_onchip_memory_0.s1** (0x0..0xffff)

2.2.2.2.7. Configuring the Reset Vector of the Nios V Processor

1. Select the Nios V Processor IP and open the IP Parameter Editor.
2. Navigate to the **Vectors** tab.
3. Configure as follows:

- **Reset Agent**: intel_onchip_memory_0.s1
- **Reset Offset**: 0x0

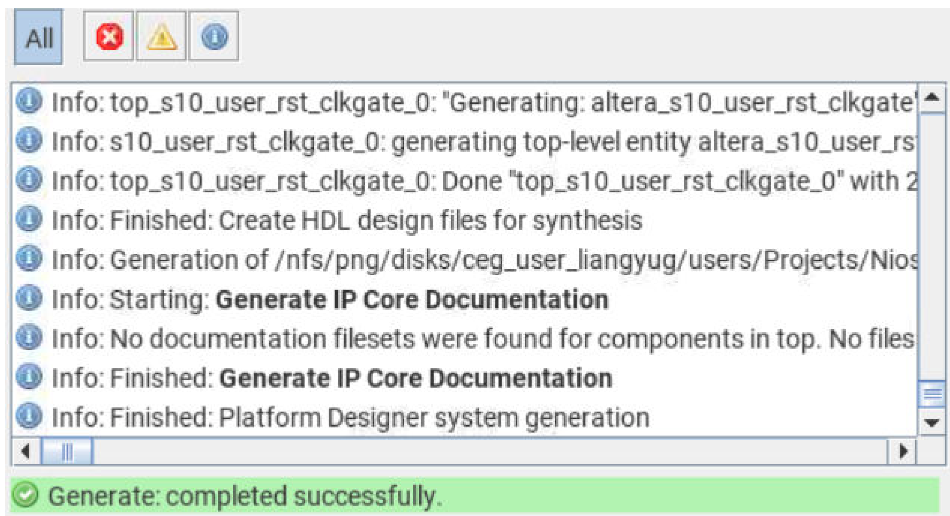
Figure 18. Reset Vector



2.2.2.2.8. Saving and Generating System HDL

1. Save the system.
2. Click **Generate HDL** at the bottom right corner of the Platform Designer.
3. Leave all settings at default, and click **Generate**.
4. Ensure that the Nios V processor hardware system is successfully generated.

Figure 19. Successful Generation



5. The Platform Designer generates a folder named `niosv_top`, which stores the system generation files.
6. Exit the Platform Designer, and return to the project front page.

Figure 20. Generated Project Files

DNI	File folder	
ip	File folder	
niosv_top	File folder	
qdb	File folder	
niosv_top.qpf	QPF File	2 KB
niosv_top.qsf	QSF File	2 KB
niosv_top.qsys	QSYS File	312 KB
niosv_top.qsys.legacy	LEGACY File	230 KB

2.2.2.3. Configuring Assignment and Constraint

2.2.2.3.1. Adding Design Files

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Settings**.
2. Navigate to **Files** category. You can find all related IP files and QSYS file added into your project.

Figure 21. Settings – Files Category

File Name	Type
ip/niosv_top/niosv_top_clock_in.ip	IP File
ip/niosv_top/niosv_top_reset_in.ip	IP File
niosv_top.qsys	Platform Designer System File
ip/niosv_top/niosv_top_intel_niosv_m_0.ip	IP File
ip/niosv_top/niosv_top_intel_onchip_memory_0.ip	IP File
ip/niosv_top/niosv_top_jtag_uart_0.ip	IP File
ip/niosv_top/niosv_top_s10_user_rst_clkgate_0.ip	IP File

3. Earlier during the new project creation, the top-level design entity is named `niosv_top`. Thus, the `niosv_top.qsys` file is automatically the top-level design entity.
4. Check the top-level design entity assignment in the **Project Navigator**.

Figure 22. Project Navigator

Project Navigator

Files
- ip/niosv_top/niosv_top_clock_in.ip
- ip/niosv_top/niosv_top_reset_in.ip
- niosv_top.qsys
- ip/niosv_top/niosv_top_intel_niosv_m_0.ip
- ip/niosv_top/niosv_top_intel_onchip_memory_0.ip
- ip/niosv_top/niosv_top_jtag_uart_0.ip
- ip/niosv_top/niosv_top_s10_user_rst_clkgate_0.ip

2.2.2.3.2. Adding Synopsys Design Constraint (SDC) File

1. Click **New**, select **Synopsys Design Constraint File** and click **OK**.
2. Add the following constraint:

```
create_clock -name clk -period 10.0 [get_ports clk_clk]
```

3. Save as `niosv_top.sdc`.

2.2.2.3.3. Pin Assignments

1. In Quartus Prime software, navigate to **Processing** menu bar and click **Start ➤ Start Analysis & Synthesis**.
2. Once the analysis is complete, navigate to **Assignments** menu bar and click **Pin Planner**. For this example, there are 2 pins assignments:
   - `clk_clk` assigned to PIN_U52 (FPGA_SYSTEM_CLK)
   - `reset_reset_n` assigned to PIN_G52 (FPGA_SYS_RESETn)
3. Close **Pin Planner**, and return to the project front page.

2.2.2.3.4. Configuration and Power Management Assignments

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Device ➤ Device and Pin Options**.
2. Navigate to **Configuration** category.
3. Select **VID mode of operation** as **PMBus Master**.
4. Click **Configuration Pin Options**, and make the following **Configuration pin** assignments,
   - **PWRMGT_SCL** as `SDM_IO0`
   - **PWRMGT_SDA** as `SDM_IO12`
   - **CONF_DONE** as `SDM_IO16`
   - Leave the rest empty.
5. Navigate to **Power Management & VID** category, and make the following settings
   - **Bus speed mode** as `100 kHz`
   - **Slave device type** as `Other`
   - **PMBus device 0 slave address** as `42`
   - **Voltage output format** as `Linear format`
   - **Linear format N** as `-13`
   - **Translated voltage value unit** as `Volts`
   - Turn off **PAGE command**.
6. Click **OK**, and return to the project front page.

2.2.2.4. Compiling the Quartus Prime Software Project

1. Navigate to **Processing** menu bar and click **Start Compilation**.
2. Make sure that the Quartus Prime project compiles successfully.

2.2.3. Building Hardware Design in Platform Designer — Board-Aware Flow

Related Information

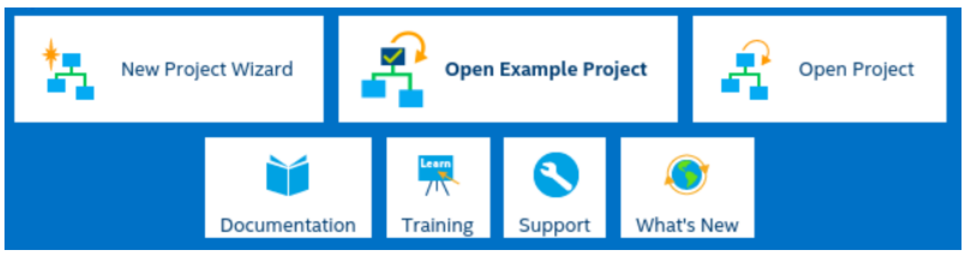
AN 988: Using the Board-Aware Flow: in the Quartus Prime Pro Edition Software
For more information on creating new Board file and new IP Presets.

2.2.3.1. Creating a New Project

Start developing the Nios V processor system by creating a new Quartus Prime software project.

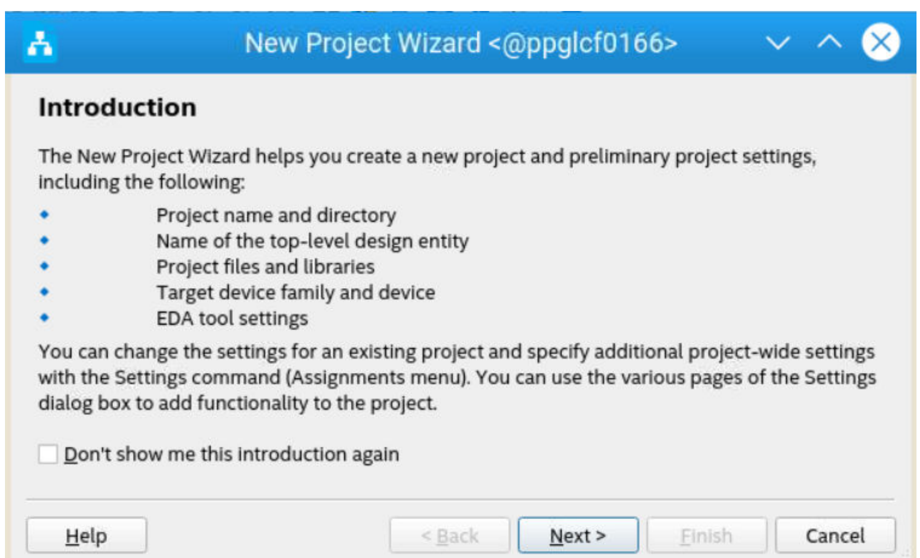
1. Launch **Quartus Prime Pro Edition** software.
2. Click **New Project Wizard** to create a new project.

Figure 23. New Project Wizard



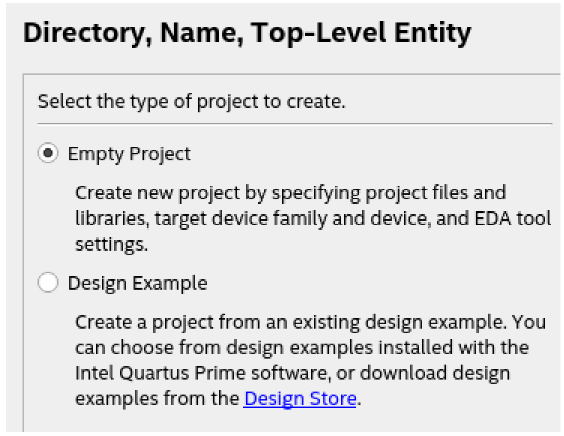
3. Read the **Introduction** and click **Next** to proceed.

Figure 24. New Project Wizard — Introduction



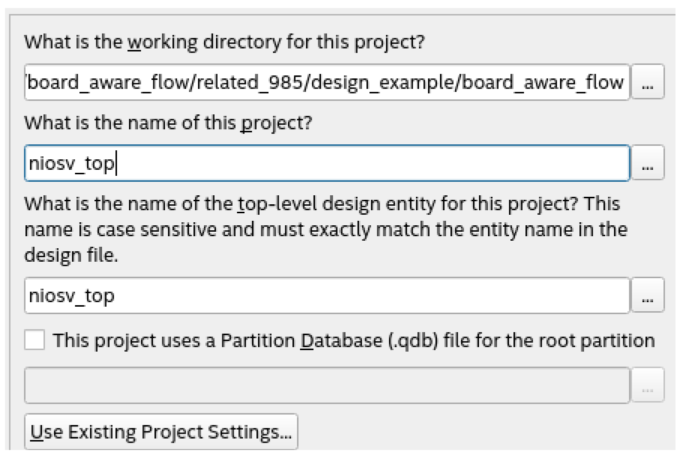
4. In **Directory, Name, Top-Level Entity** window,
   a. Select **Empty project**.

Figure 25. Selecting Project Settings



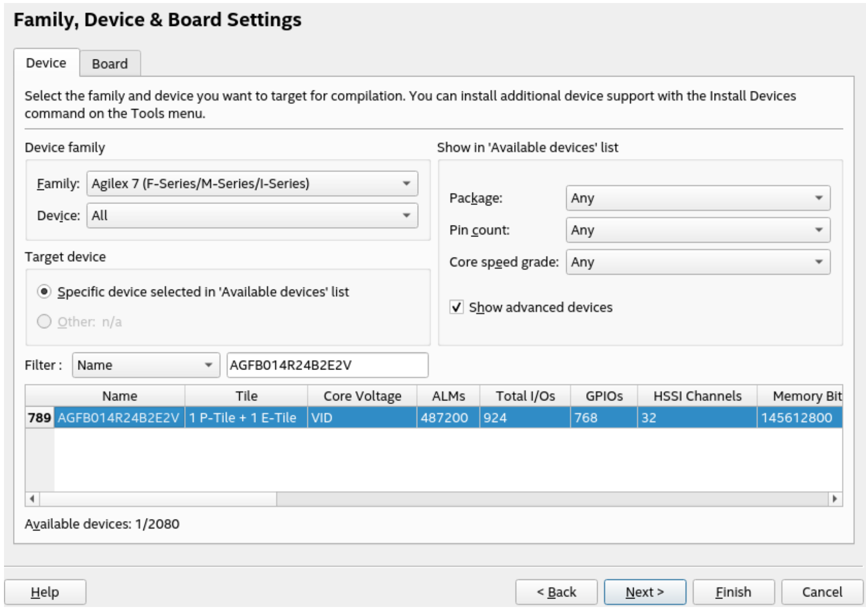
b. Enter your working directory, define the name of the project, and enter the top-level design entity as `niosv_top`. Click **Next** to proceed.

Figure 26. Specifying the Properties of the Project



c. Filter **Device Name** as **AGFB014R24B2E2V** which is for Agilex 7 FPGA F-Series Transceiver-SoC Development Kit.

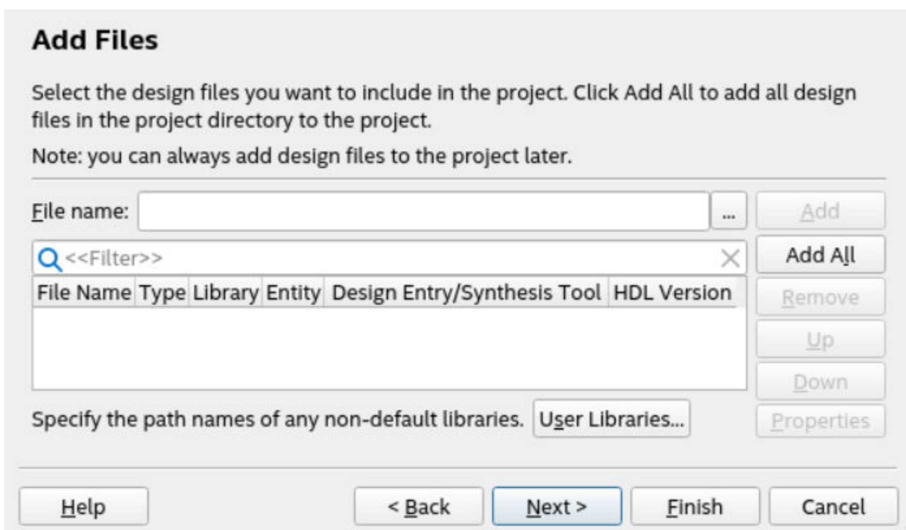
Figure 27. Device Tab



d. Click **Next** to proceed.

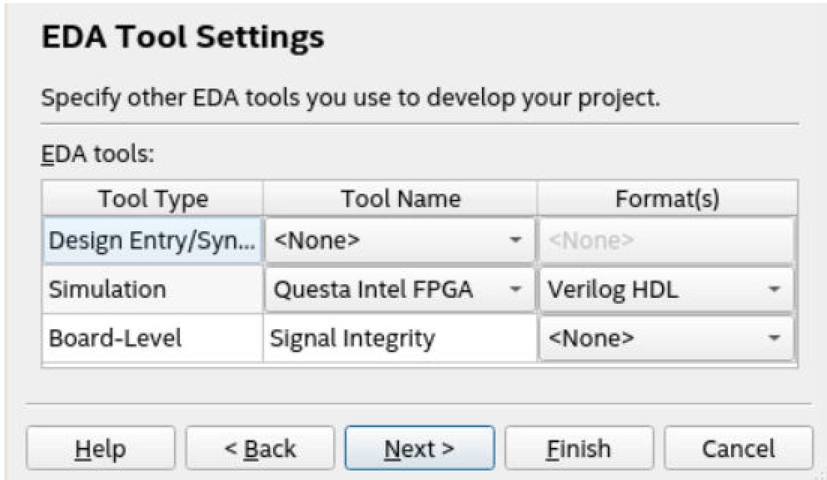
5. In **Add Files** window, leave it empty and click **Next** to proceed.

Figure 28. Add Files Window



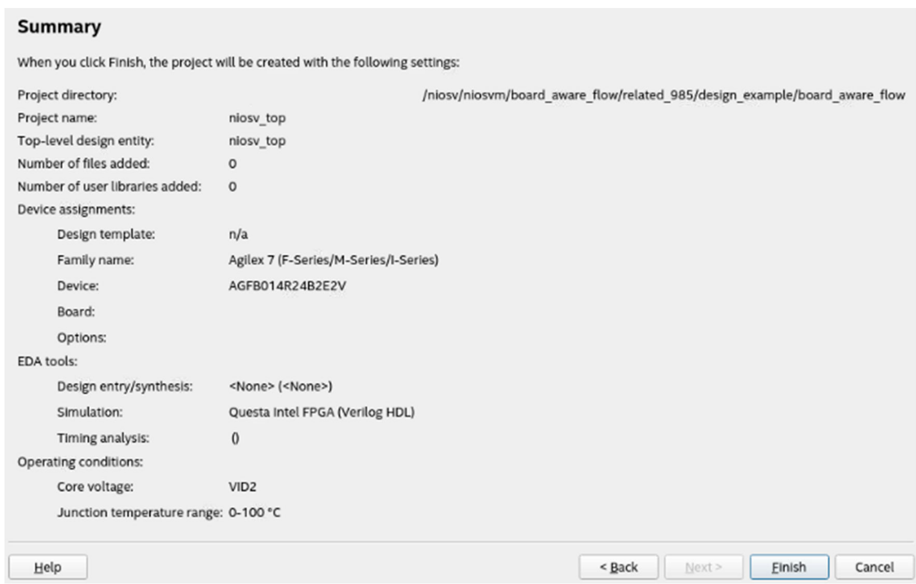
6. In **EDA Tool Settings** windows, select **Questa IP**. The selection is for simulation in the topic *Simulating the Nios V Processor*.

Figure 29. EDA Tool Settings



7. Click **Next** to review the summary of the new project. Then click **Finish**.

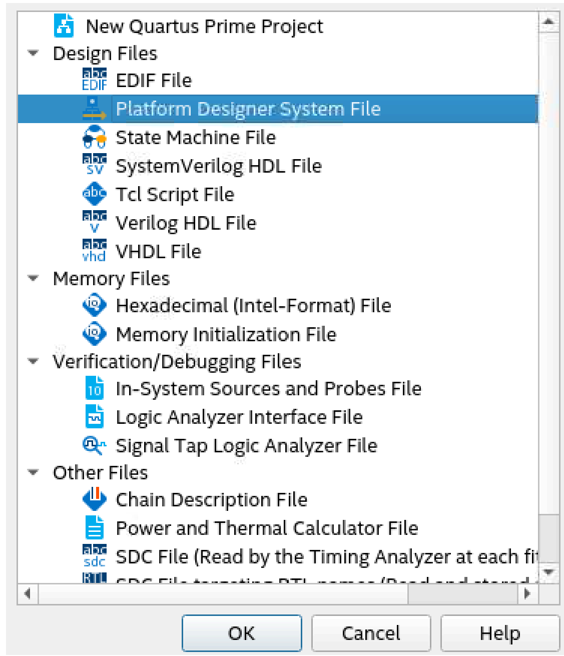
Figure 30. Summary



2.2.3.2. Creating a Platform Designer System

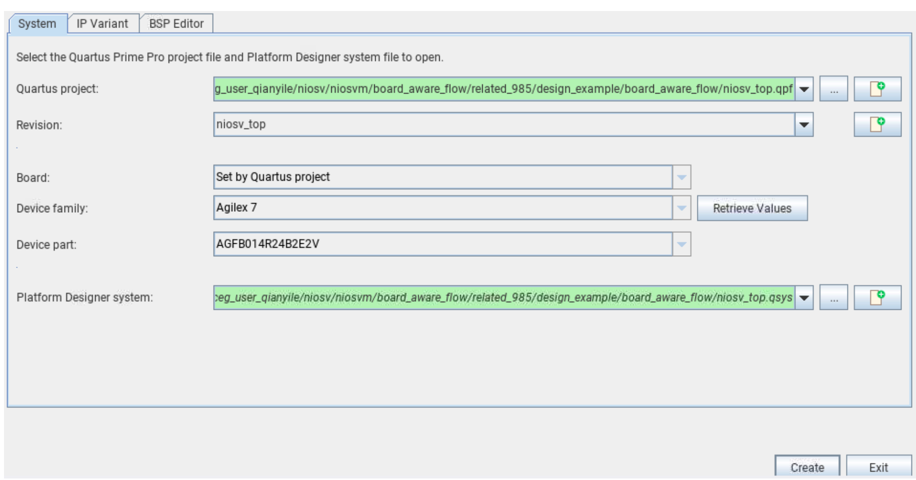
1. Click **New**, select **Platform Designer System File**, and click **OK**.

Figure 31. New Window



2. In the **Open System** window, check the project information.
    a. **Quartus project**: `<Working directory>/niosv_top.qpf`
    b. **Revision**: `niosv_top`
    c. **Device family**: Agilex 7
    d. **Device part**: AGFB014R24B2E2V
    e. **Platform Designer system**: Click **Create new Platform Designer System** icon, and name the QSYS file as `niosv_top`.

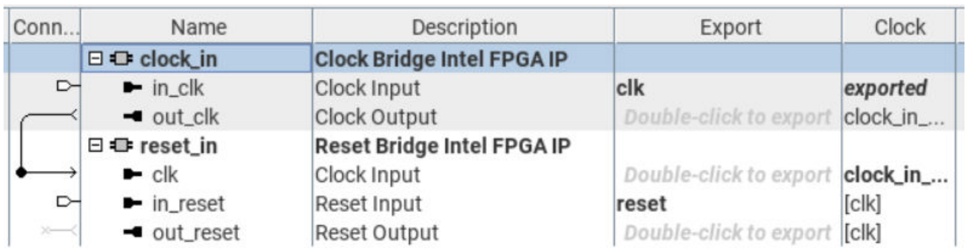
Figure 32. Open System Window



3. Click **Create**.

4. In the Platform Designer system, the software instantiates the Clock Bridge and Reset Bridge IP by default.
5. In Clock Bridge IP, configure **Explicit clock rate** as `100000000` Hz (100 MHz).
6. In Reset Bridge IP, enable it as an **Active low reset**.

Figure 33. Default Platform Designer System



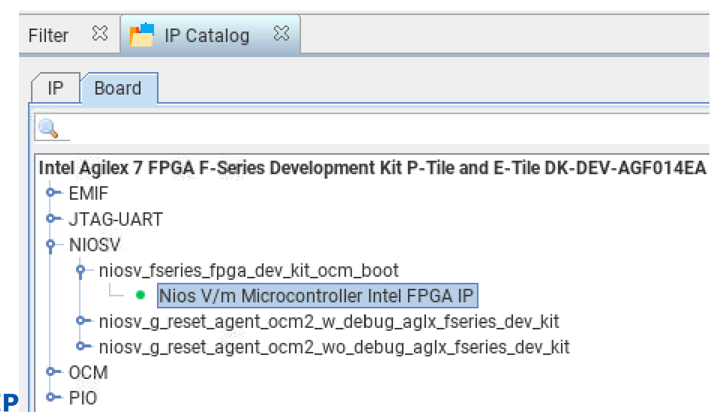
2.2.3.2.1. Adding Nios V/m Processor IP

1. Navigate to **IP Catalog** and click **Board**.
2. Expand **NIOSV** and **niosv_fseries_fpga_dev_kit_ocm_boot**.

Figure 34. IP Catalog - Board

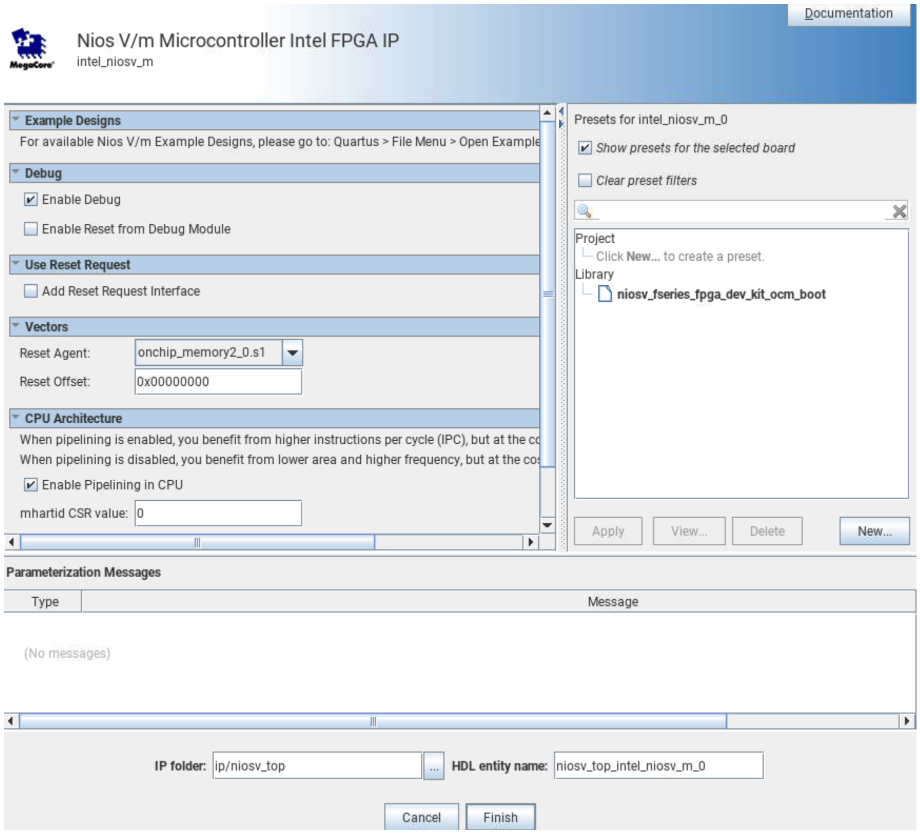
3. Double-click **Nios V/m Microcontroller IP**. The New IP Variation window appears.

Figure 35. Nios V/m Processor IP



4. Click **Finish** to instantiate the processor. Leave it at the default settings.

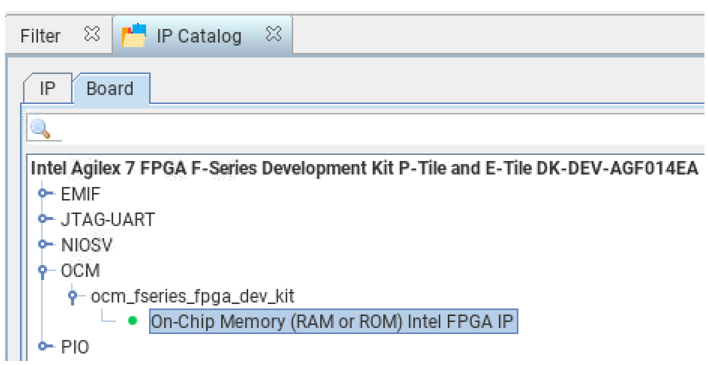
Figure 36. Nios V/m Processor IP Parameter Editor



2.2.3.2.2. Adding On-Chip Memory (RAM or ROM) IP

1. In **IP Catalog**, expand **OCM** and **ocm_fseries_fpga_dev_kit**.
2. Double click **On-Chip Memory (RAM or ROM) IP**. The New IP Variation window appears.

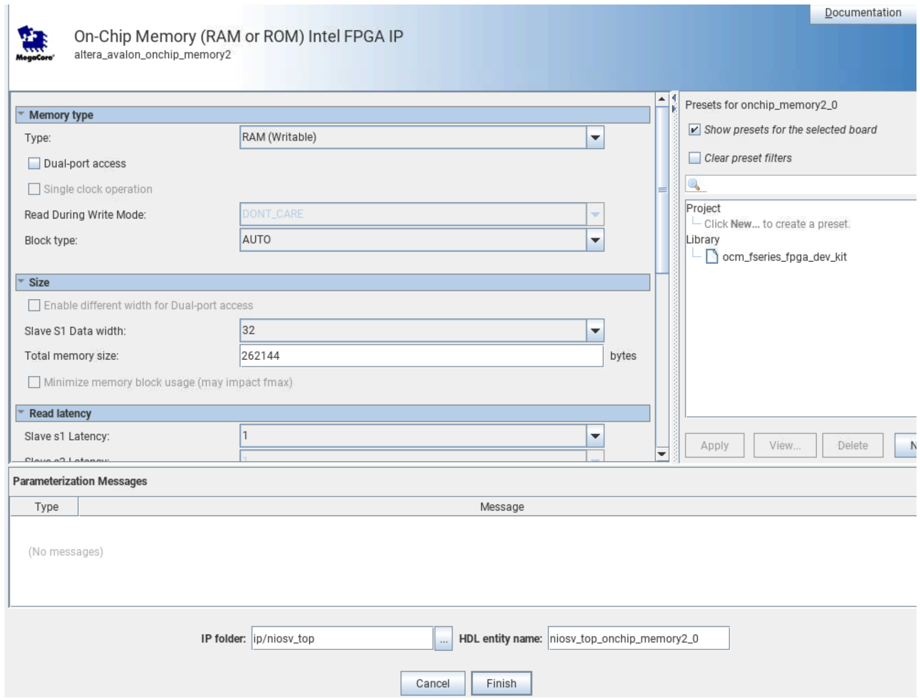
Figure 37. On-Chip Memory (RAM or ROM) IP



3. Configure the **Total memory size** as **262144**.

4. Navigate to **Memory initialization**, enable **Initialize memory content** and **Enable non-default initialization file**. Provide the filename `hello.hex`.
5. Leave other settings at default.
6. Click **Finish** to instantiate the peripheral.

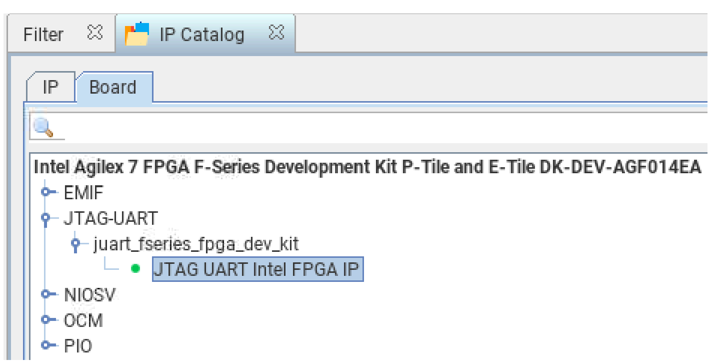
Figure 38. On-Chip Memory (RAM or ROM) IP Parameter Editor



2.2.3.2.3. Adding JTAG UART IP

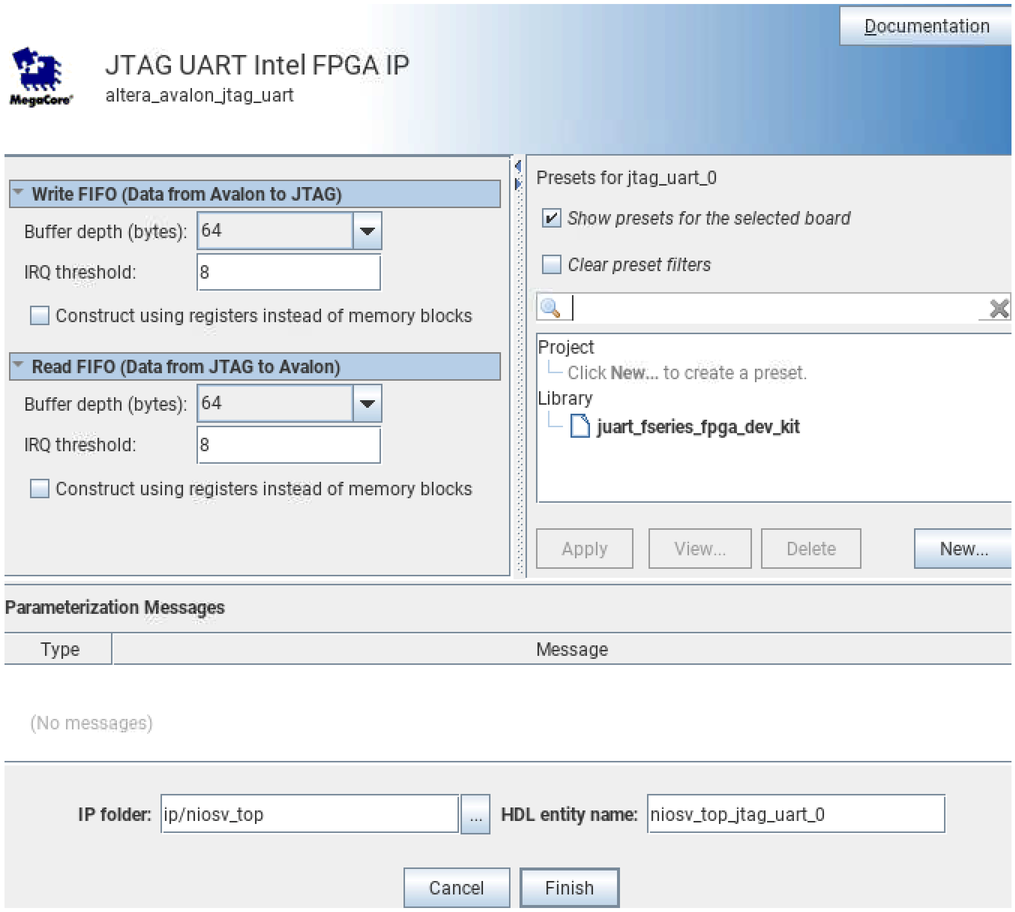
1. In **IP Catalog**, expand **JTAG-UART** and **juart_fseries_fpga_dev_kit**.
2. Double-click **JTAG UART IP**. The New IP Variation window appears.

Figure 39. JTAG UART IP



3. Click **Finish** to instantiate the processor. Leave it at the default settings.

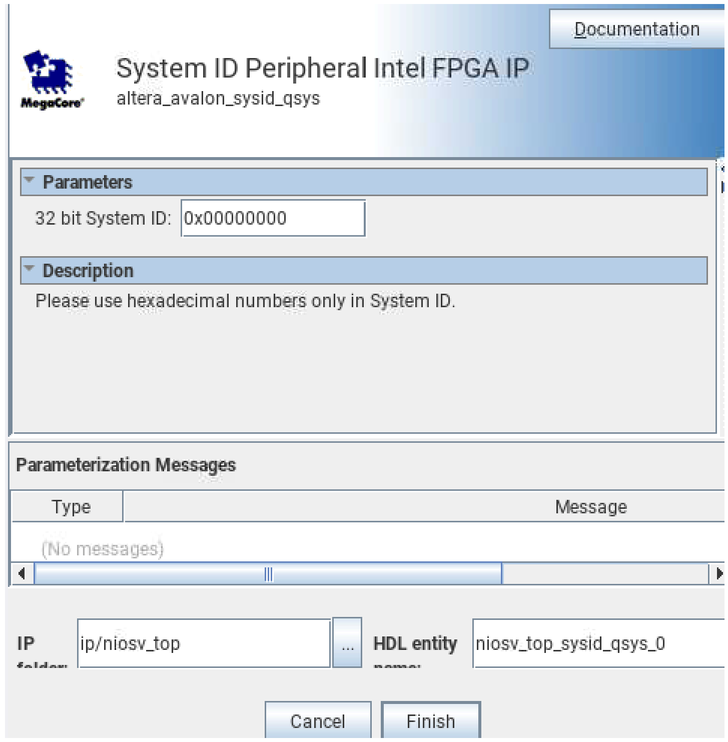
Figure 40. JTAG UART IP Parameter Editor



2.2.3.2.4. Adding System ID Peripheral IP

1. Search for **System ID Peripheral IP** in the **IP Catalog**.
2. Add the **System ID Peripheral IP** under **Basic Functions ➤ Simulation; Debug and Verification ➤ Debug and Performance** section. The New IP Variation window appears.
3. Click **Finish** to instantiate the processor. Leave it at the default settings.

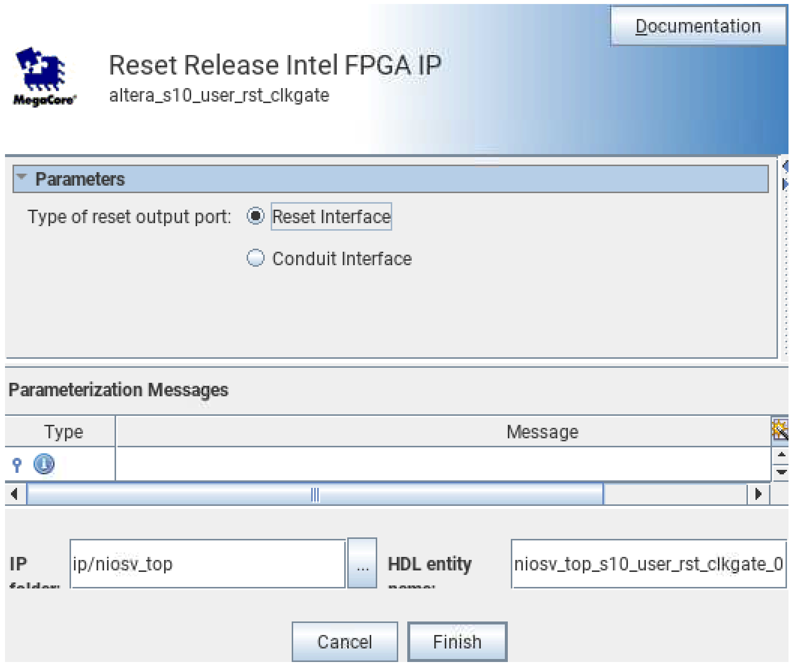
Figure 41. System ID Peripheral IP Parameter Editor



2.2.3.2.5. Adding Reset Release IP

1. Search for **Reset Release IP** in the **IP Catalog**.
2. Add the **Reset Release IP** under **Basic Functions ➤ Configuration and Programming** section. The New IP Variation window appears.
3. Select the type of reset output port as **Reset Interface**.
4. Click **Finish** to instantiate the processor.

Figure 42. Reset Release IP Parameter Editor



2.2.3.2.6. Connect Interfaces and Signals

Connect the Clock Bridge IP, Reset Bridge, Reset Release IP and Nios V/m processor to the peripheral.

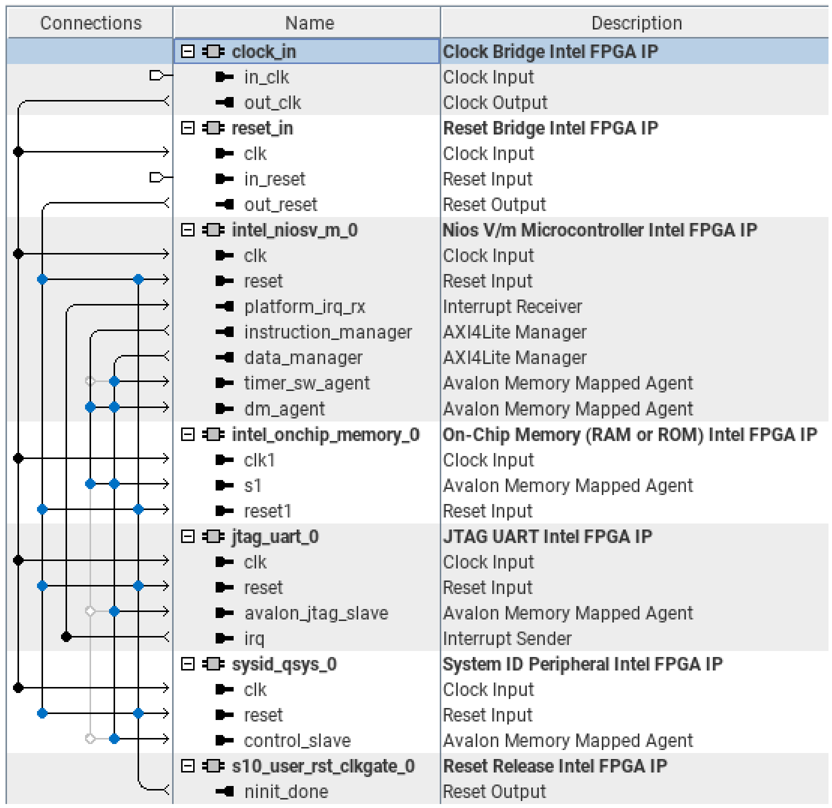
Table 3. Connection between Host and Agent

IP	Host	Peripheral
Clock Bridge IP	`out_clk`	`intel_niosv_m_0.clk`
		`onchip_memory2_0.clk1`
		`sysid_qsys_0.clk`
		`jtag_uart_0.clk`
Reset Bridge IP	`out_reset`	`intel_niosv_m_0.reset`
		`onchip_memory2_0.reset1`
		`sysid_qsys_0.reset`
		`jtag_uart_0.reset`
Reset Release IP	`ninit_done`	`intel_niosv_m_0.reset`
		`onchip_memory2_0.reset1`
		`sysid_qsys_0.reset`
		`jtag_uart_0.reset`
Nios V/m Processor IP	`platform_irq_rx`	`jtag_uart_0.irq`
	`instruction_manager`	`onchip_memory2_0.s1`

continued...

IP	Host	Peripheral
	data_manager	onchip_memory2_0.s1
		sysid_qsys_0.control_slave
		jtag_uart_0.avalon_jtag_slave

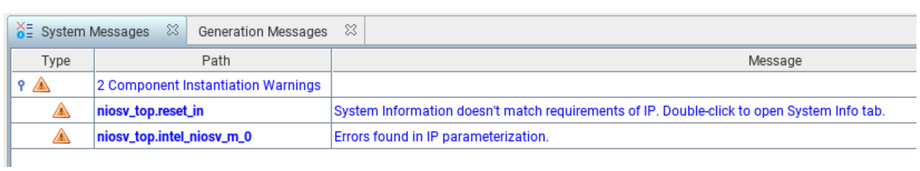
Figure 43. Full System Connection



2.2.3.2.7. Clear System Warnings and Errors

1. Navigate to the **System** menu bar.
2. Click **Assign Base Addresses**.

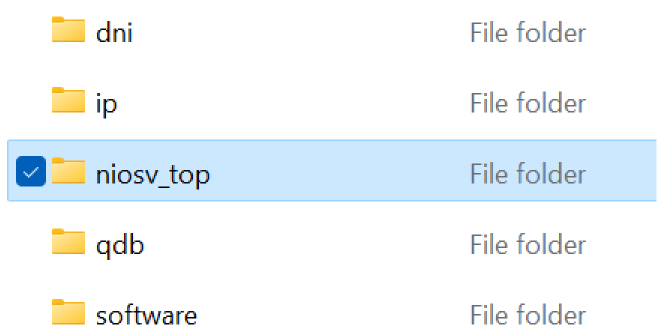
Figure 44. Example of System Connectivity Issues



2.2.3.2.8. Saving and Generating System HDL

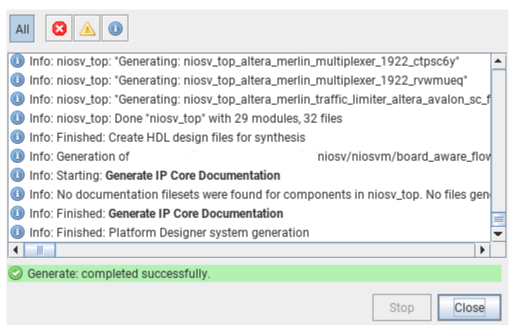
1. Save the system.
2. Click **Generate HDL** at the bottom right corner of the Platform Designer.
3. Leave all settings at default and click **Generate**.
4. Exit the Platform Designer and return to the project front page.

Figure 45. Generated Project Files



Ensure that the Nios V processor hardware system is successfully generated.

Figure 46. Successful Generation



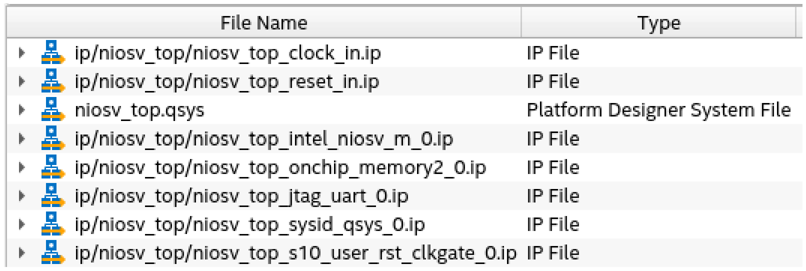
The Platform Designer generates a folder name `niosv_top`, which stores the system generation files.

2.2.3.3. Configuring Assignment and Constraint

2.2.3.3.1. Adding Design Files

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Settings**.
2. Navigate to **Files** category. You can find all related IP files and QSYS file added into your project.

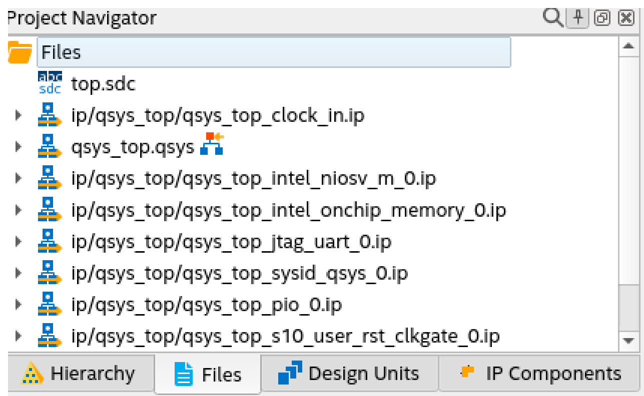
Figure 47. Settings – Files Category



File Name	Type
ip/niosv_top/niosv_top_clock_in.ip	IP File
ip/niosv_top/niosv_top_reset_in.ip	IP File
niosv_top.qsys	Platform Designer System File
ip/niosv_top/niosv_top_intel_niosv_m_0.ip	IP File
ip/niosv_top/niosv_top_onchip_memory2_0.ip	IP File
ip/niosv_top/niosv_top_jtag_uart_0.ip	IP File
ip/niosv_top/niosv_top_sysid_qsys_0.ip	IP File
ip/niosv_top/niosv_top_s10_user_rst_clkgate_0.ip	IP File

3. Earlier during the project creation, the top-level design entity is named `top`. Thus, the `niosv_top.qsys` file is automatically the top-level design entity.
4. Check the top-level design entity assignment in the **Project Navigator**.

Figure 48. Project Navigator



2.2.3.3.2. Adding Synopsys Design Constraint (SDC) File

1. Click **New**, select **Synopsys Design Constraint File** and click **OK**.
2. Add the following constraint:

```
create_clock -name clk -period 10.0 [get_ports clk_clk]
```

3. Save as `niosv_top.sdc`.

2.2.3.3.3. Pin Assignments

1. In Quartus Prime software, navigate to **Processing** menu bar and click **Start ➤ Start Analysis & Synthesis**.
2. Once the analysis is complete, navigate to **Assignments** menu bar and click **Pin Planner**. For this example, there is 2 pins assignment:
   - `clk_clk` assigned to PIN_U52 (FPGA_SYSTEM_CLK)
   - `reset_reset_n` assigned to PIN_G52 (FPGA_SYS_RESETn)
3. Close **Pin Planner** and return to the project front page.

2.2.3.3.4. Configuration and Power Management Assignments

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Device ➤ Device and Pin Options**.
2. Navigate to **Configuration** category.
3. Select **VID mode of operation** as **PMBus Master**.
4. Click **Configuration Pin Options**, and configure the following **Configuration pin** assignments,
    - **PWRMGT_SCL** as `SDM_IO0`
    - **PWRMGT_SDA** as `SDM_IO12`
    - **CONF_DONE** as `SDM_IO16`
    - Leave the rest empty
5. Navigate to **Power Management & VID** category, and configure the following settings:
    - **Bus speed mode** as `100 kHz`
    - **Slave device type** as `Other`
    - **PMBus device 0 slave address** as `42`
    - **Voltage output format** as `Linear format`
    - **Linear format N** as `-13`
    - **Translated voltage value** unit as Volts
    - Turn off **PAGE command**
6. Click **OK** and return to the project front page.

2.2.3.4. Compiling the Quartus Prime Software Project

1. Navigate to **Processing** menu bar and click **Start Compilation**.
2. Make sure that the Quartus Prime project compiles successfully.

2.2.4. Building Hardware Design in Platform Designer — Configurable Example Design

2.2.4.1. Creating a New Project

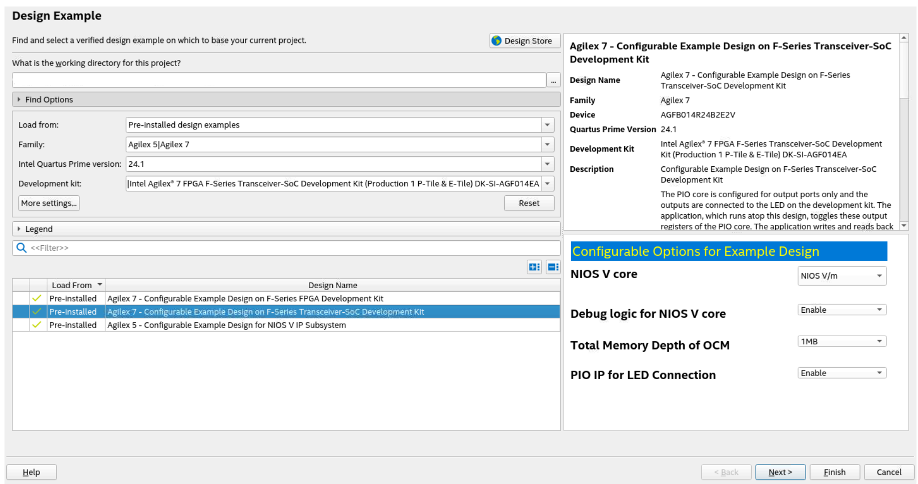
Start developing the Nios V processor system by choosing the example project in Quartus Prime software project.

1. Launch **Quartus Prime Pro Edition** software.
2. Click **Open Example Project** to select the configurable example design.

Figure 49. New Project Wizard

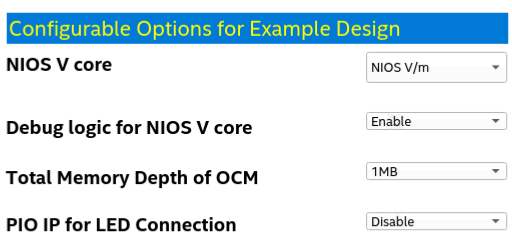
3. Select **Agilex 7 - Configurable Example Design on F-series Transceiver-SoC Development Kit**.

Figure 50. Open Example Project



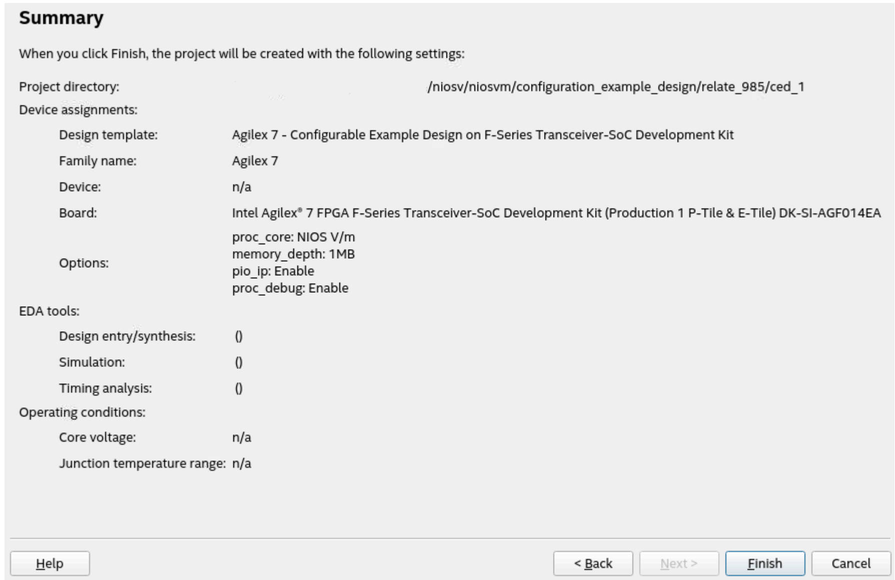
4. In **Configurable Options for Example Design**, disable **PIO IP for LED Connection**.

Figure 51. Configurable Options for Example Design



5. Click **Next** to review the summary of the new project. Then click **Finish**.

Figure 52. Summary



2.2.4.2. Creating a Platform Designer System

1. In **Project Navigator**, click **Files** and double-click on the QSYS file to open the system hardware.

Figure 53. Project Navigator

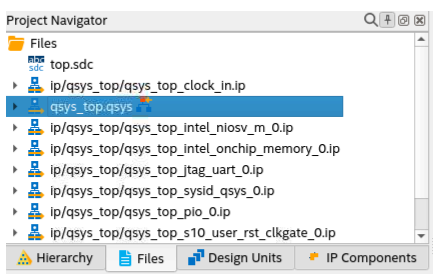
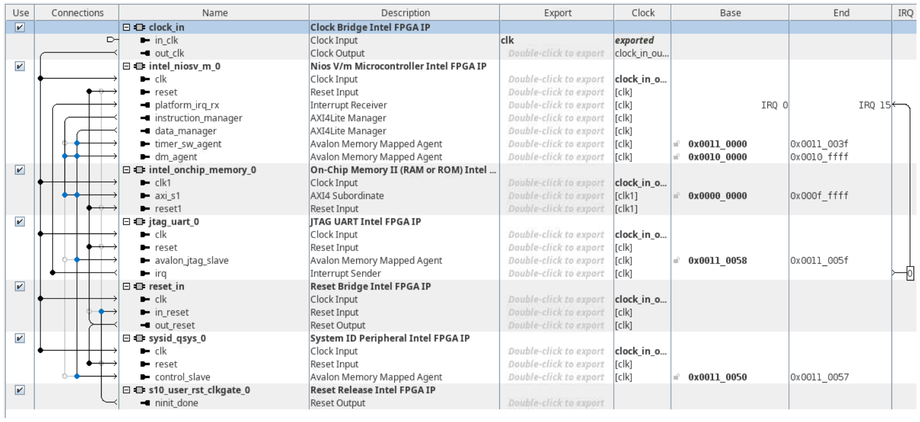


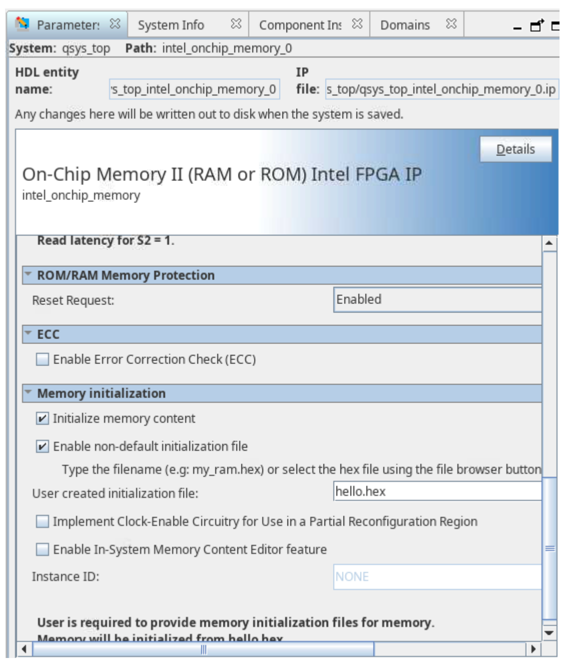
Figure 54. Full System Connection

The full system connection created in configurable example design.



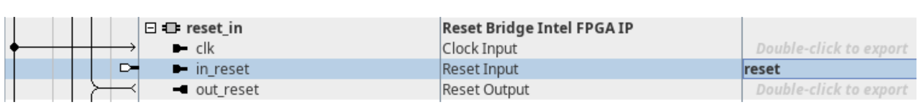
2. Click **On-Chip Memory II (RAM or ROM) IP** in the **System View**.
3. In the **Parameter** window, navigate to the **Memory Initialization**, enable **Initial memory content** and **Enable non-default initialization file**. Provide the filename `hello.hex`.

Figure 55. On-Chip Memory II (RAM or ROM) IP Parameter Editor



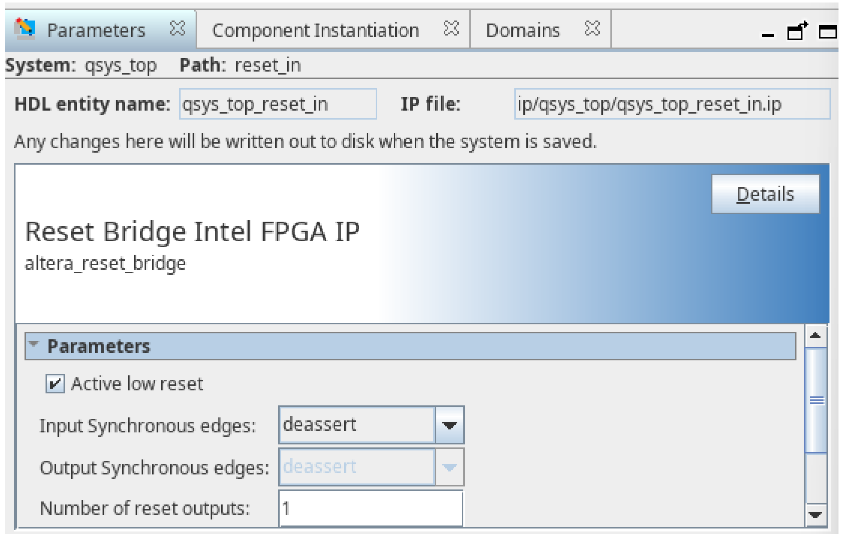
4. Click **Reset Bridge IP** in the **System View**.
5. Navigate to **Export** column, double-click on `in_reset` to export the reset pin.

Figure 56. Export Reset Pin



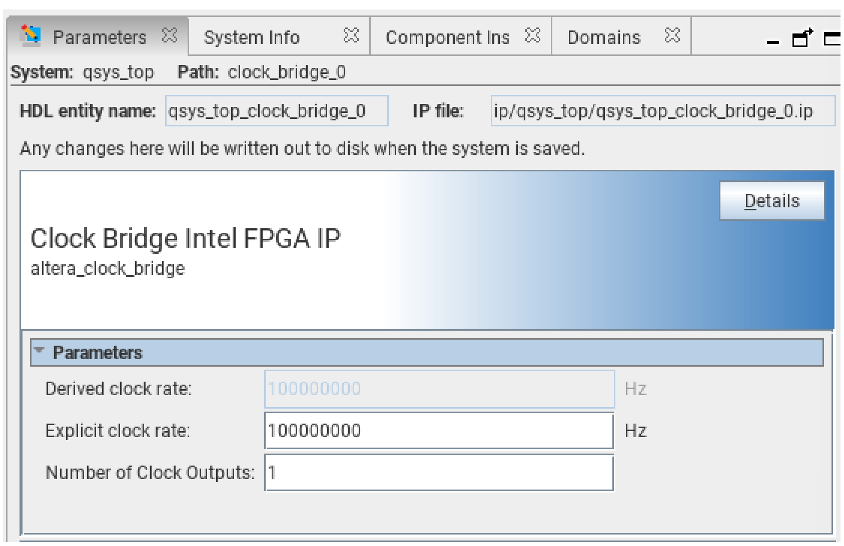
6. In the **Parameter** window, navigate to **Parameters** and enable **Active low reset**.

Figure 57. Reset Bridge IP Parameter Editor



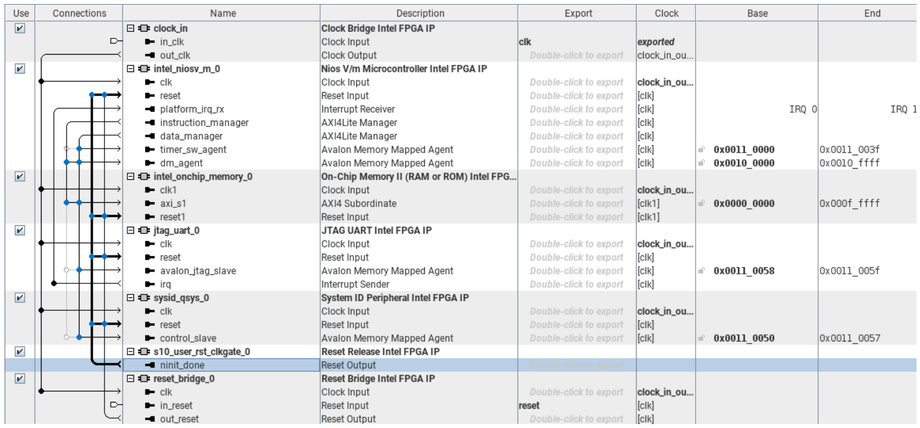
7. Click **Clock Bridge IP** in the **System View**.
8. In the **Parameter** window, set the **Explicit clock rate** to **100MHz**.

Figure 58. Clock Bridge IP Parameter Editor



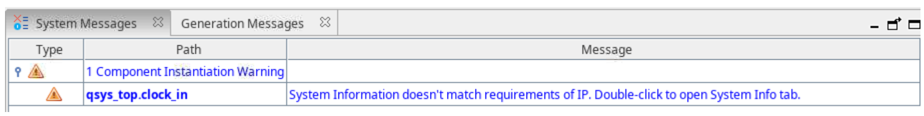
9. Connect all `reset` to `ninit_done` in **Reset Release IP.**

Figure 59. Full System Connection



10. Click **Sync System Info** at the bottom right corner of the Platform Designer.

Figure 60. Example of System Connectivity Issues



2.2.4.2.1. Saving and Generating System HDL

1. Save the system.
2. Click **Generate HDL** at the bottom right corner of the Platform Designer.
3. Leave all settings at default and click **Generate**.

Figure 61. Generation Window

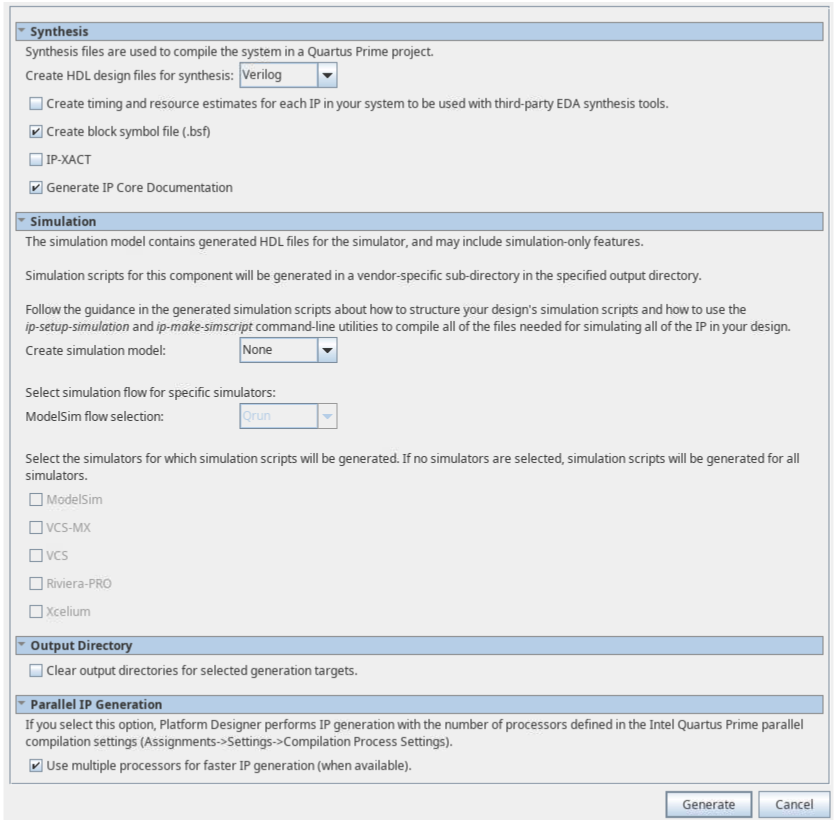
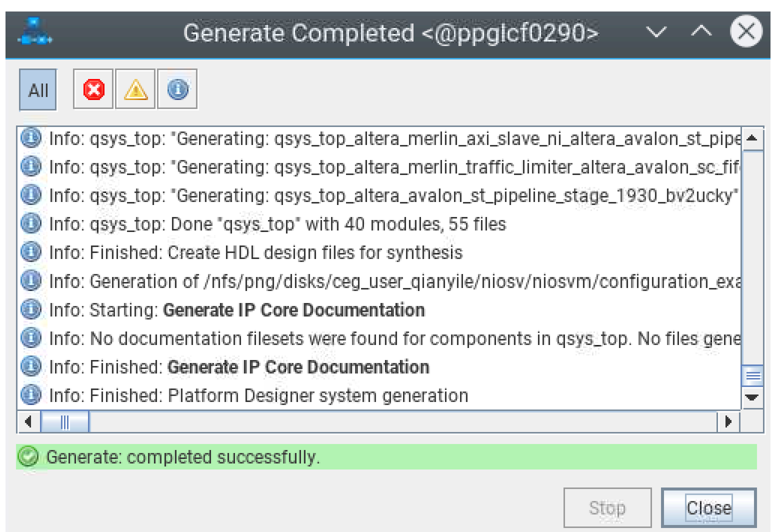
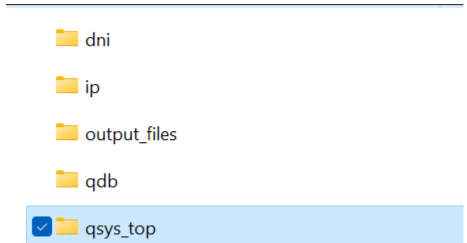


Figure 62. Successful Generation



4. Exit the Platform Designer and return to the project front page.

Figure 63. Generated Project Files

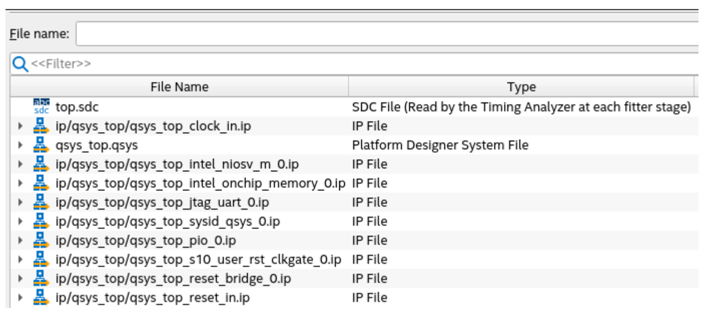


2.2.4.3. Configuring Assignment and Constraint

2.2.4.3.1. Adding Design Files

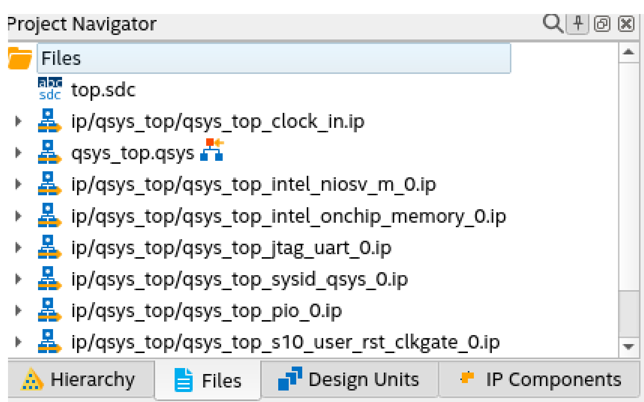
1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Settings**.
2. Navigate to **Files** category. You can find all related IP files and QSYS files added to your project.

Figure 64. Settings – Files Category



3. During the project creation, the top-level design entity is named `top`. Thus, the `qsys_top.qsys` file is automatically the top-level design entity.
4. Check the top-level design entity assignment in the **Project Navigator**.

Figure 65. Project Navigator



2.2.4.3.2. Pin Assignments

1. In Quartus Prime software, navigate to **Processing** menu bar and click **Start ➤ Start Analysis & Synthesis**.
2. Once the analysis is complete, navigate to **Assignments** menu bar and click **Pin Planner**. For this example, there are 2 pins assignments:
   - `clk_clk` assigned to PIN_U52 (FPGA_SYSTEM_CLK)
   - `reset_reset_n` assigned to PIN_G52 (FPGA_SYS_RESETn)
3. Close **Pin Planner**, and return to the project front page.

2.2.4.3.3. Configuration and Power Management Assignments

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Device ➤ Device and Pin Options**.
2. Navigate to **Configuration** category.
3. Select **VID mode of operation** as **PMBus Master**.
4. Click **Configuration Pin Options**, and make the following **Configuration pin** assignments,
   - **PWRMGT_SCL** as `SDM_IO0`
   - **PWRMGT_SDA** as `SDM_IO12`
   - **CONF_DONE** as `SDM_IO16`
   - Leave the rest empty.
5. Navigate to **Power Management & VID** category, and make the following settings
   - **Bus speed mode** as `100 kHz`
   - **Slave device type** as `Other`
   - **PMBus device 0 slave address** as `42`
   - **Voltage output format** as `Linear format`
   - **Linear format N** as `-13`
   - **Translated voltage value unit** as `Volts`
   - Turn off **PAGE command**.
6. Click **OK**, and return to the project front page.

2.2.4.4. Compiling the Quartus Prime Software Project

1. Navigate to **Processing** menu bar and click **Start Compilation**.
2. Make sure that the Quartus Prime project compiles successfully.

2.2.5. Building Software Design with Ashling RiscFree IDE for Altera FPGAs

After the processor system is ready, you may begin building the software design using Ashling RiscFree IDE for Altera FPGAs software. It consists of the following steps:

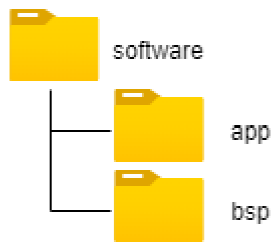
1. Create a board support package (BSP) project.
2. Create a Nios V processor application project with Hello World source code.
3. Import both projects into RiscFree IDE's workspace.
4. Build the Hello World application.

To ensure a streamlined build flow, you are encouraged to create similar directory tree in your design project. The following software design flow is based on this directory tree.

To create the software project directory tree, follow these steps:

1. In your design project folder, create a folder called `software`.
2. In the `software` folder, create two folders called `app` and `bsp`

Figure 66. Software Project Directory Tree

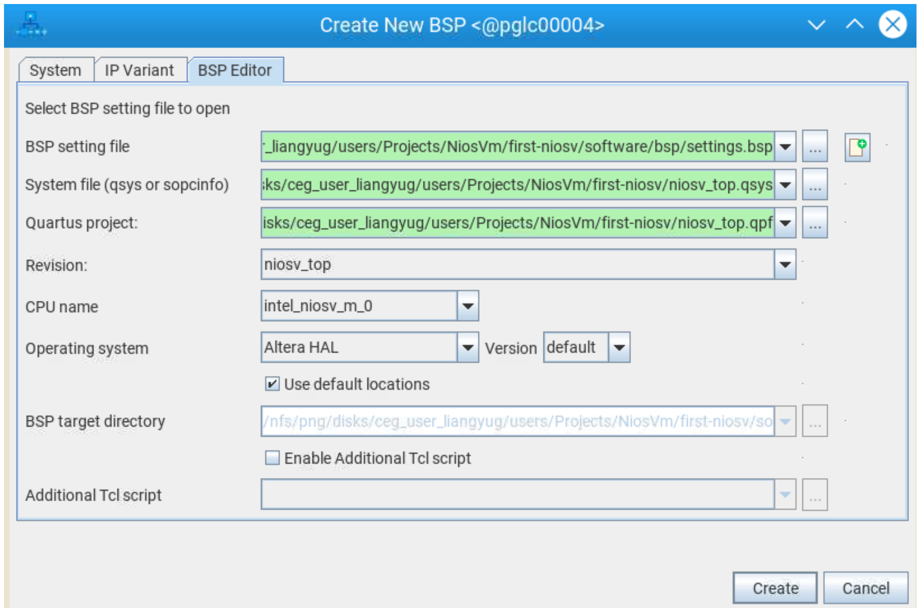


2.2.5.1. Creating a Board Support Package

Board Support Package (BSP) provides a software runtime environment for embedded systems, such as Nios V/m processor systems. Platform Designer includes the BSP Editor tool, to generate and configure BSP contents.

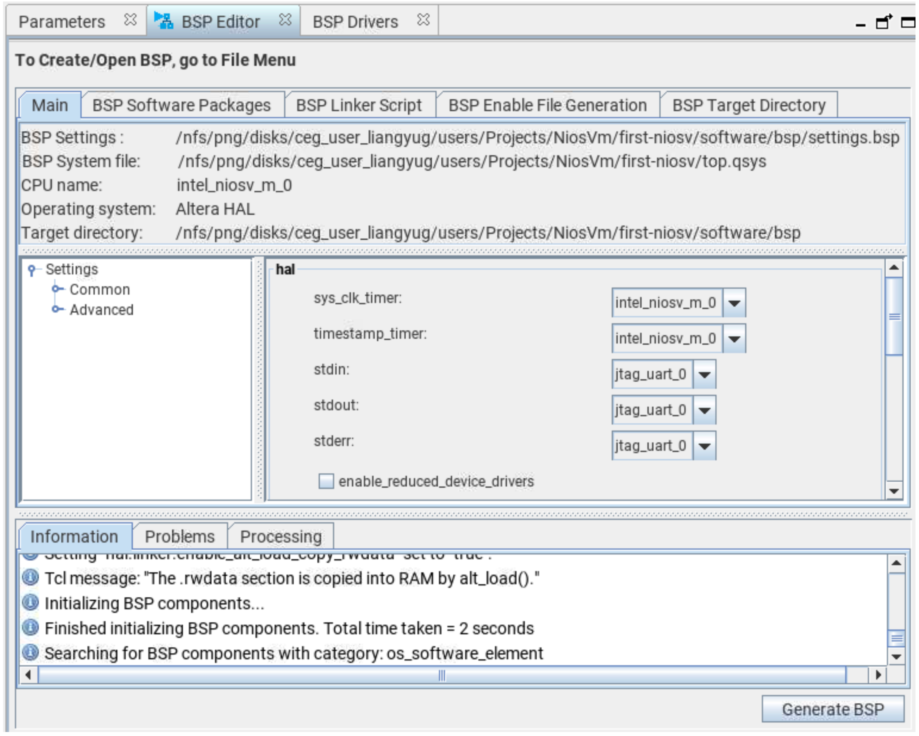
1. In the Quartus Prime software, go to **Tools ➤ Platform Designer**.
2. In the Platform Designer window, go to **File ➤ New BSP**.
3. The **Create New BSP** window appears.
4. For **BSP setting file**, create a BSP file (`settings.bsp`) in `<Working directory>/software/bsp/settings.bsp`.
5. For **System file (qsys or sopcinfo)**, select the Nios V/m processor Platform Designer system (`niosv_top.qsys`).
6. For **Quartus project**, select the example design Quartus Project File (`niosv_top.qpf`).
7. For **Revision**, select `niosv_top`.
8. For **CPU name**, select `intel_niosv_m_0`.
9. For **Operating system**, select `Altera HAL`.
10. Click **Create** to create the BSP file.

Figure 67. Create New BSP window



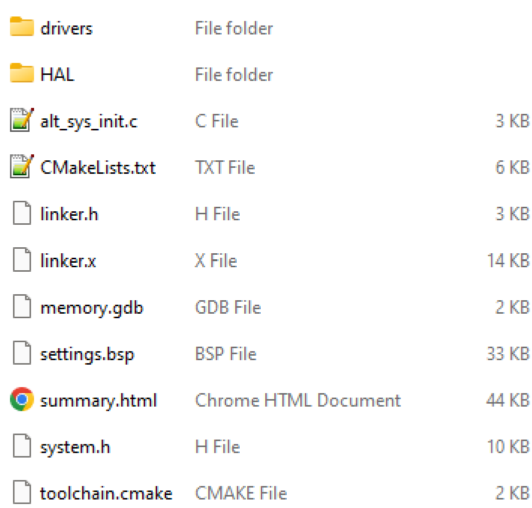
11. The **BSP Editor** tab appears.
12. You do not need to modify further. This example design uses the default BSP settings.
13. Click **Generate BSP** to generate the BSP file.

Figure 68. BSP Editor Tab



14. The BSP Editor generates the BSP files in `<Working directory>/software/bsp` folder.

Figure 69. Generated BSP Files



Alternatively, you can generate the BSP files using CLI.

a. Search Nios V Command Shell in the **Start Menu** and launch it.

b. Launch the Nios V Command Shell.

```
$ niosv-shell
```

c. Execute the command below to generate a BSP folder.

```
$ niosv-bsp -c -p=niosv_top.qpf -s=niosv_top.qsys -t=hal <Working
directory>/software/bsp/settings.bsp
```

2.2.5.2. Creating an Application Project File

Application Project (app) provides the software application for Nios V/m processor system.

1. In `<Working directory>/software/app` folder, create a C source code. Name it as `hello.c`.
2. In `hello.c`, copy and paste the Hello World application code below

```
#include <stdio.h>
#include <unistd.h>

void looper() {
    for (int i = 0; i < 1000; ++i) {
        printf("Hello world, this is the Nios V/m cpu checking in %d...\n",
i);
    }
}

int main() {
    looper();
    usleep(1000000);
    printf("Bye world!\n");
    fflush(stdout);
    return 0;
}
```

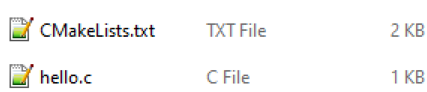
3. Launch the Nios V Command Shell.

```
$ niosv-shell
```

4. Execute the command below to generate an application `CMakeLists.txt`.

```
$ niosv-app --bsp-dir=software/bsp --app-dir=software/app \
--srcs=software/app/hello.c --elf-name=hello.elf
```

Figure 70. Generated APP Files



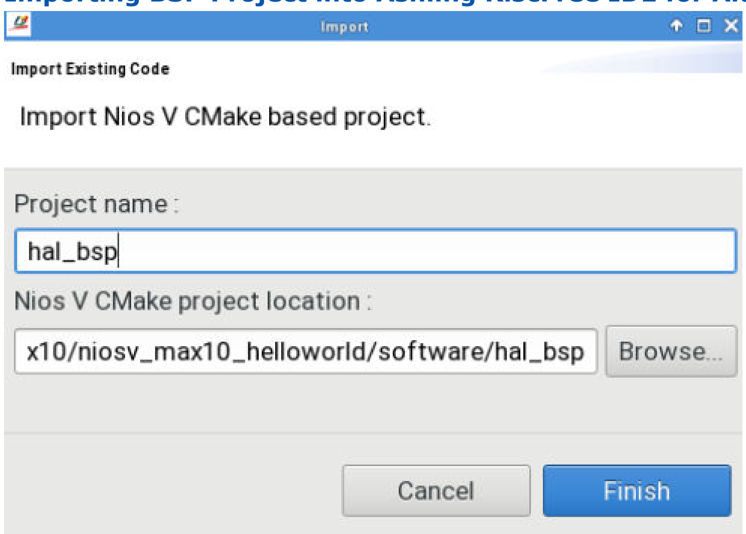
2.2.5.3. Importing Projects into Ashling RiscFree IDE for Altera FPGAs

2.2.5.3.1. Importing Nios V Processor BSP Project

Follow these steps to import the Nios V processor BSP:

1. Search Ashling RiscFree IDE for Altera FPGAs in **Start Menu** and open it.
2. Set the working directory as the **Workspace**.
3. Open the **Import** wizard, click **File ➤ Import Nios V CMake Project**.
4. In the Import window, browse and select the location of the Nios V processor BSP project.
5. The **Project name** is automatically fill according to the name of BSP project.
6. Click **Finish**. The BSP project is added to the Project Explorer.

Figure 71. Importing BSP Project into Ashling RiscFree IDE for Altera FPGAs



2.2.5.3.2. Importing Application Project

Follow these steps to import the application project:

1. Continue with the same Workspace from Importing Nios V Processor BSP Project.
2. Open the **Import** wizard, click **File ➤ Import Nios V CMake Project**.
3. In the **Import** window, browse and select the location of the Nios V processor APP project.
4. The **Project name** is automatically filled according to the name of the APP project.
5. Click **Finish**. The APP project is added to the **Project Explorer**.

2.2.5.4. Building the Hello World Application

You can browse the BSP and APP project files in Ashling RiscFree IDE for Altera FPGAs **Project Explorer** tab.

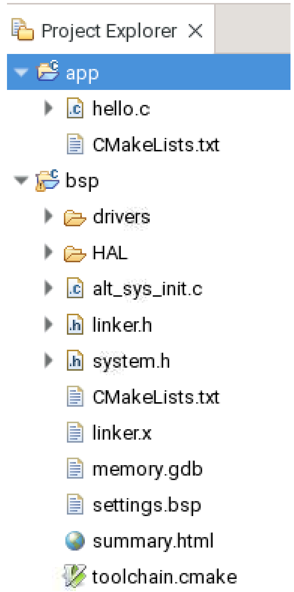


Figure 72. Project Explorer Tab (Before Build Project)

1. Go to the **Project** in the menu tab and click **Build Project**.
2. When the build is complete, you can find the console prints as shown in the examples below.
3. In addition to the console prints, you can find the generated ELF file in the APP project folder.

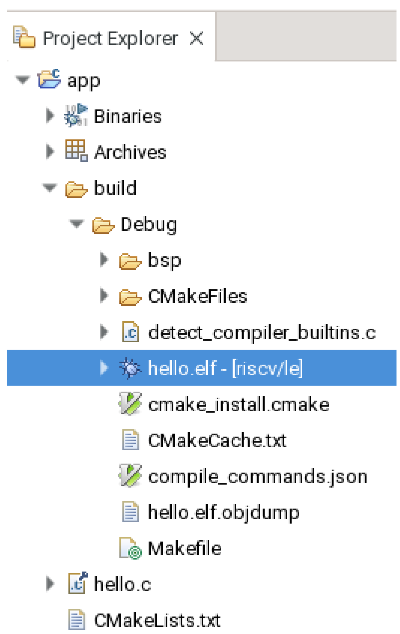


Figure 73. Project Explorer Tab (After Build Project)

Example 1. CMake Console Print

```
cmake -DCMAKE_BUILD_TYPE:STRING=Debug -DCMAKE_EXPORT_COMPILE_COMMANDS:BOOL=ON -G
"Unix Makefiles" <Working directory>/software/bsp
-- The ASM compiler identification is GNU
-- Found assembler: <Disk drive>/riscfree/toolchain/riscv32-unknown-elf/bin/
riscv32-unknown-elf-gcc
-- The C compiler identification is GNU 12.1.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: <Disk drive>/riscfree/toolchain/riscv32-unknown-
elf/bin/riscv32-unknown-elf-gcc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- The CXX compiler identification is GNU 12.1.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: <Disk drive>/riscfree/toolchain/riscv32-
unknown-elf/bin/riscv32-unknown-elf-g++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to: <Working directory>/software/bsp/build/Debug
```

Example 2. Make Console Print

```
/usr/bin/gmake -j all
Scanning dependencies of target hal2_bsp
[  3%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/crt0.S.obj
[ 14%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/machine_trap.S.obj
[ 20%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dma_txchan_open.c.obj
[ 21%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/alt_log_macro.S.obj
[ 30%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_find_dev.c.obj
[ 32%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dev_llist_insert.c.obj
[ 32%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_busy_sleep.c.obj
[ 45%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_icache_flush.c.obj
[ 48%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_irq_handler.c.obj
[ 53%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_ioctl.c.obj
[ 58%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_printf.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_alarm_start.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_write.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dcache_flush_no_writeback.c.obj
[ 98%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/alt_mcount.S.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_close.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_env_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_do_dtors.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_dev.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_errno.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dma_rxchan_open.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_exit.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_do_ctors.c.obj
[ 25%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dcache_flush_all.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_execve.c.obj
[ 26%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_dcache_flush.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fcntl.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_environ.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_find_file.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fd_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fd_unlock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fs_reg.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_getchar.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_gettod.c.obj
[ 38%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fork.c.obj
```

```
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_flash_dev.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_iic.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_instruction_exception_register.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_gmon.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fstat.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_get_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_getpid.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_io_redirect.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_iic_isr_register.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_icache_flush_all.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_kill.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_link.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putchar.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_main.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_open.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_remap_uncached.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_remap_cached.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_log_printf.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_isatty.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_sbrk.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_read.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_release_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putstr.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_rename.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_malloc_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_load.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_settod.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_stat.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putcharbuf.c.obj
[ 77%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_tick.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_usleep.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_times.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_uncached_free.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_write.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_wait.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_lseek.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_tls.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
intel_fpga_api_niosv.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_unlink.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
intel_fpga_platform_api_niosv.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/alt_sys_init.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_ioctl.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/mtimer.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_init.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_read.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/intel_niosv_irq.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_uncached_malloc.c.obj
[100%] Linking C static library libhal2_bsp.a
[100%] Built target hal2_bsp
```

Alternatively, you can build the application project using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to build the application.

```
$ cmake -G "Unix Makefiles" -DCMAKE_BUILD_TYPE=Debug -B software/app/debug -S
software/app
$ make -C software/app/debug
```

2.3. Simulating the Nios V Processor System

Before experimenting with the designs in actual hardware, Altera recommends you to simulate the whole processor system and evaluate its intended functions. The benefits of simulation is not only for resource utilization but also to protect the actual hardware from unexpected damage due to mishandling within the hardware or software design.

2.3.1. Preparing Hardware Design for Simulation

Note: Before continuing the simulation, you must successfully build the hardware SOF file (with the memory-initialization feature enabled).

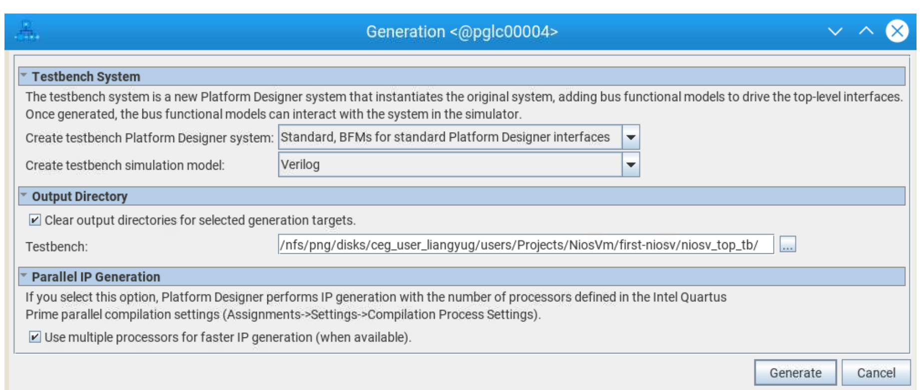
To generate hardware simulation files, perform the following steps:

1. Launch the Quartus Prime software and open the **Platform Designer** from the **Tools** menu.
2. Open the `niosv_top.qsys` file.
3. In Platform Designer, navigate to **Generate ➤ Generate Testbench System**.
4. On the **Generation** window, set the following parameters to these values:
    a. Create testbench Platform Designer system— *Standard, BFMs for standard Platform Designer interfaces*.
    b. Create testbench simulation model—*Verilog*
    c. Turn on *Use multiple processors for faster IP generation (when available)*.
5. Click **Generate**, and **Save**, if prompted.

    Altera IP cores and Platform Designer systems generate simulation setup scripts. Modify these scripts to set up supported simulators. You can find the script, `msim_setup.tcl` in the following path:

    ```
    <Working directory>/niosv_top_tb/sim/mentor
    ```

Figure 74. Testbench Generation



2.3.2. Preparing Software Design for Simulation

Note: Before continuing the simulation, you must successfully build the software ELF file.

To generate software memory initialization file, perform the following command:

```
elf2hex <Working directory>/software/app/build/Default/hello.elf \
-b 0x0 -w 32 -e 0x26fff \
<Working directory>/software/app/build/Default/hello.hex
```

The command converts the software ELF into a HEX memory initialization file for the On-Chip RAM. The RAM have a data width of 32 bits, which starts from base address of 0x0 and ends at 0x26FFF.

2.3.3. Checking Simulation Files

At this point in the design flow, you have generated your system and created all the files necessary for simulation listed in the table below.

Table 4. Simulation Files Generated

File	Description
`<Working directory>/niosv_top_tb/*`	Platform Designer generates a testbench system when you enable the **Create testbench Platform Designer** system option.
`<Working directory>/niosv_top_tb/niosv_top_tb/sim/mentor/msim_setup.tcl`	Sets up a Questa simulation environment and creates alias commands to compile the required device libraries and system design files in the correct order and loads the top-level design for simulation.
`<Working directory>/software/app/build/Default/hello.hex`	Memory Initialization Files (`.hex`) is required to initialize memory components in your system.

2.3.4. Running System Simulation

The `msim_setup.tcl` script in the package generated creates alias commands for each step. For the list of commands, refer to the following table:

Macros	Description
`dev_com`	Compile device library files.
`com`	Compiles the design files in correct order.
`elab`	Elaborates the top-level design.
`elab_debug`	Elaborates the top-level design with the `novopt` option.
`ld`	Compiles all the design files and elaborates the top-level design.
`ld_debug`	Compiles all the design files and elaborates the top-level design with the **vopt** option. *Note:* The **vopt** option is to run optimization before elaborating the top-level design in the simulator.

You can run the simulation in the Questa simulator by performing the following steps,

1. Launch the Nios V Command Shell.
2. Open the **Questa for Intel FPGA** simulator using the command `vsim`.
3. In the Questa **Transcript** window, change your working directory to the `mentor` folder.

```
cd <Working directory>/niosv_top_tb/niosv_top_tb/sim/mentor
```

4. Copy the memory initialization file into the `mentor` folder.

```
file copy -force \
<Working directory>/software/app/build/Default/hello.hex ./
```

5. Run the `msim_setup.tcl`.

```
do msim_setup.tcl
```

6. Compiles all the design files and elaborates the top-level design with **vopt** option.

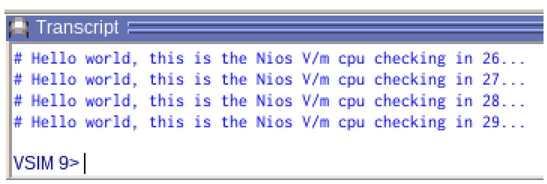
```
ld_debug
```

7. Run the simulation for more than 5 milliseconds.

```
run 10ms
```

At the end of the simulation, you can find the following message prints in the Questa **Transcript** window: `Hello world, this is the Nios V/m cpu checking in <loop number>`.

Figure 75. Questa Transcript Window Message



You can observe the simulation results from the waveform viewer:

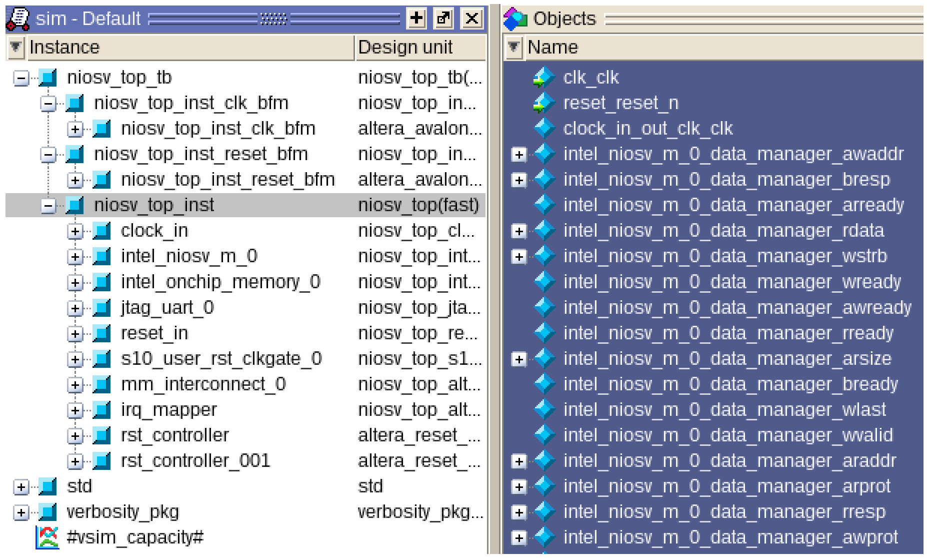
1. In Questa for Intel FPGA simulator, navigate to the **Instance** window.
2. Unroll `niosv_top_tb`.
3. Select `niosv_top_inst`, and the simulator populates a list of signals in the **Objects** window.
4. Right-click any of the signals, and click **Add Waves** to add them into the **Wave** window.
5. Restart and run the simulation for more than 5 milliseconds.

```
restart
```

```
run 10ms
```

6. Wait for the simulator to complete the given time.

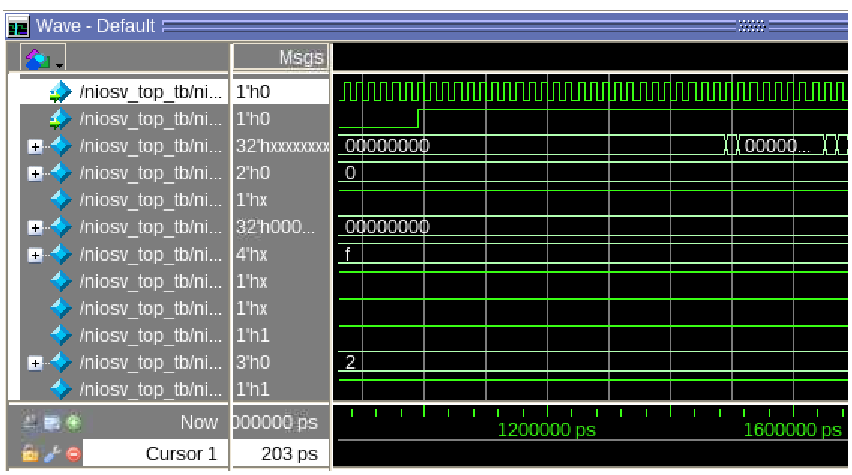
Figure 76. Questa Instance and Objects Windows



The following figure shows the simulated wave result.

- First signal: Clock Input
- Second signal: Reset Input
- Other signals: Any one or more signals within the `niosv_top_inst`

Figure 77. Waveform Viewer



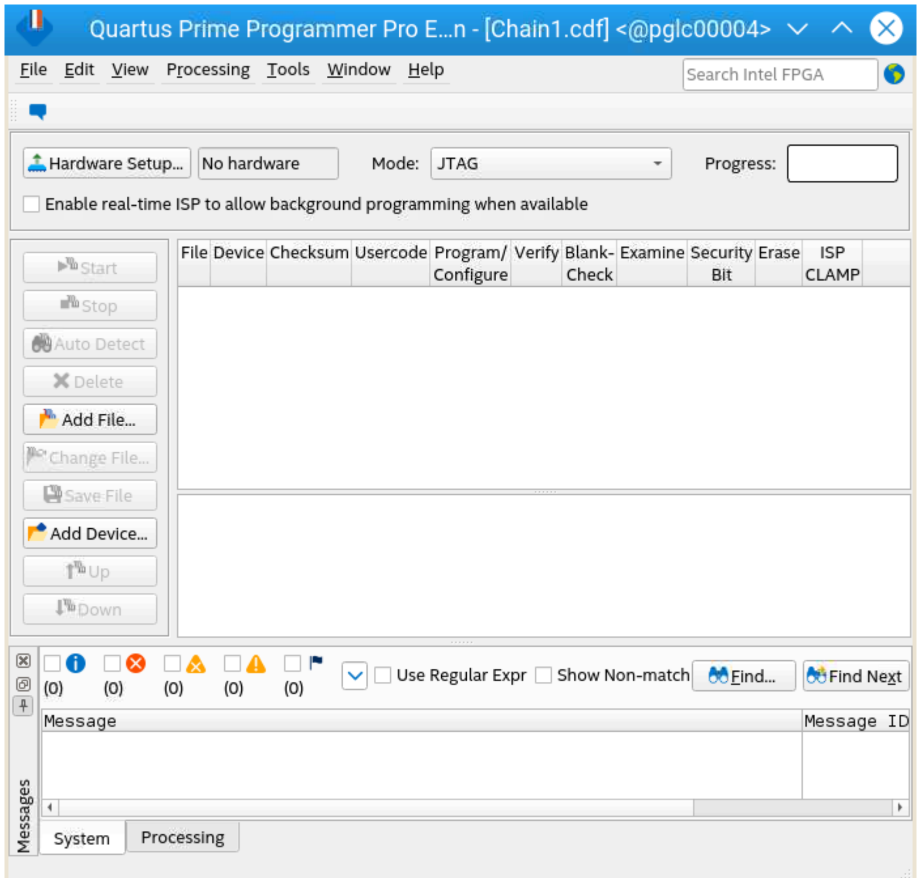
2.4. Programming the System

Use the Quartus Prime Programmer tool and the Ashling RiscFree IDE for Altera FPGAs to program the Nios V processor-based system into the FPGA and to run your application.

2.4.1. Programming Hardware SOF File

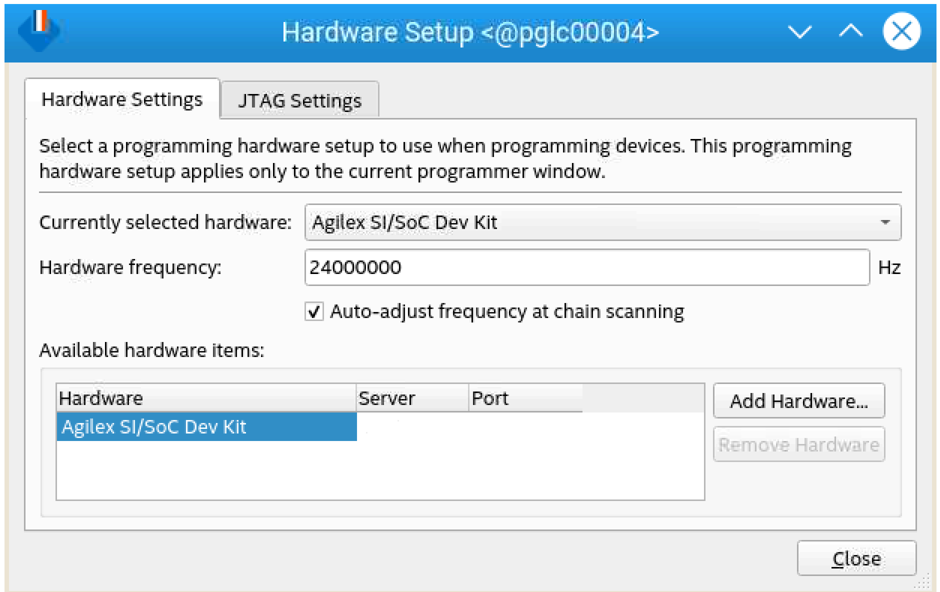
1. Connect the Agilex 7 FPGA F-Series Transceiver-SoC Development Kit to the host PC using the USB Blaster II.
2. Open the Quartus Prime Programmer.

Figure 78. Quartus Prime Programmer



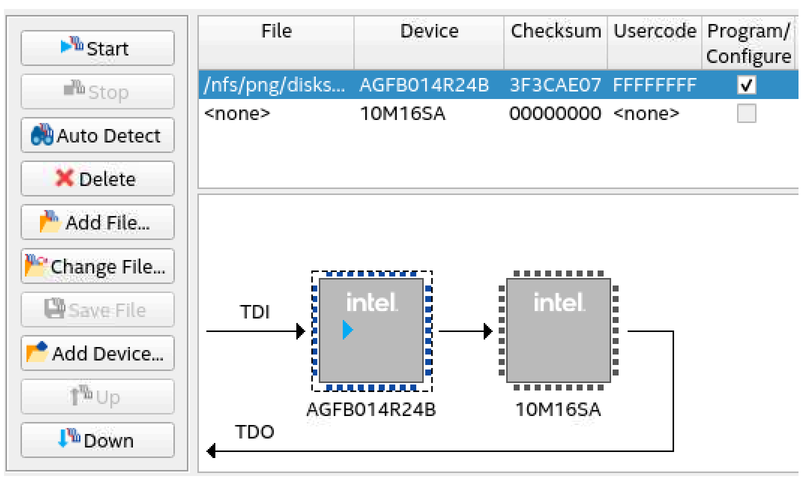
3. Click **Hardware Setup**.
4. Check the availability of the development kit in **Available hardware items**.
   a. If available, select the development kit in **Currently selected hardware**.
   b. If not available, check the cable connection, and the JtagServer driver installation.

Figure 79. Hardware Setup



5. Click **Auto Detect** and select the appropriate device OPN.
6. Select the Agilex device, click **Change File** and select `niosv_top.sof` file.
7. Once the SOF file is ready, check **Program/Configure** and click **Start**.

Figure 80. List of Devices with JTAG Chain



8. Wait until the **Progress** bar reaches 100% (Successful).

Figure 81. Progress Bar

9. You have successfully configured the development kit with the processor hardware system.

   Alternatively, you can program the hardware SOF file through CLI.

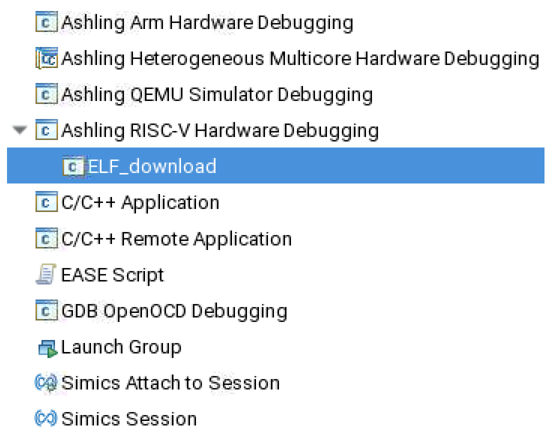
Execute the following command to program the SOF file.

```
$ quartus_pgm -c 1 -m JTAG -o p;<Working directory>/output_files/
niosv_top.sof@1
```

2.4.2. Downloading the Software ELF File

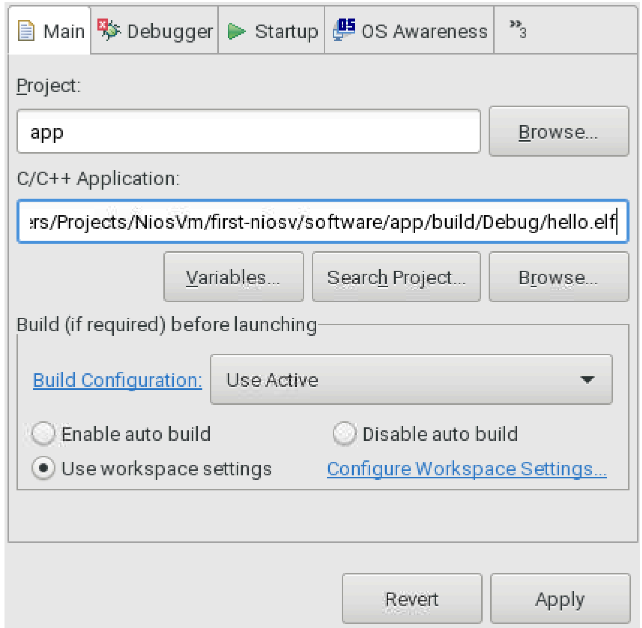
1. Ensure that the development kit is successfully configured with the processor system.
2. Launch the Ashling RiscFree IDE for Altera FPGAs.
3. Navigate to **Run ➤ Run Configurations**.
4. In **Run Configuration** window, double click **Ashling RISC-V Hardware Debugging** and name it as `ELF_download`.

Figure 82. Run Configuration



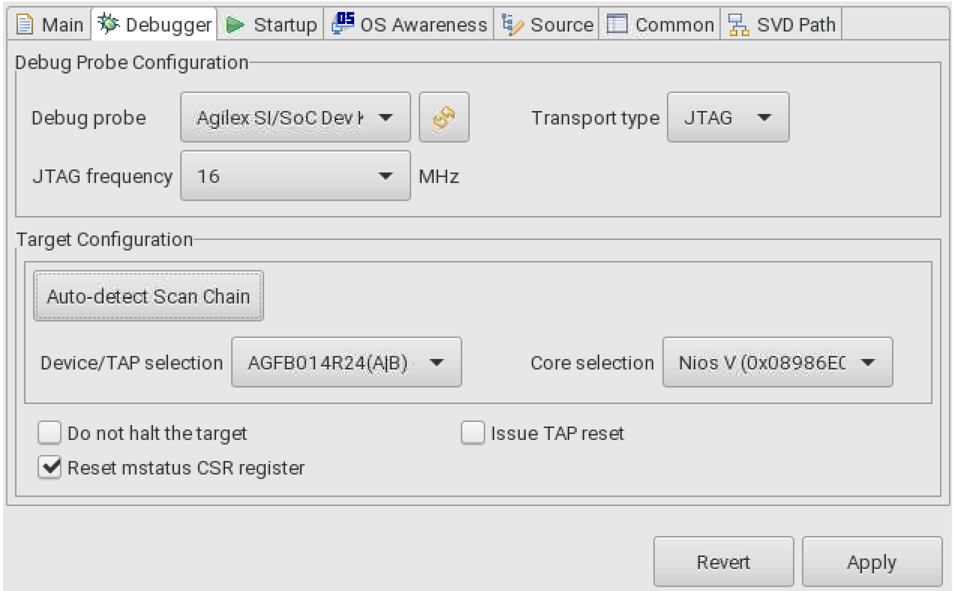
5. In the **Main** tab, make the following settings:
   a. **Project**: app
   b. **C/C++ Application**: `<Working directory>/software/app/build/Debug/hello.elf`

Figure 83. Main Tab



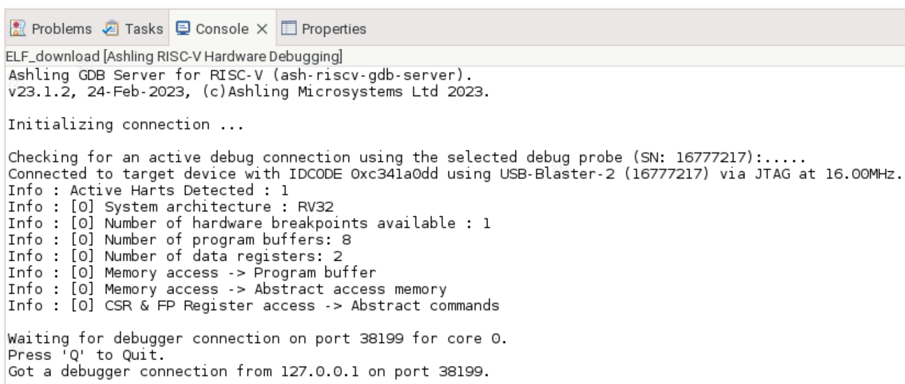
6. In the **Debugger** tab, make the following settings:
   a. **Debug Probe Configuration**:
      i. **Debug probe**: Agilex development kit
      ii. **Transport type**: JTAG
      iii. **JTAG frequency**: 16 MHz
   b. **Target Configuration**: Click **Auto-detect Scan Chain** to list all possible cores. Select the appropriate **Device/TAP** and **Nios V Processor Core**.

Figure 84. Debugger Tab



7. Click **Apply** and **Run**. Ashling RiscFree IDE for Altera FPGAs prints the following message in its **Console**.

Figure 85. ELF_download Message (Console tab)



Alternatively, you can download the software ELF file using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to download the ELF file.

```
$ niosv-download <Working directory>/software/app/debug/hello.elf -g -r
```

2.4.3. Applying External Tool Configuration

External tool configuration is a generic feature where you can configure the Ashling RiscFree IDE for Altera FPGAs to include external tools. An example of such tools is the `juart-terminal` executable, which is used to display the Hello World message through the JTAG UART IP.

To perform external tool configuration for `juart-terminal`, follow these steps:

1. Go to **Run ➤ External Tools ➤ External Tools Configurations**.
2. Double click **Program** to open a **New_configuration** window.
3. Rename the configuration as `Nios V JTAG UART Output`.
4. In **Location**, click **Browse File system**.
5. Browse and select the `juart-terminal` file in the following paths:

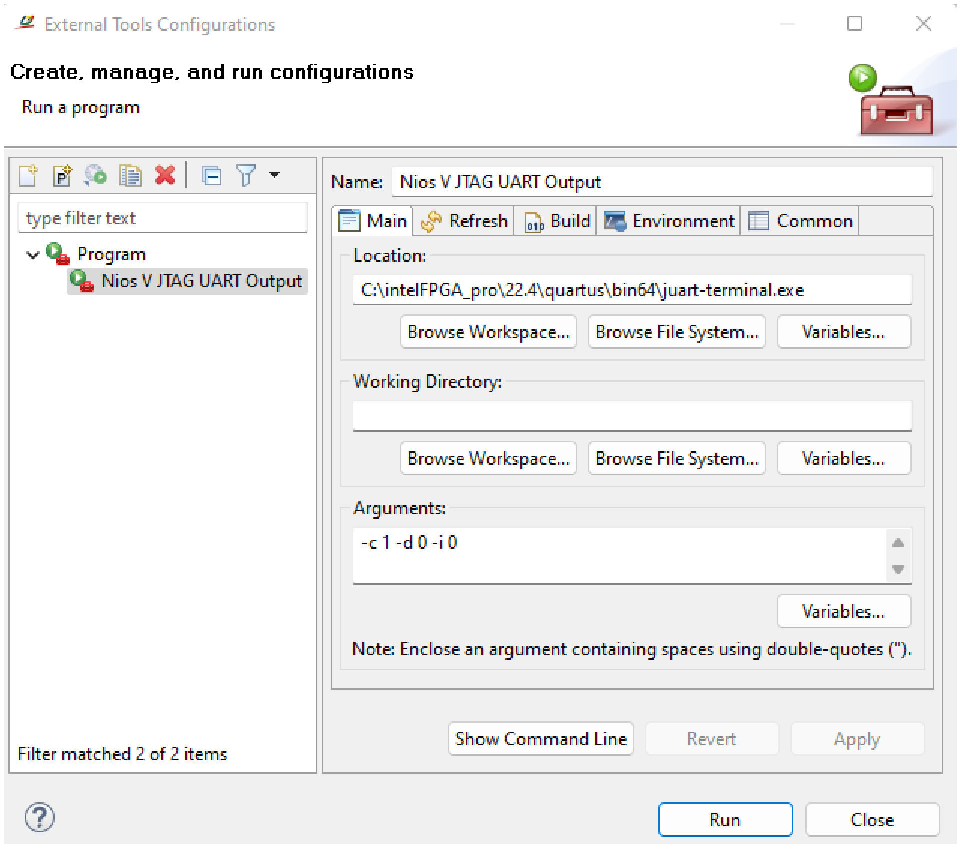
Table 5. Path

Operating System	Path
Windows*	`<Quartus Prime installation directory>/quartus/bin64/juart-terminal.exe`
Linux*	`<Quartus Prime installation directory>/quartus/linux64/juart-terminal`

6. Set the **Arguments** as "-c 1 -d 1 -i 0". This configures the JTAG UART connection is towards the JTAG UART IP at cable 1, device 1, and instance 0.
7. Click **Apply**.

Note: JTAG device chain index varies according to your setup. Run `jtagconfig --debug` to obtain the latest device chain index numbering.

Figure 86. External Tools Configuration

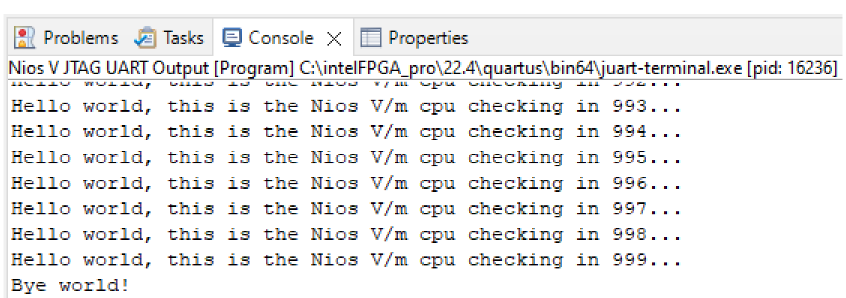


2.4.4. Run Hello World Application

With the **External Tool Configuration (Nios V JTAG UART Output)** defined successfully, you may proceed to use it to display the Hello World application.

1. Go to **Run ➤ External Tools ➤ External Tools Configurations**.
2. Select **Nios V JTAG UART Output**.
3. Click **Run**.
4. The Hello World message is printed on the **Console** tab.

Figure 87. Hello World Message (Console tab)



Alternatively, you can display the Hello World application using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to display the Hello World application.

```
$ juart-terminal -c 1 -d 0 -i 0
```

2.5. Debugging the System

Occasionally, the generated system might fail to run or display unexpected behaviors. You can debug the hardware and software system to pinpoint the root cause and ultimately return the system to normal. Optionally, you can also apply the debug tools to view a processor system's inner workings and running mechanisms while on-the-fly.

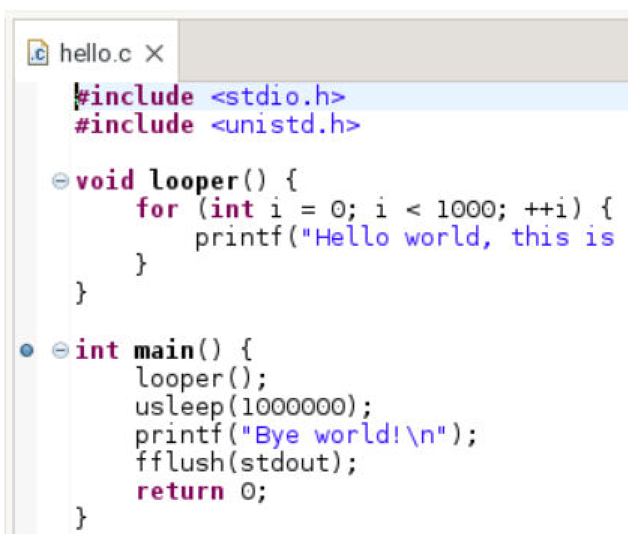
2.5.1. Software Debugging

2.5.1.1. Setting Initial Breakpoint

Before debugging any system, you need to have at least one breakpoint to suspend the execution inside the program.

1. Open the Ashling RiscFree IDE for Altera FPGAs.
2. Open the Nios V processor software project (hello.c).
3. Set a breakpoint at `main()` by double-clicking on the left-margin of the line containing `main()`.
4. Ensure that a blue circle have appeared beside `main()`.

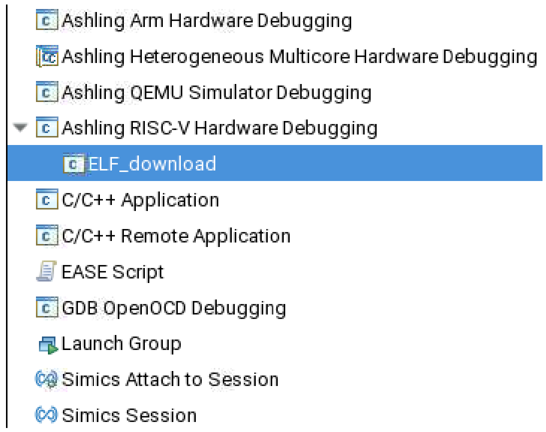
Figure 88. Initial Breakpoint at main()



2.5.1.2. Starting the Debugger

1. In the **Run** menu tab, select **Debug Configurations**.
2. Under **Ashling RISC-V Hardware Debugging**, find `ELF_download` (Created in Downloading Software ELF File.
3. Select `ELF_download` to start the Debugger using the same settings.

Figure 89. Debug Configuration



4. In the **Main** tab, check the following settings
   a. **Project**: `app`
   b. **C/C++ Application**: `<Working directory>/software/app/build/Debug/hello.elf`
5. In the **Debugger** tab, check the following settings:
   a. For **Debug Probe Configuration**,
      i. **Debug probe**: Agilex development kit
      ii. **Transport type**: JTAG
      iii. **JTAG frequency**: 16 MHz
   b. **Target Configuration**: Click **Auto-detect Scan Chain** to list all possible cores. Select the appropriate **Device/TAP** and **Nios V Processor Core**.

Figure 90. Main Tab

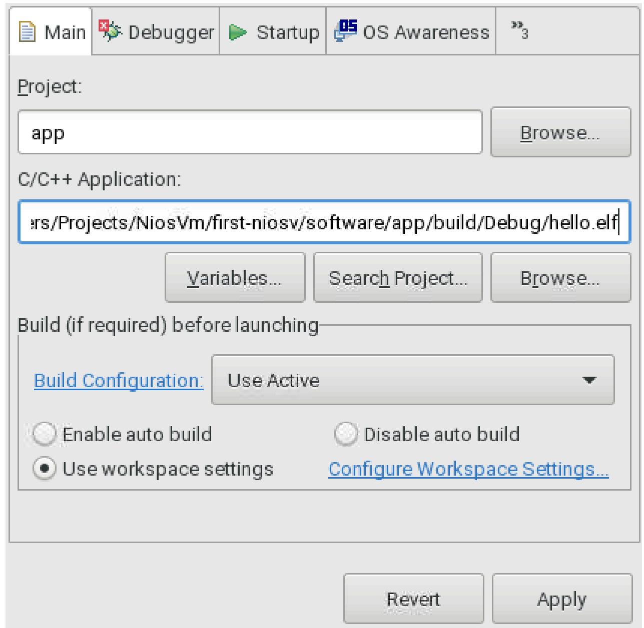
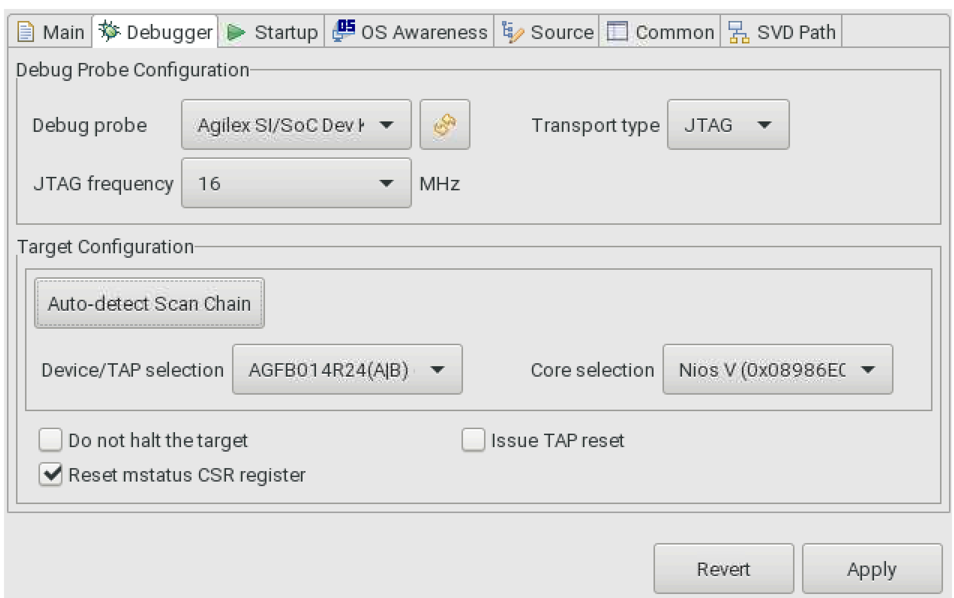
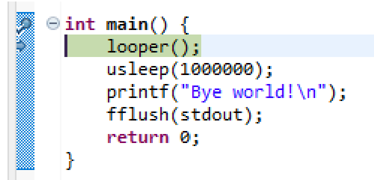


Figure 91. Debugger Tab



6. Click **Apply** and **Debug**.
7. The Ashling RiscFree IDE for Altera FPGAs switches to the **Debug Perspective**.
8. The program begins execution, and suspends at the initial breakpoint, `main()`.

Figure 92. Suspended at Initial Breakpoint



2.5.1.3. View Global Variables

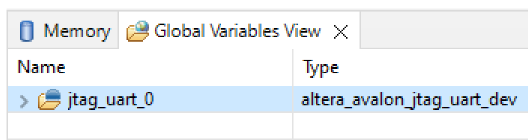
Ashling RiscFree IDE for Altera FPGAs has dedicated global variables.

To view the global variables in the application (`.elf`), follow these steps:

1. Go to **Window ➤ Show View ➤ Other... ➤ Debug ➤ Global Variables View**.
2. Right-click in the **Global Variables View** tab, and select the **Add Global Variables** icon.
3. Select the required variables to view.

Note: You can change the value of a variable. Right-click the variable and select **Change Value**.

Figure 93. Global Variables View



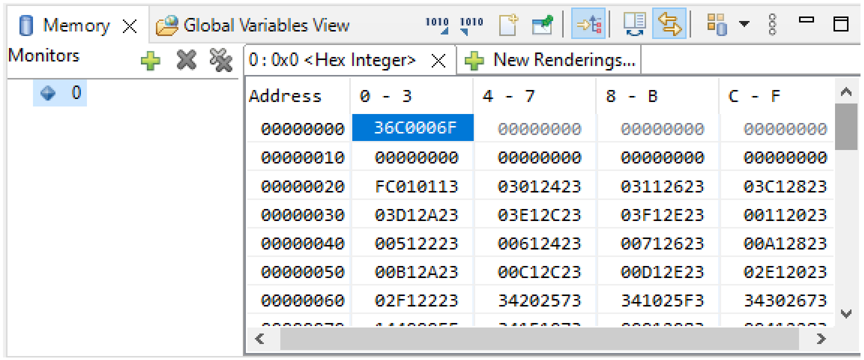
2.5.1.4. View Memory

Ashling RiscFree IDE for Altera FPGAs supports memory browser. You can view the content of On-Chip Memory (RAM) or other memory devices. This example targets the start address of On-Chip Memory (RAM), which the Hello World application begins.

To launch the memory browser, follow these steps:

1. Go to **Window ➤ Show View ➤ Memory Browser**.
2. Select **Add Memory Monitor**.
3. Provide the memory address 0x0, and click **OK**.

Figure 94. Memory Browser at Address 0x0



4. Go to `<Working directory>/software/app/build/Debug` folder.
5. Open the `hello.elf.objdump` file.
6. Search for `Disassembly of section .entry`.
7. The disassembly shows that the information at starting address 0 is 0x36c006f, which is exactly the same as in the **Memory Browser**.
8. You can continue to verify section `.exceptions`.

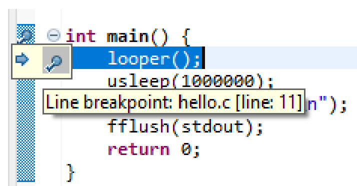
Figure 95. Disassembly of Hello World Application

```
Disassembly of section .entry:

00000000 <__reset>:
 * Jump to the _start entry point in the .text section if reset code
 * is allowed or if optimizing for RTL simulation.
 */

    /* Jump to the _start entry point in the .text section. */
    tail _start
   0:   36c0006f            j   36c <_start>
    ...
```

2.5.1.5. Setting a Second Breakpoint

1. Open the Nios V processor software project (`hello.c`).
2. Set a second breakpoint at `looper()` by double-clicking on the left-margin of the line containing `looper()`.
3. Ensure that a blue circle has appeared beside `looper()`.

Figure 96. Second Breakpoint at `looper()`



2.5.1.6. Control Debugger

Using the two breakpoints, you can control the debugging process using the Debug bar.

Figure 97. Execution Control (Debug bar)

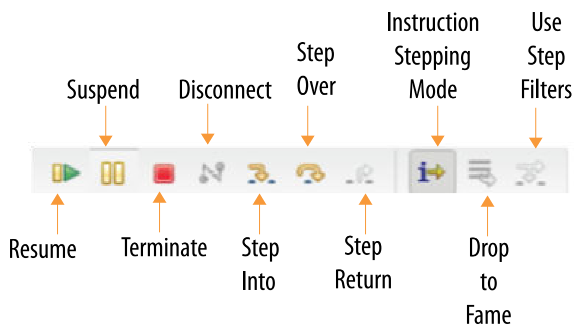


Table 6. Description of Execution Control

Function	Description
Resume	Continues the debugging process until the next breakpoint, or until the end of the debugging process.
Suspend	Halts the running code while debugging.
Terminate	Terminates the debug session. You need to relaunch the debug session after terminated.
Disconnect	Disconnects the debug tool connected to the RiscFree* IDE. Note: You are required to terminate the debug session before disconnecting the debug tool.
Step Into	Steps into the next method call (entering it) at the currently executing line of code.
Step Over	Steps over the next method call (without entering it) at the currently executing line of code. The method is still executed.
Step Return (Step Out)	Returns from a method which has been stepped into (exiting it). The remainder of the code that was skipped by returning is still executed.
Instruction Stepping Mode	Switches to instruction stepping mode.
Drop to Frame	Pops the current stack frame and puts control back out to the calling method, resetting any local variables.
Use Step Filters	Changes whether step filters should be used in the current Debug View.

3. Hello World on MAX® 10 FPGA 10M50 Evaluation Kit

3.1. Hardware and Software Requirements

Hardware Requirements

- Any Altera FPGA Development Kits

  Note: This tutorial uses MAX® 10 10M50 Evaluation Kit (DK-DEV-10M50F484-B).
- Power adapter
- USB Blaster II

Software Requirements

- Quartus Prime Standard Edition Software
- Ashling RiscFree IDE for Altera FPGAs
- Questa Intel FPGA Edition

Note: You need to acquire the license for the Nios V processor to compile the design in Quartus Prime software.

3.2. Generating the System

The implementation of Altera FPGA devices requires a hardware system developed using the Quartus Prime software. The Nios V processor requires an additional software system that complements the processor hardware system. You can develop a Nios V processor software system that is compatible to your Nios V processor hardware system using Ashling RiscFree IDE for Altera FPGAs.

3.2.1. Building Hardware Design in Platform Designer

Begin designing the system hardware by instantiating the Nios V processor and its peripherals into Platform Designer. After configuring the system assignments and constraints, complete the hardware design by performing a successful compilation.

Table 7. Component Description

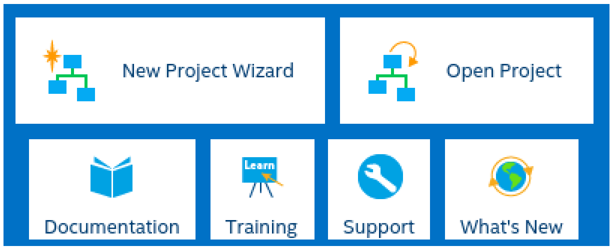
Components	Description
Nios V/m Processor IP	Runs application by executing instructions.
JTAG UART IP	Enables serial character communication between Nios V/m processor and host computer
On-Chip Memory II IP	Stores data and instructions.
System ID Peripheral IP	Verify that an executable program is compiled.

3.2.1.1. Creating a New Project

Start developing the Nios V processor system by creating a new Quartus Prime software.

1. Launch **Quartus Prime Standard Edition** software.
2. Click **New Project Wizard** to create a new project.

Figure 98. New Project Wizard



3. Read the **Introduction** and click **Next** to proceed.

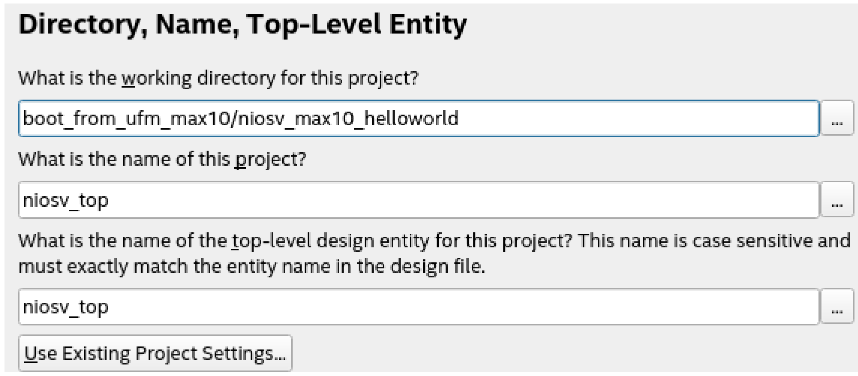
Figure 99. New Project Wizard—Introduction

4. In **Directory, Name, Top-Level Entity** window, follow these steps:

a. Enter your working directory, define the name of the project and enter the top-level design entity as `niosv_top`.

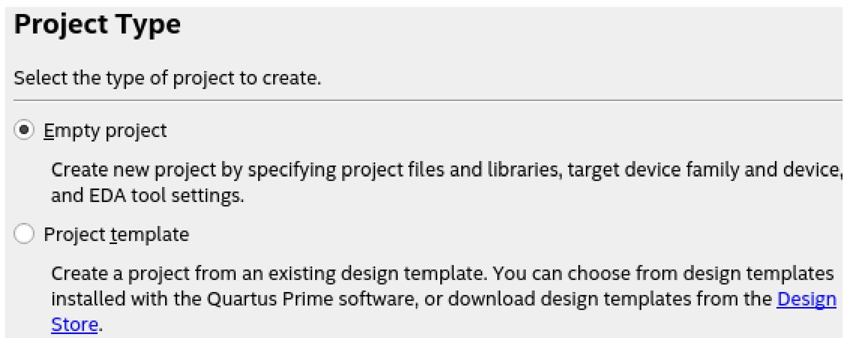
b. Click **Next** to proceed.

Figure 100. Specifying the Properties of the Project



5. In **Project Type** window, select **Empty Project** and click **Next**.

Figure 101. Selecting Project Settings



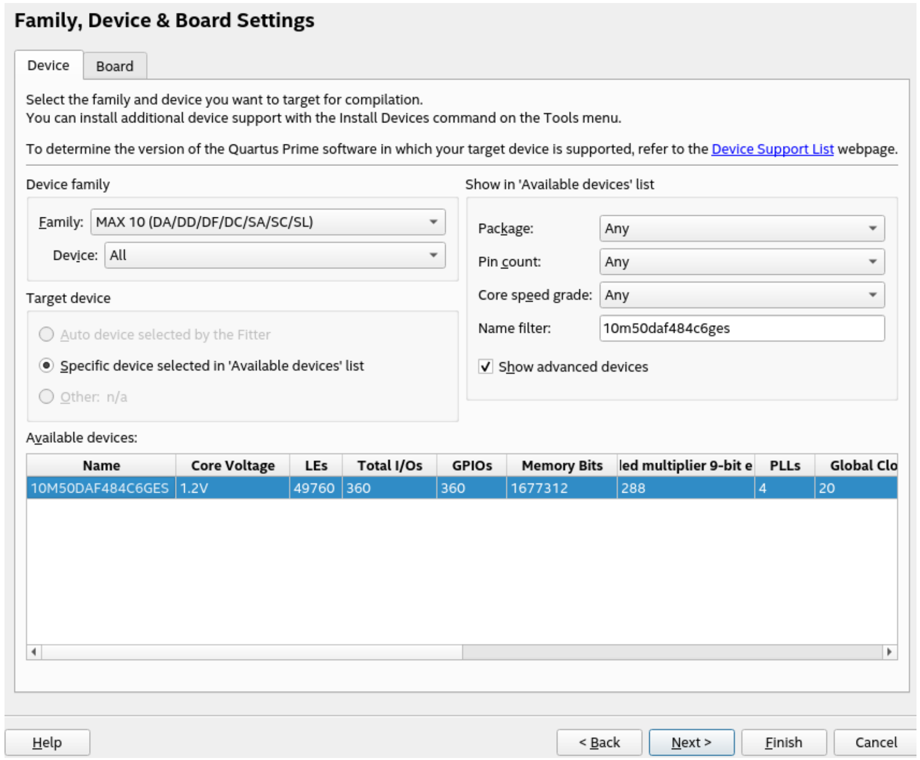
6. In **Add Files** window, leave it empty and click **Next** to proceed.

Figure 102. Add Files Window

7. In **Family, Device & Board Settings** window, select the following values:
   a. **Family:** MAX10 (DA/DD/DF/DC/SA/SC/SL)
   b. **Name filter:** 10M50DAF484C6GES

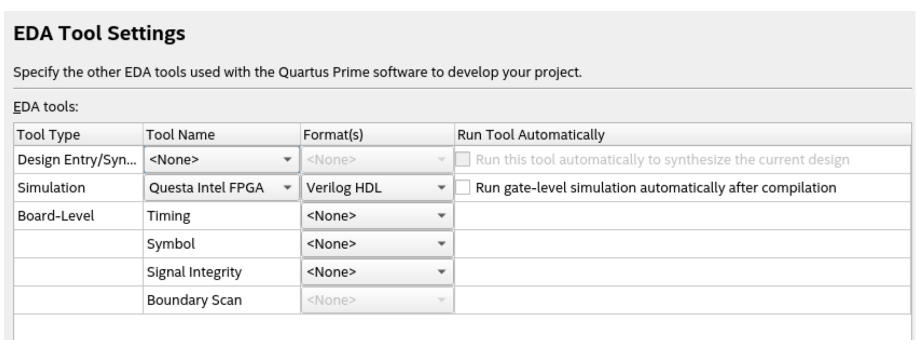
   Click **Next**.

Figure 103. Device Tab



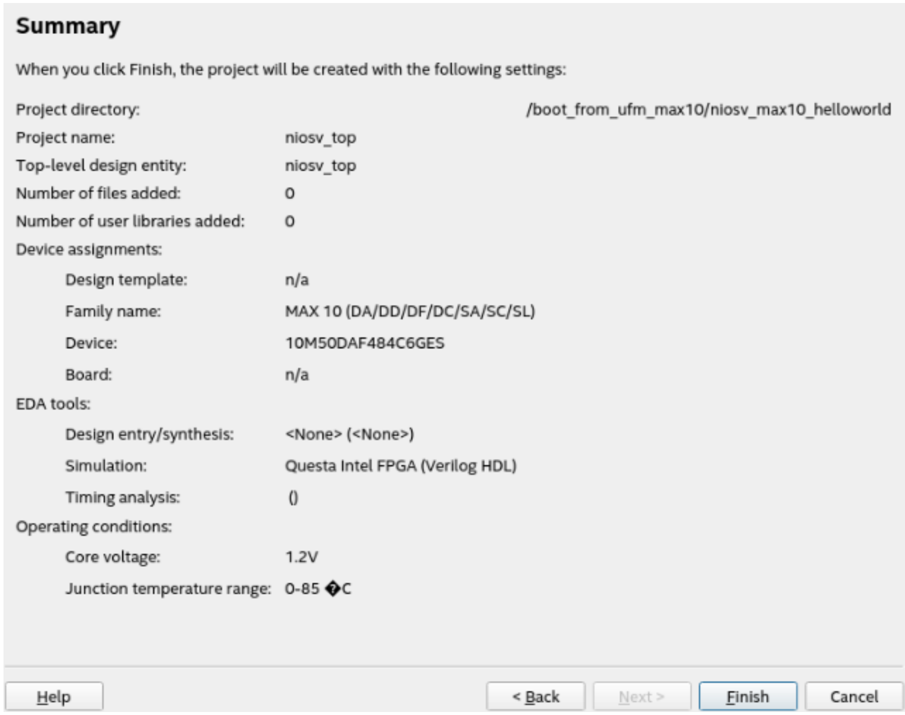
8. In **EDA Tool Settings** window, select **Questa IP**. (This is for simulation in later chapter.)

Figure 104. EDA Tool Settings



9. Click **Next** to review the summary of the new project. Then click **Finish**.

Figure 105. Summary



3.2.1.2. Creating a Platform Designer System

1. Click **New**, select **Qsys System File** and click **OK**.

Figure 106. New Window

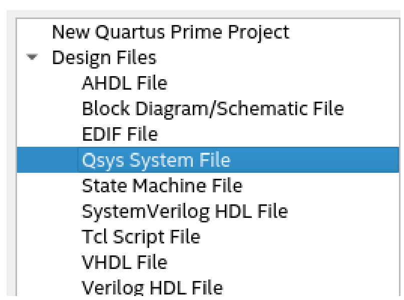
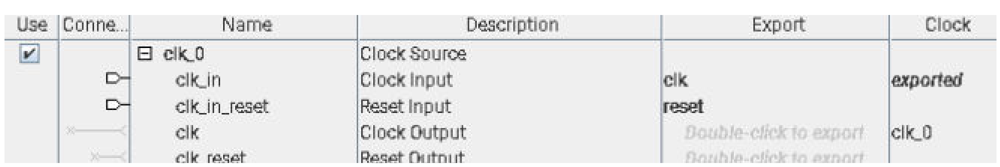


Figure 107. Default Platform Designer System

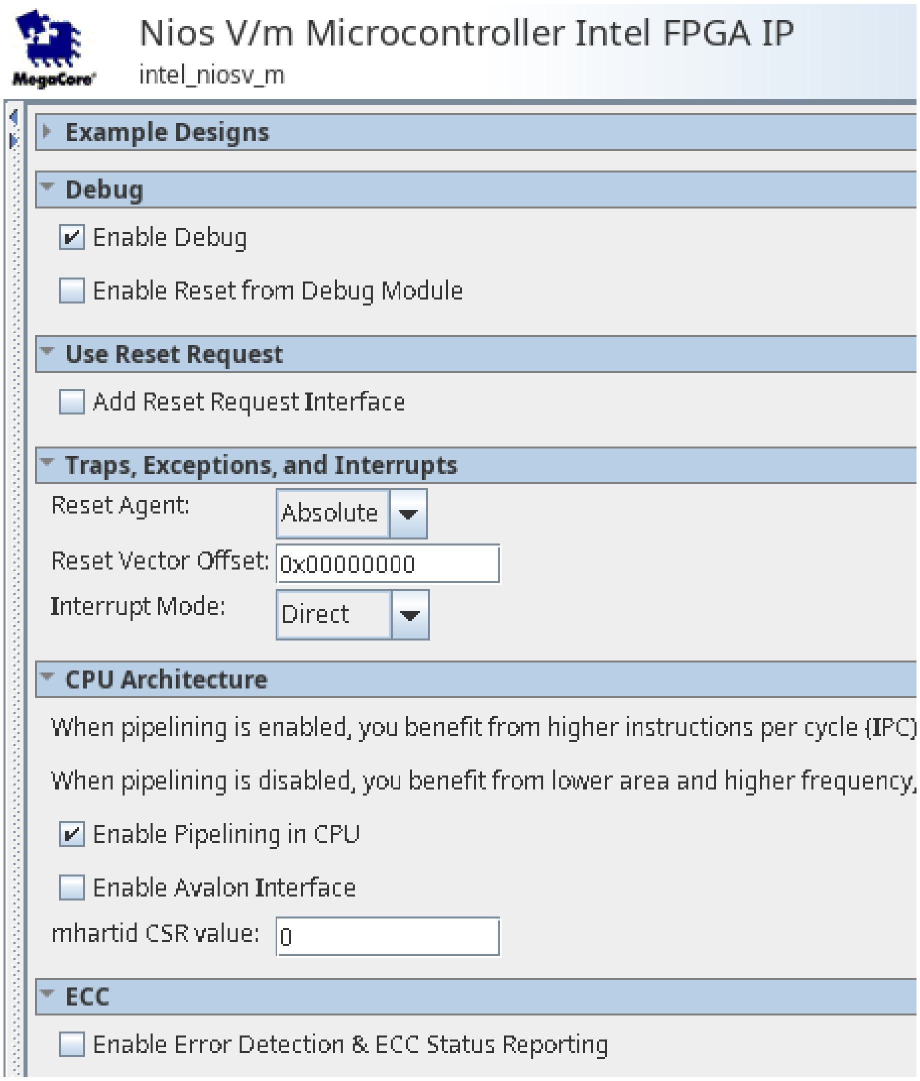


2. In Clock Source IP, configure the **Clock frequency** as `50000000` Hz (50 MHz).

Adding Nios V/m Processor IP

1. Search for **Nios V/m Processor** in the **IP Catalog**.
2. Add the **Nios V/m Microcontroller IP** under **Processors and Peripherals ➤ Embedded Processors** section. The New IP Variation window appears.
3. Click **Finish** to instantiate the processor. Leave it at the default settings.

Figure 108. Nios V/m Processor IP Parameter Editor

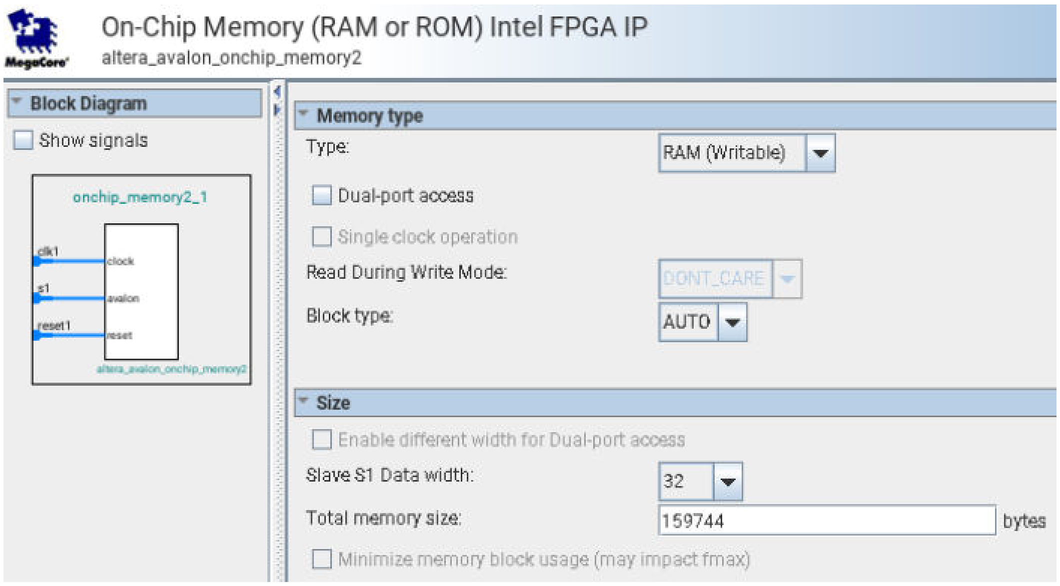


Adding On-Chip Memory (RAM or ROM) IP

1. Search for **On-Chip Memory** in the **IP Catalog**.
2. Add the **On-Chip Memory (RAM or ROM) IP** under **Basic Functions ➤ On Chip Memory**. The New IP Variation window appears.
3. Configure the **Total memory size** as **159744**.

4. Turn on the option **Enable non-default initialization file** and provide the filename `helloworld.hex`.
5. Leave other settings at default.
6. Click **Finish** to instantiate the peripheral.

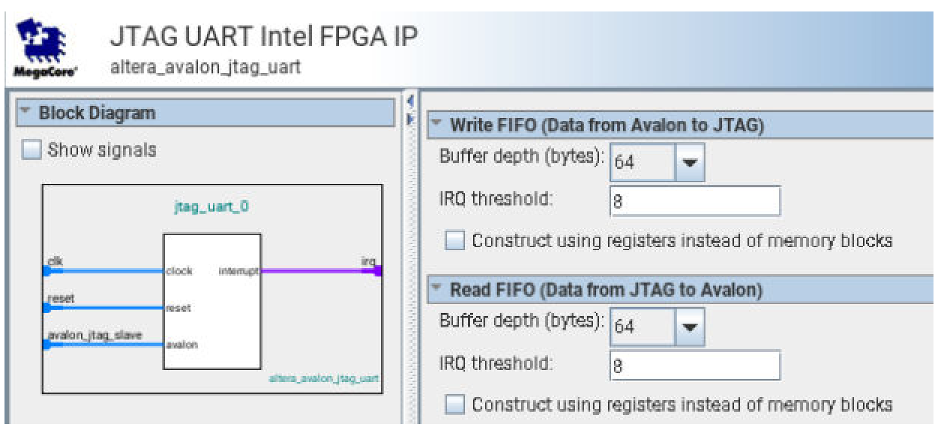
Figure 109. On-Chip Memory (RAM or ROM) IP Parameter Editor



Adding JTAG UART IP

1. Search for **JTAG UART** in the **IP Catalog**.
2. Add the **JTAG UART IP** under **Interface Protocols ➤ Serial** section. The **New IP Variation** window appears.
3. Click **Finish** to instantiate the peripheral. Leave it at the default settings.

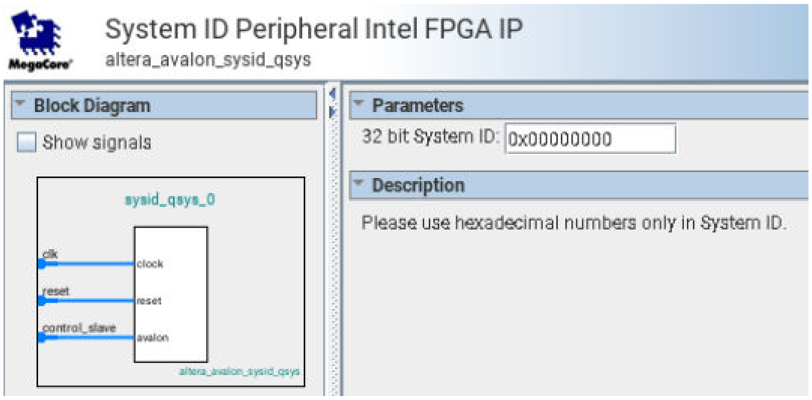
Figure 110. JTAG UART Parameter Editor



Adding System ID Peripheral IP

1. Search for **System ID** in the **IP Catalog**.
2. Add the System ID Peripheral IP under **Basic Function ➤ Simulation; Debug and Verification ➤ Debug and Performance** section. The New IP Variation window appears.
3. Click **Finish** to instantiate the peripheral. Leave it at the default settings.

Figure 111. System ID Parameter Editor



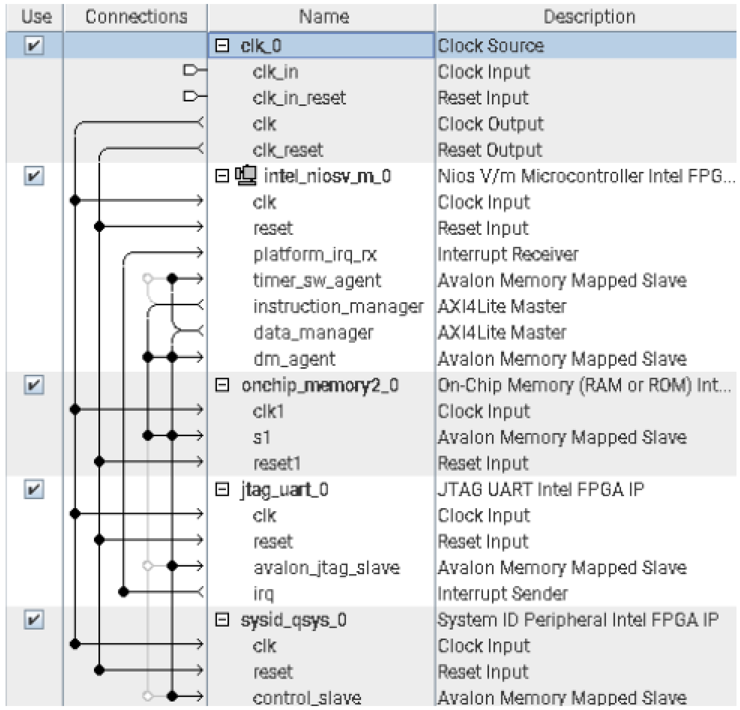
Connect Interfaces and Signals

Connect the clock source and Nios V processor to the peripherals.

Table 8. Connection between Host and Agent

IP	Host	Peripheral
Clock Source IP	`clk`	`intel_niosv_m_0.clk`
		`onchip_memory2_0.clk`
		`jtag_uart_0.clk`
		`sysid_qsys_0.clk`
	`clk_reset`	`intel_niosv_m_0.reset`
		`onchip_memory2_0.reset1`
		`jtag_uart_0.reset`
		`sysid_qsys_0.reset`
Nios V/m Processor IP	`platform_irq_x`	`jtag_uart_0.irq`
	`instruction_manager`	`onchip_memory2_0.s1`
	`data_manager`	`onchip_memory2_0.s1`
		`jtag_uart_0.avalon_jtag_slave`
		`sysid_qsys_0.control_slave`

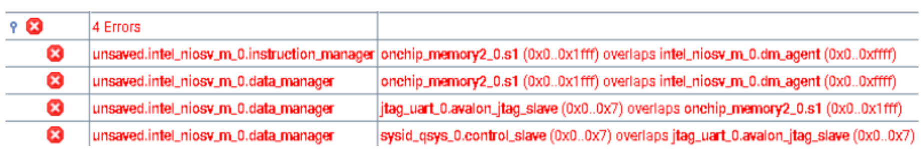
Figure 112. Full System Connection



Clear System Warnings and Errors

1. Navigate to the **System** menu bar.
2. Click **Assign Base Addresses**.

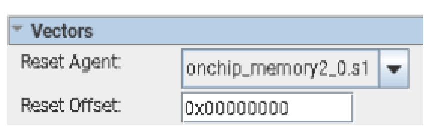
Figure 113. Example of System Connectivity Issue



Configuring the Reset Vector of the Nios V Processor

1. Double click on **Nios V Processor IP** to open the **IP Parameter Editor**.
2. Navigate to the **Vectors** tab.
3. Configure as follows:
   a. **Reset Agent**: onchip_memory1_0.s1
   b. **Reset Offse**t: 0x0

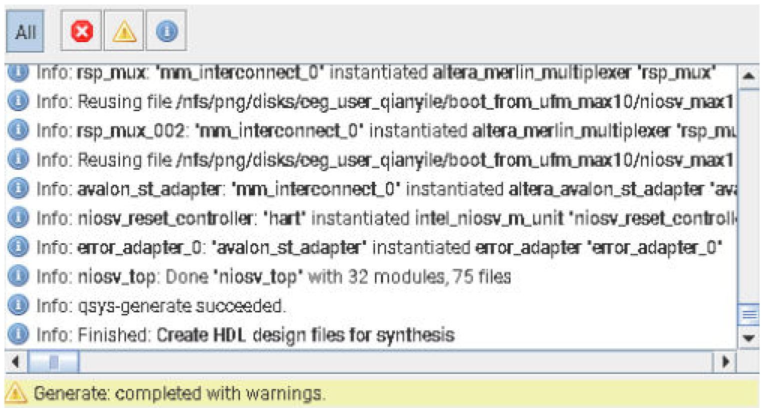
Figure 114. Reset Vector



Saving and Generating System HDL

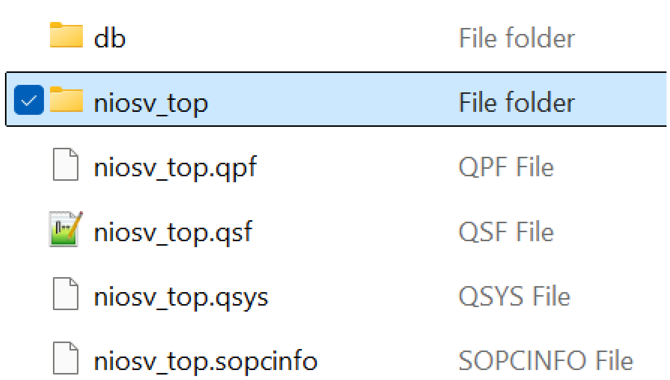
1. Navigate to the **File** menu bar.
2. Click **Save**.
3. Save your QSYS design as `niosv_top.qsys` and click **Save**.
4. Click **Generate HDL** at the bottom right corner of the Platform Designer.
5. Ensure that the Nios V processor hardware system is successfully generated.

Figure 115. Successful Generation



6. The Platform Designer generates a folder named `niosv_top`, which stores the system generation files.
7. Exit the Platform Designer and return to the project front page.

Figure 116. Generated Project Files



3.2.1.3. Configuring Assignment and Constraint

Add Synopsys Design Constraint (SDC) File

1. Click **New**, select **Synopsys Design Constraint File** and click **OK**.
2. Add the following constraint:

```
create_clock -name {altera_reserved_tck} -period 62.500 -waveform { 0.000
31.250 }[get_ports {altera_reserved_tck}]
create_clock -name {clk} -period 20.0 [get_ports clk_clk]
```

```
set_clock_groups -asynchronous -group [get_clocks {altera_reserved_tck}]
```

3. Save as `niosv_top.sdc`.

Adding Design Files

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Settings**.
2. Navigate to **Files** category. Click **Add All** to add the QSYS file to your project.
3. Click **OK** to exit the Settings window.

Figure 117. Settings – Files Category

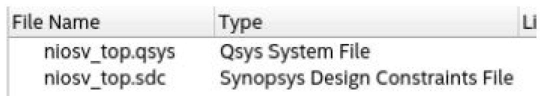
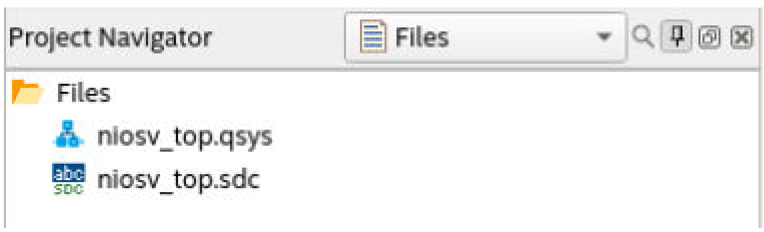


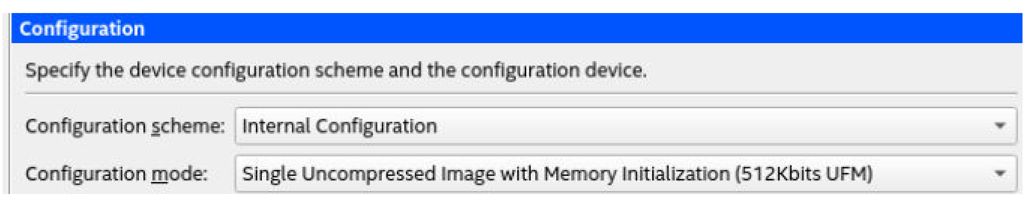
Figure 118. Project Navigator



Configuration Mode

1. In Quartus Prime software, navigate to **Assignments** menu bar and click **Device**.
2. Click **Device and Options**.
3. Navigate to **Configuration** category, set the **Configuration mode** to **Single Uncompressed Image with Memory Initialization**.

Figure 119. Configuration Mode



4. Click **OK** to exit **Device and Options** window.
5. Click **OK** to exit **Device** window.

Pin Assignment

1. In Quartus Prime software, navigate to **Processing** menu bar and click **Start ➤ Start Analysis & Elaboration**.
2. Once the analysis is complete, navigate to **Assignments** menu bar and click **Pin Planner**. For this example, there are 2 pins assignments:

- `clk_clk` assigned to `PIN_J10`
- `reset_reset_n` assigned to `PIN_D9`

3. Close **Pin Planner** and return to the project front page.

3.2.1.4. Compiling the Quartus Prime Software Project

1. Navigate to **Processing** menu bar and click **Start Compilation**.
2. Make sure that the Quartus Prime project compiles successfully.

3.2.2. Building Software Design with Ashling RiscFree IDE for Altera FPGAs

After the processor system is ready, you may begin building the software design using Ashling RiscFree IDE for Altera FPGAs software. It consists of the following steps:

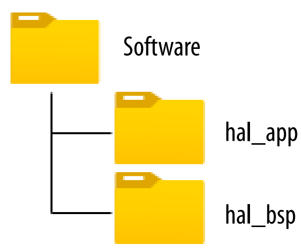
1. Create a board support package (BSP) project.
2. Create a Nios V processor application project with Hello World source code.
3. Import both projects into RiscFree IDE's workspace.
4. Build the Hello World application.

To ensure a streamlined build flow, you are encouraged to create similar directory tree in your design project. The following software design flow is based on this directory tree.

To create the software project directory tree, follow these steps:

1. In your design project folder, create a folder called `software`.
2. In the `software` folder, create two folders called `hal_app` and `hal_bsp`

Figure 120. Software Project Directory Tree



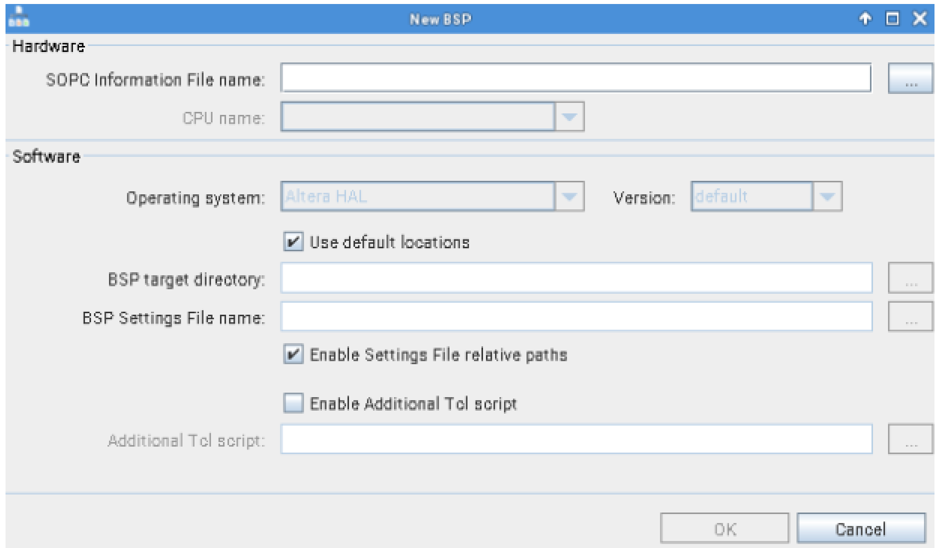
3.2.2.1. Creating a Board Support Package

Board Support Package (BSP) provides a software runtime environment for embedded systems, such as Nios V/m processor systems. Platform Designer includes the **BSP Editor** tool, to generate and configure BSP contents.

To launch the **BSP Editor**, follow these steps:

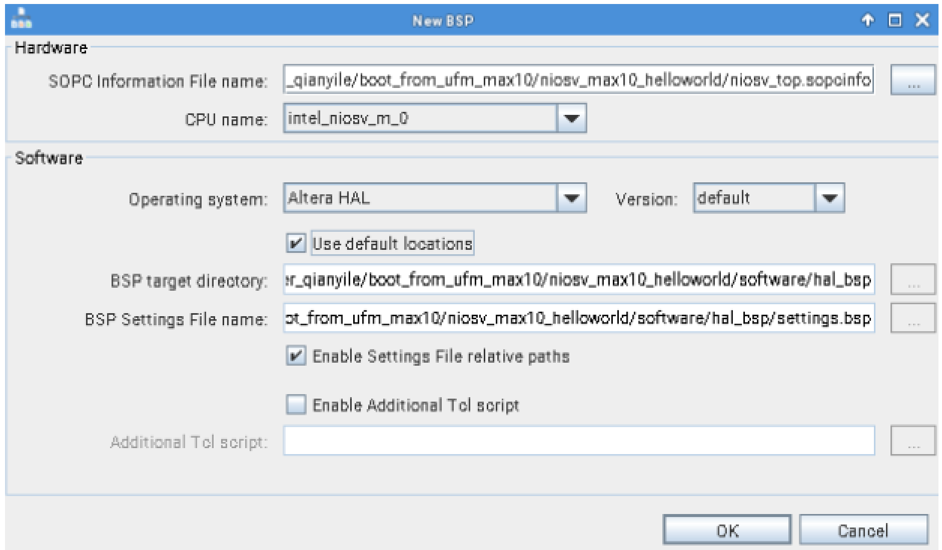
1. Enter the Nios V Command Shell.
2. Invoke the **BSP Editor** with `niosv-bsp-editor` command.
3. In the **BSP Editor**, click **File ➤ New BSP** to start your BSP project.

Figure 121. Create New BSP Window



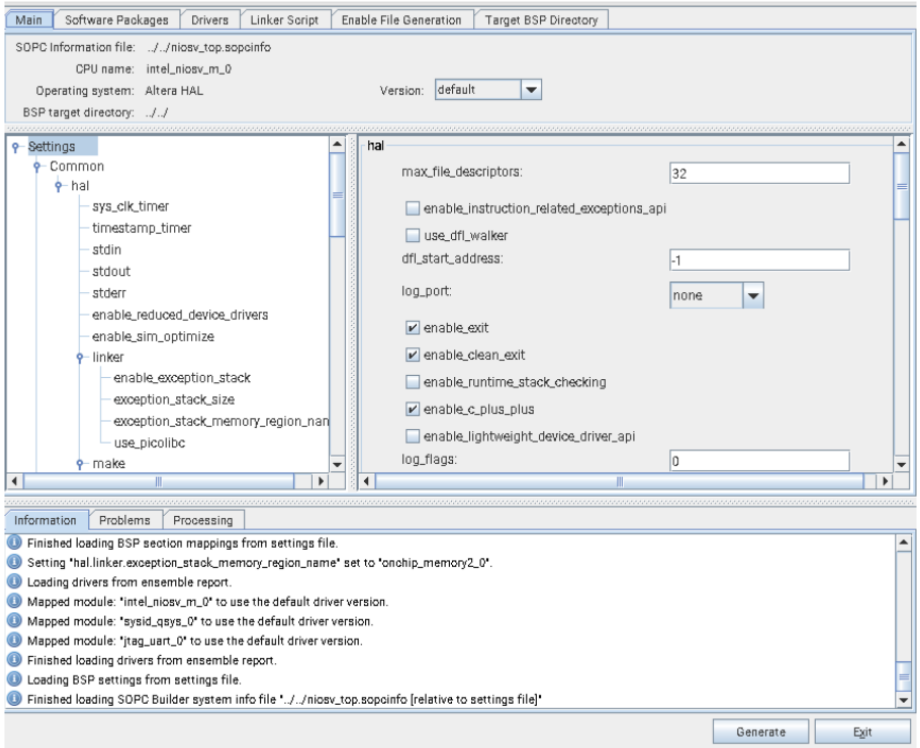
4. Configure the following settings:
   - **SOPC Information File name**: `niosv_top.sopcinfo`
   - **CPU name**: `intel_niosv_m_0`
   - **Operating system**: `Altera HAL`
   - **Version**: Leave as default.
   - **BSP target directory**: `<Working directory>/software/hal_bsp`
   - **BSP Settings File name**: `<Working directory>/software/hal_bsp/settings.bsp`
   - **Additional Tcl scripts**: Provide a BSP Tcl script by enabling **Enable Additional Tcl script**.

Figure 122. Configure New BSP



5. Click **OK**.
6. You do not need to modify further. This example design uses the default BSP settings.
7. Click **Generate BSP** to generate the BSP file.
8. Click **Exit** when the BSP is generated successfully.

Figure 123. BSP Editor Tab



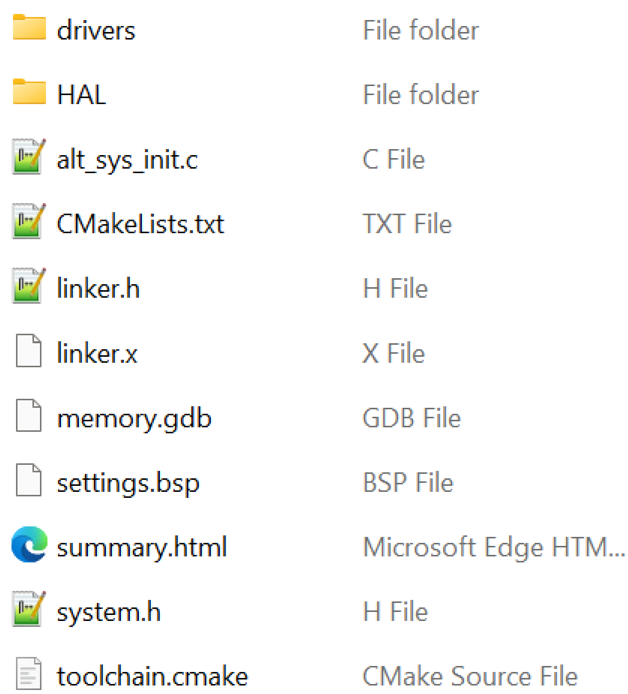
9. The BSP Editor generates the BSP files in `<Working directory>/software/hal_bsp`. Alternatively, you can generate the BSP files using CLI.
    a. Search Nios V Command Shell in the **Start Menu** and launch it.
    b. Launch the Nios V Command Shell.

    ```
    $ niosv-shell
    ```

    c. Execute the command below to generate a BSP folder.

    ```
    $ niosv-bsp -c  -s=niosv_top.sopcinfo -t=hal
    <Working directory>/software/hal_bsp/settings.bsp
    ```

Figure 124. Generated BSP Files



3.2.2.2. Creating an Application Project Files

Application Project (app) provides the software application for Nios V/m processor system.

Follow these steps to create an Application Project Files:

1. In `<Working directory>/software/hal_app` folder, create a C source code. Name it as `helloworld.c`.
2. In `helloworld.c`, copy and paste the Hello World application code below.

```
#include <stdio.h>
#include <unistd.h>

void looper() {
    for (int i = 0; i < 1000; ++i) {
     printf("Hello world, this is the Nios V/m cpu checking in %d...\n",i);
    }
}

int main() {
    looper();
    usleep(1000000);
    printf("Bye world!\n");
    fflush(stdout);
    return 0;
}
```

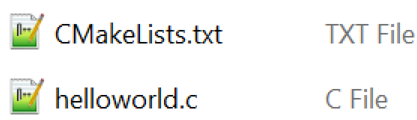
3. Launch the Nios V Command Shell.

```
$ niosv-shell
```

4. Execute the command below to generate a BSP folder.

```
$ niosv-app --bsp-dir=software/hal_bsp --app-dir=software/hal_app \
--srcs=software/hal_app/helloworld.c --elf-name=helloworld.elf
```

Figure 125. Generated APP Files



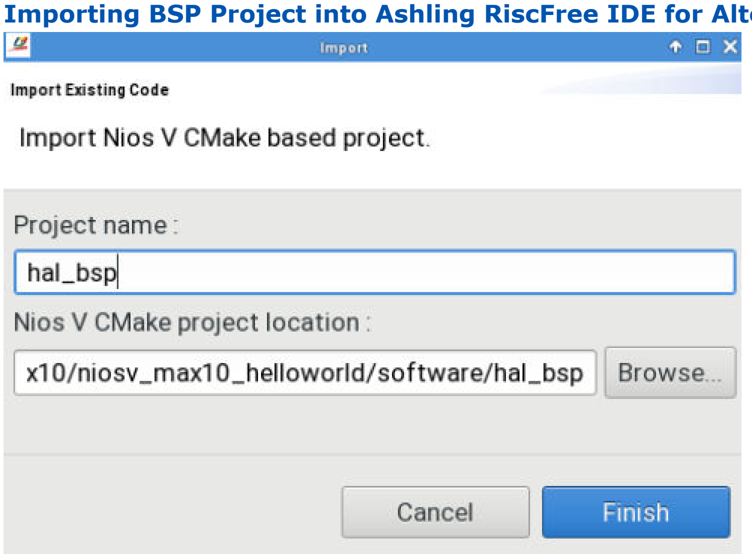
3.2.2.3. Importing Projects into Ashling RiscFree IDE for Altera FPGAs

3.2.2.3.1. Importing Nios V Processor BSP Project

Follow these steps to import the Nios V processor BSP:

1. Search Ashling RiscFree IDE for Altera FPGAs in **Start Menu** and open it.
2. Set the working directory as the **Workspace**.
3. Open the **Import** wizard, click **File ➤ Import Nios V CMake Project**.
4. In the Import window, browse and select the location of the Nios V processor BSP project.
5. The **Project name** is automatically fill according to the name of BSP project.
6. Click **Finish**. The BSP project is added to the Project Explorer.

Figure 126. Importing BSP Project into Ashling RiscFree IDE for Altera FPGAs



3.2.2.3.2. Importing Application Project

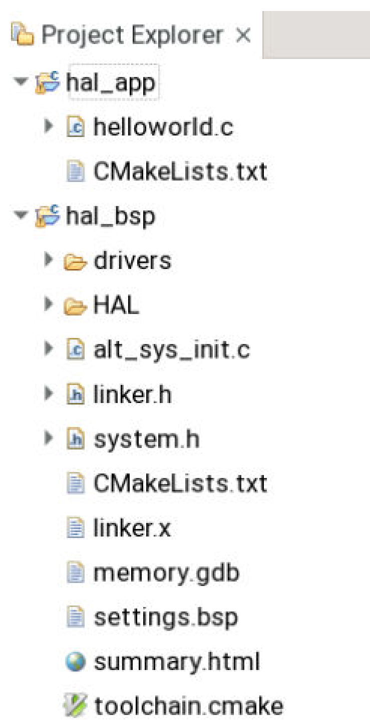
Follow these steps to import the application project:

1. Continue with the same Workspace from Importing Nios V Processor BSP Project.
2. Open the **Import** wizard, click **File ➤ Import Nios V CMake Project**.
3. In the **Import** window, browse and select the location of the Nios V processor APP project.
4. The **Project name** is automatically filled according to the name of the APP project.
5. Click **Finish**. The APP project is added to the **Project Explorer**.

3.2.2.4. Building the Hello World Application

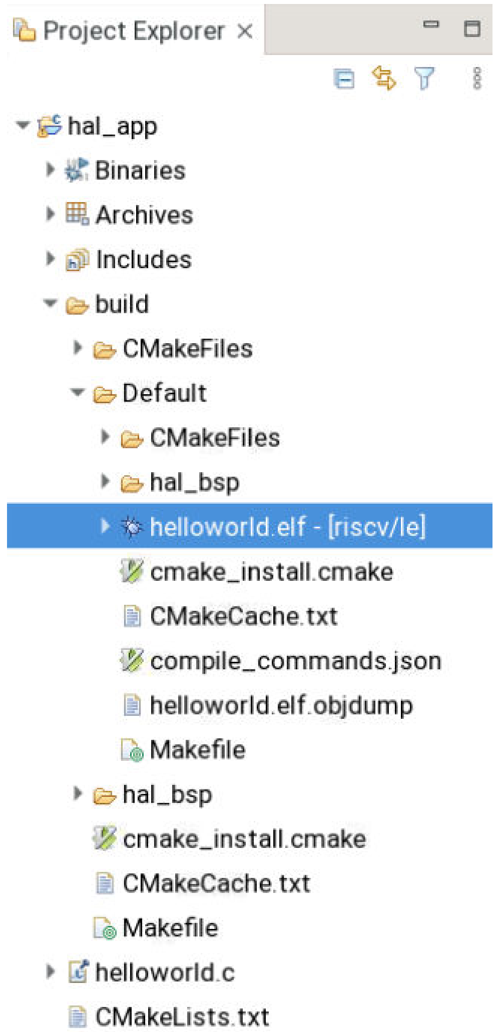
You can browse the BSP and APP project files in Ashling RiscFree IDE for Altera FPGAs **Project Explorer** tab.

Figure 127. Project Explorer Tab (Before Build Project)



1. Go to the **Project** in the menu tab and click **Build Project**.
2. When the build is complete, you can find the console prints as shown in the examples below.
3. In addition to the console prints, you can find the generated ELF file in the APP project folder.

Figure 128. Project Explorer Tab (After Build Project)



Example 3. CMake Console Print

```
cmake -DCMAKE_EXPORT_COMPILE_COMMANDS:BOOL=ON -G "Unix Makefiles" <Working
Directory>/niosv_max10_helloworld/software/hal_app
-- Defaulting build type to Debug.
-- The ASM compiler identification is GNU
-- Found assembler: <Disk Drive>/riscfree/toolchain/riscv32-unknown-elf/bin/
riscv32-unknown-elf-gcc
-- The C compiler identification is GNU 13.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: <Disk Drive>/riscfree/toolchain/riscv32-unknown-
elf/bin/riscv32-unknown-
elf-gcc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- The CXX compiler identification is GNU 13.2.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: <Disk Drive>/riscfree/toolchain/riscv32-
unknown-elf/bin/riscv32-unknown-
```

```
elf-g++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (2.4s)
-- Generating done (0.4s)
-- Build files have been written to: <Working Directory>/software/hal_app/build/
Default
```

Example 4. Make Console Print

```
/usr/bin/gmake -j all
Scanning dependencies of target hal2_bsp
[ 3%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/crt0.S.obj
[ 14%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/machine_trap.S.obj
[ 20%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dma_txchan_open.c.obj
[ 21%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/alt_log_macro.S.obj
[ 30%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_find_dev.c.obj
[ 32%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dev_llist_insert.c.obj
[ 32%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_busy_sleep.c.obj
[ 45%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_icache_flush.c.obj
[ 48%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_irq_handler.c.obj
[ 53%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_ioctl.c.obj
[ 58%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_printf.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_alarm_start.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_write.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dcache_flush_no_writeback.c.obj
[ 98%] Building ASM object CMakeFiles/hal2_bsp.dir/HAL/src/alt_mcount.S.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_close.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_env_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_do_dtors.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_dev.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_errno.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dma_rxchan_open.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_exit.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_do_ctors.c.obj
[ 25%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_dcache_flush_all.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_execve.c.obj
[ 26%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_dcache_flush.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fcntl.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_environ.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_find_file.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fd_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fd_unlock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fs_reg.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_getchar.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_gettod.c.obj
[ 38%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fork.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_flash_dev.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_iic.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_instruction_exception_register.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_gmon.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_fstat.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_get_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_getpid.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_io_redirect.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_iic_isr_register.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_icache_flush_all.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_kill.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_link.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putchar.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_main.c.obj
```

```
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_open.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_remap_uncached.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_remap_cached.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_log_printf.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_isatty.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_sbrk.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_read.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_release_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putstr.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_rename.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_malloc_lock.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_load.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_settod.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_stat.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_putcharbuf.c.obj
[ 77%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_tick.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_usleep.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_times.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_uncached_free.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_write.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_wait.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_lseek.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_tls.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
intel_fpga_api_niosv.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/alt_unlink.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
intel_fpga_platform_api_niosv.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/alt_sys_init.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_fd.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_ioctl.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/mtimer.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_init.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/drivers/src/
altera_avalon_jtag_uart_read.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/intel_niosv_irq.c.obj
[ 98%] Building C object CMakeFiles/hal2_bsp.dir/HAL/src/
alt_uncached_malloc.c.obj
[100%] Linking C static library libhal2_bsp.a
[100%] Built target hal2_bsp
```

Alternatively, you can build the application project using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to build the application.

```
$ cmake -G "Unix Makefiles" -B software/hal_app/build -S software/hal_app
$ make -C software/hal_app/build
```

3.3. Simulating the Nios V Processor System

Before experimenting with the designs in actual hardware, Altera recommends you to simulate the whole processor system and evaluate its intended functions. The benefits of simulation is not only for resource utilization but also to protect the actual hardware from unexpected damage due to mishandling within the hardware or software design.

3.3.1. Preparing Hardware Design for Simulation

Note: Before continuing the simulation, you must successfully build the hardware SOF file (with the memory-initialization feature enabled).

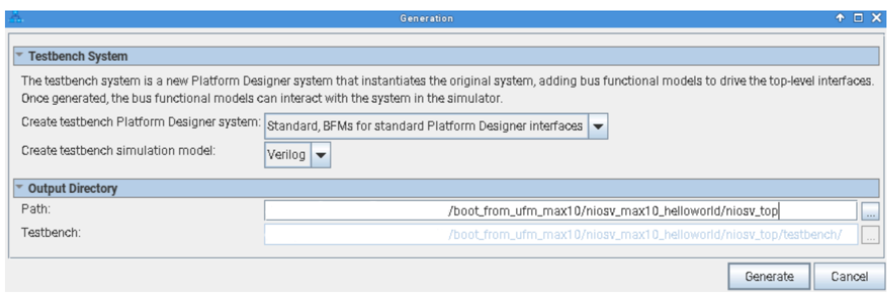
To generate hardware simulation files, perform the following steps:

1. Launch the Quartus Prime software and open the **Platform Designer** from the **Tools** menu.
2. Open the `niosv_top.qsys` file.
3. In Platform Designer, navigate to **Generate ➤ Generate Testbench System**.
4. On the **Generation** window, set the following parameters to these values:
   a. Create testbench Platform Designer system— *Standard, BFMs for standard Platform Designer interfaces*.
   b. Create testbench simulation model—*Verilog*
   c. Turn on *Use multiple processors for faster IP generation (when available)*.
5. Click **Generate**, and **Save**, if prompted.

   Altera IP cores and Platform Designer systems generate simulation setup scripts. Modify these scripts to set up supported simulators. You can find the script, `msim_setup.tcl` in the following path:

```
<Working directory>/niosv_top/testbench/mentor
```

Figure 129. Testbench Generation



3.3.2. Preparing Software Design for Simulation

Note: Before continuing the simulation, you must successfully build the software ELF file.

To generate software memory initialization file, perform the following command:

```
elf2hex <Working directory>/software/app/build/Default/hello.elf \
-b 0x0 -w 32 -e 0x26fff \
<Working directory>/software/app/build/Default/hello.hex
```

The command converts the software ELF into a HEX memory initialization file for the On-Chip RAM. The RAM have a data width of 32 bits, which starts from base address of 0x0 and ends at 0x26FFF.

3.3.3. Checking Simulation Files

At this point in the design flow, you have generated your system and created all the files necessary for simulation listed in the table below.

Table 9. Simulation Files Generated

File	Description
`<Working directory>/niosv_top/testbench/*`	Platform Designer generates a testbench system when you enable the **Create testbench Platform Designer** system option.
`<Working directory>/niosv_top_tb/niosv_top/testbench/mentor/msim_setup.tcl`	Sets up a Questa simulation environment and creates alias commands to compile the required device libraries and system design files in the correct order and loads the top-level design for simulation.
`<Working directory>/software/hal_app/build/helloworld.hex`	Memory Initialization Files (`.hex`) is required to initialize memory components in your system.

3.3.4. Running System Simulation

The `msim_setup.tcl` script in the package generated creates alias commands for each step. For the list of commands, refer to the following table:

Macros	Description
`dev_com`	Compile device library files.
`com`	Compiles the design files in correct order.
`elab`	Elaborates the top-level design.
`elab_debug`	Elaborates the top-level design with the `novopt` option.
`ld`	Compiles all the design files and elaborates the top-level design.
`ld_debug`	Compiles all the design files and elaborates the top-level design with the **vopt** option. *Note:* The **vopt** option is to run optimization before elaborating the top-level design in the simulator.

You can run the simulation in the Questa simulator by performing the following steps,

1. Launch the Nios V Command Shell.
2. Open the **Questa for Intel FPGA** simulator using the command `vsim`.
3. In the Questa **Transcript** window, change your working directory to the `mentor` folder.

```
cd <Working directory>/niosv_top_tb/niosv_top_tb/sim/mentor
```

4. Copy the memory initialization file into the `mentor` folder.

```
file copy -force \
<Working directory>/software/app/build/Default/hello.hex ./
```

5. Run the `msim_setup.tcl`.

```
do msim_setup.tcl
```

6. Compiles all the design files and elaborates the top-level design with **vopt** option.

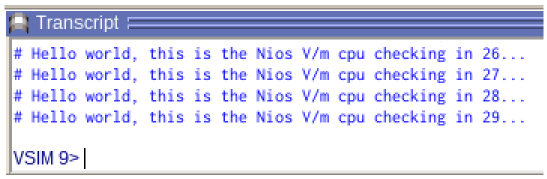
```
ld_debug
```

7. Run the simulation for more than 5 milliseconds.

```
run 10ms
```

At the end of the simulation, you can find the following message prints in the Questa **Transcript** window: `Hello world, this is the Nios V/m cpu checking in <loop number>`.

Figure 130. Questa Transcript Window Message



You can observe the simulation results from the waveform viewer:

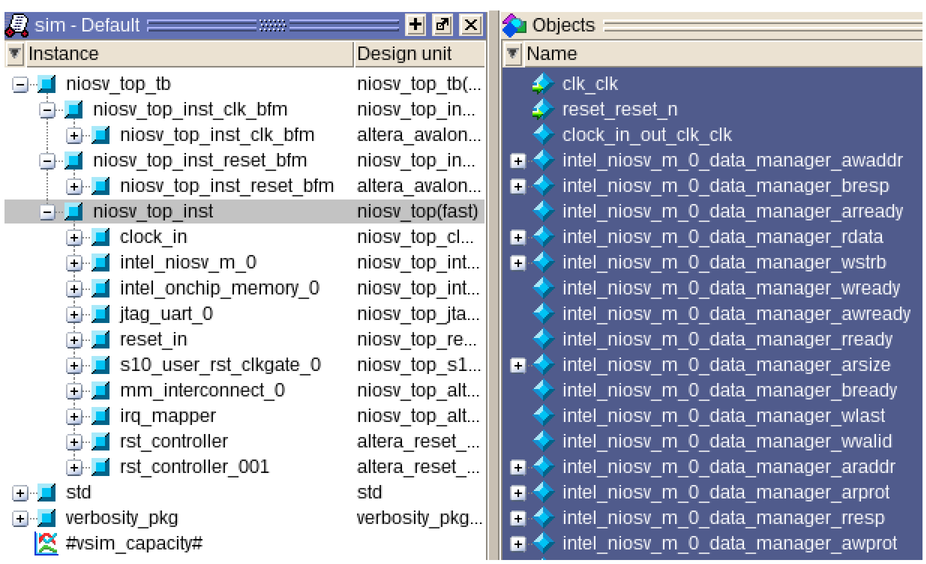
1. In Questa for Intel FPGA simulator, navigate to the **Instance** window.
2. Unroll `niosv_top_tb`.
3. Select `niosv_top_inst`, and the simulator populates a list of signals in the **Objects** window.
4. Right-click any of the signals, and click **Add Waves** to add them into the **Wave** window.
5. Restart and run the simulation for more than 5 milliseconds.

```
restart
```

```
run 10ms
```

6. Wait for the simulator to complete the given time.

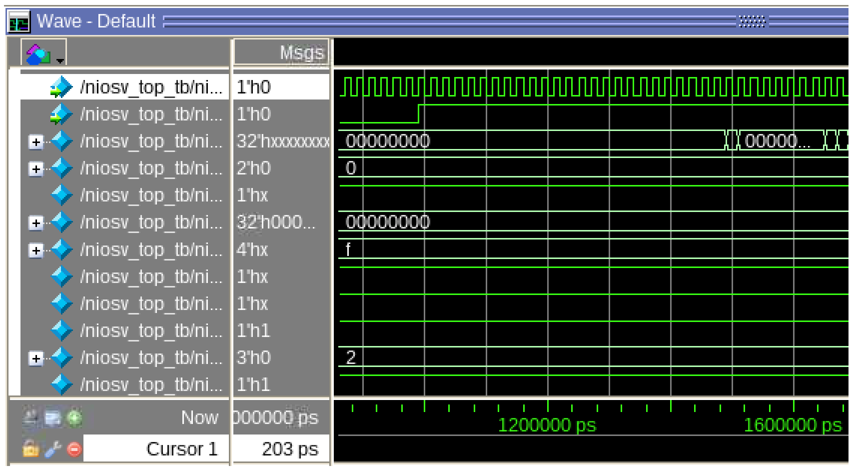
Figure 131. Questa Instance and Objects Windows



The following figure shows the simulated wave result.

- First signal: Clock Input
- Second signal: Reset Input
- Other signals: Any one or more signals within the `niosv_top_inst`

Figure 132. Waveform Viewer



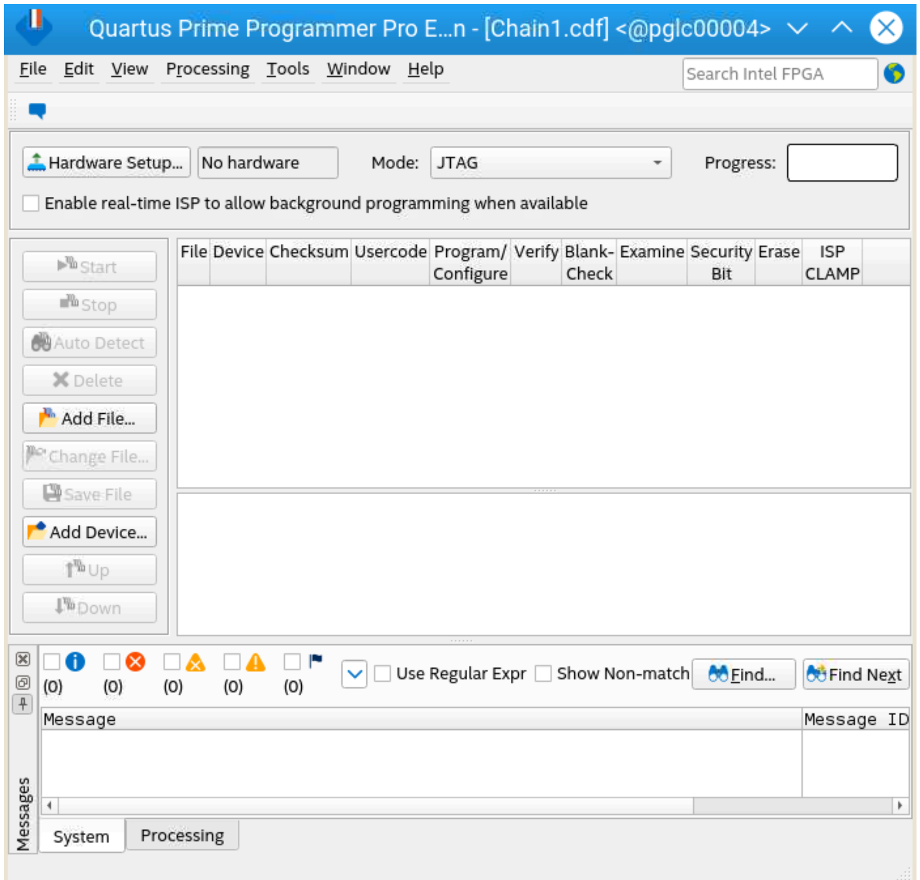
3.4. Programming the System

Use the Quartus Prime Programmer tool and the Ashling RiscFree IDE for Altera FPGAs to program the Nios V processor-based system into the FPGA and to run your application.

3.4.1. Programming Hardware SOF File

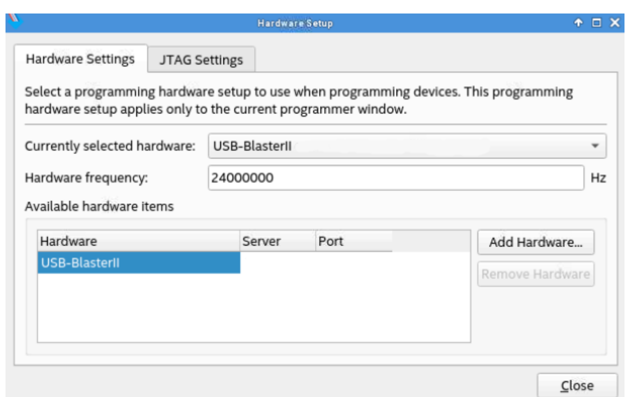
1. Connect the MAX 10 FPGA 10M50 Evaluation Kit to the host PC using the USB Blaster II.
2. Open the Quartus Prime Programmer.

Figure 133. Quartus Prime Programmer



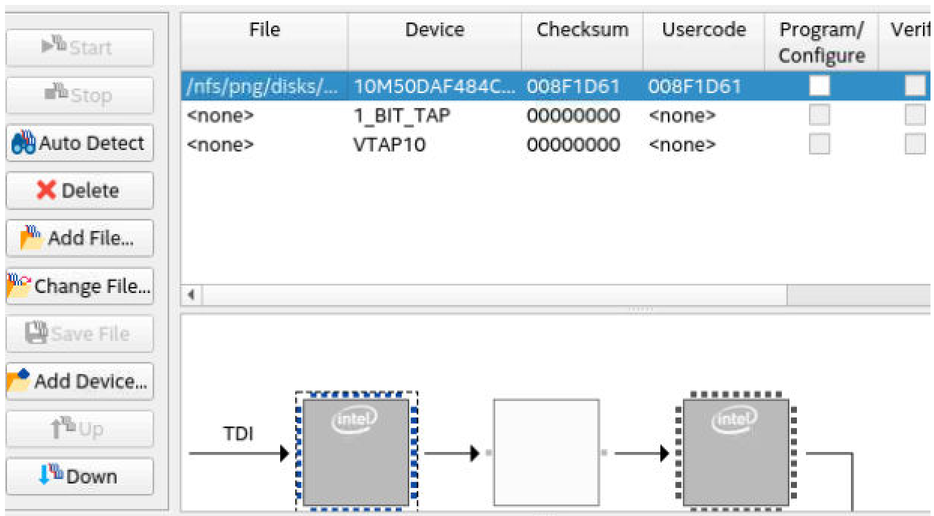
3. Click **Hardware Setup**.
4. Check the availability of the development kit in **Available hardware items**.
   a. If available, select the development kit in **Currently selected hardware**.
   b. If not available, check the cable connection, and the JtagServer driver installation.

Figure 134. Hardware Setup



5. Click **Auto Detect** and select the appropriate device OPN.
6. Select the MAX 10 device, click **Change File** and select `niosv_top.sof` file.
7. Once the SOF file is ready, check **Program/Configure** and click **Start**.

Figure 135. List of Devices with JTAG Chain



8. Wait until the **Progress** bar reaches 100% (Successful).

Figure 136. Progress Bar

9. You have successfully configured the development kit with the processor hardware system.

   Alternatively, you can program the hardware SOF file through CLI.

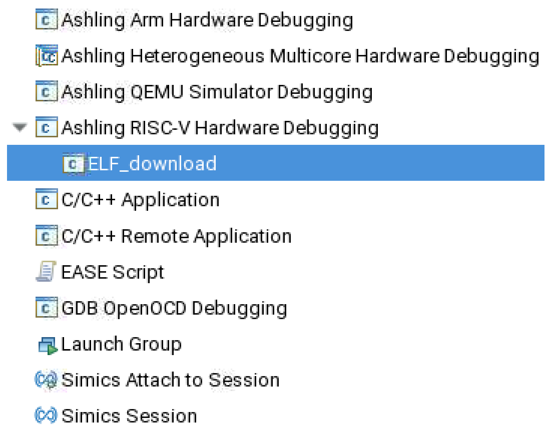
   Execute the following command to program the SOF file.

```
$ quartus_pgm -c 1 -m JTAG -o p;<Working directory>/output_files/
niosv_top.sof@1
```

3.4.2. Downloading the Software ELF File

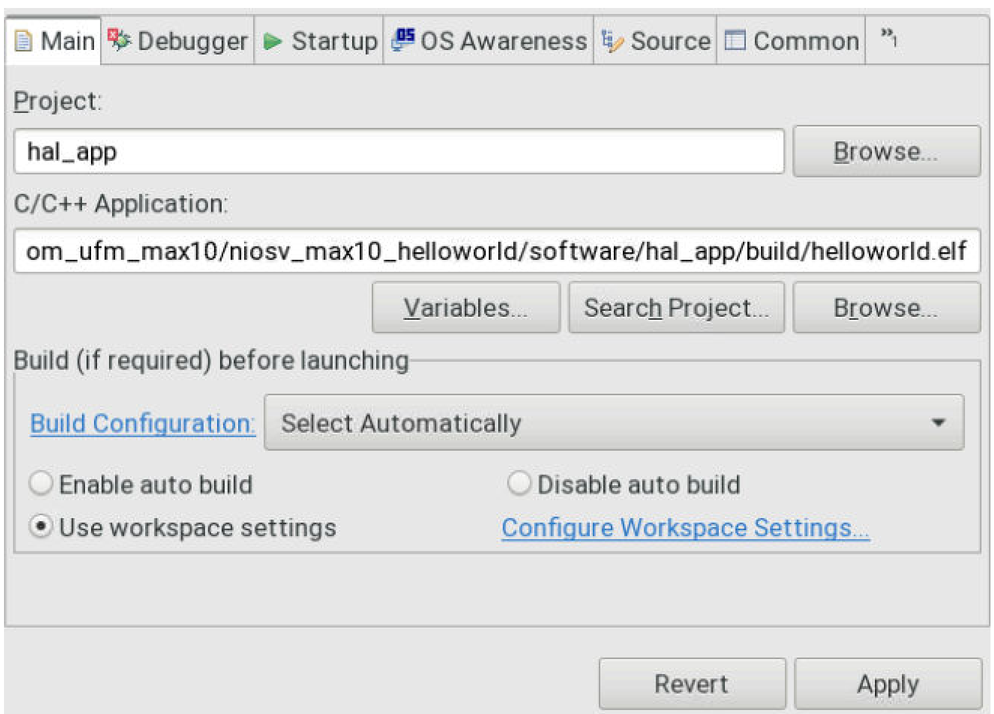
1. Ensure that the development kit is successfully configured with the processor system.
2. Launch the Ashling RiscFree IDE for Altera FPGAs.
3. Navigate to **Run ➤ Run Configurations**.
4. In **Run Configuration** window, double click **Ashling RISC-V Hardware Debugging** and name it as `ELF_download`.

Figure 137. Run Configuration



5. In the **Main** tab, make the following settings:
   a. **Project**: hal_app
   b. **C/C++ Application**: <Working directory>/software/hal_app/build/helloworld.elf

Figure 138. Main Tab



6. In the **Debugger** tab, make the following settings:
   a. **Debug Probe Configuration**:

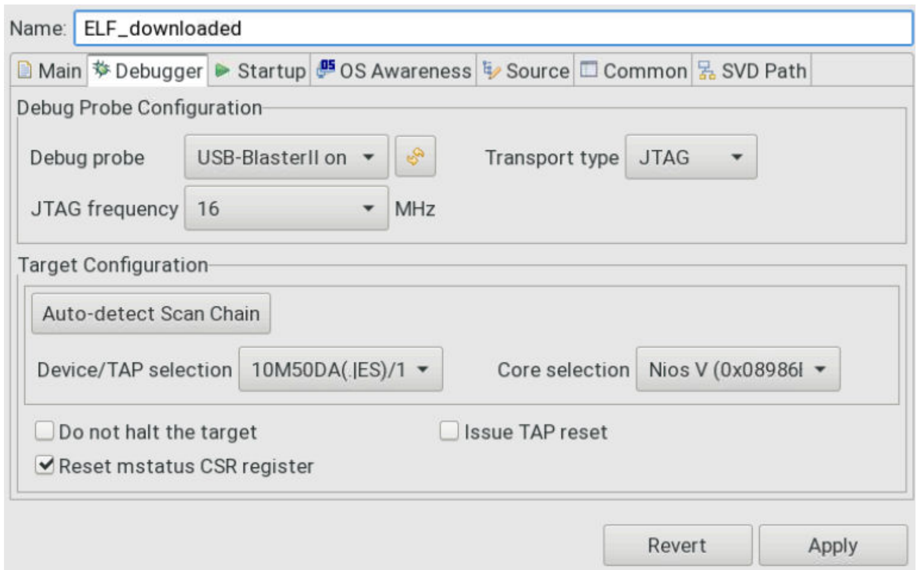
i. **Debug probe**: USB-Blaster II

ii. **Transport type**: JTAG

iii. **JTAG frequency**: 16 MHz

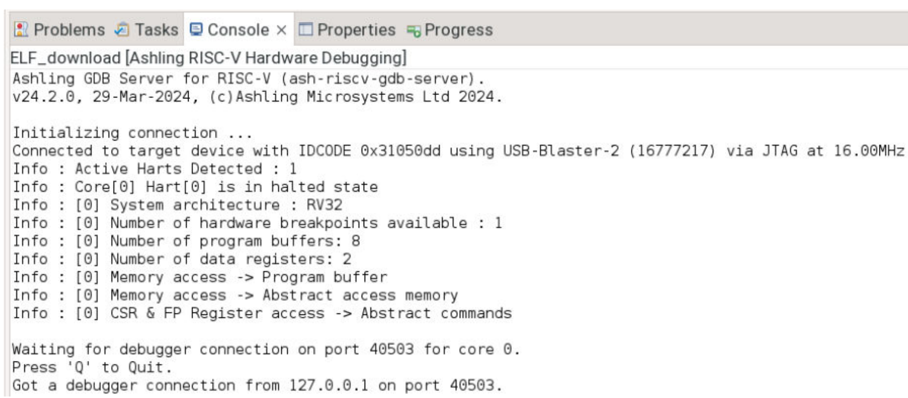
b. **Target Configuration**: Click **Auto-detect Scan Chain** to list all possible cores. Select the appropriate **Device/TAP** and **Nios V Processor Core**.

Figure 139. Debugger Tab



7. Click **Apply** and **Run**. Ashling RiscFree IDE for Altera FPGAs prints the following message in its **Console**.

Figure 140. ELF_download Message (Console tab)



```
Problems  Tasks  Console ×  Properties  Progress
ELF_download [Ashling RISC-V Hardware Debugging]
Ashling GDB Server for RISC-V (ash-riscv-gdb-server).
v24.2.0, 29-Mar-2024, (c)Ashling Microsystems Ltd 2024.

Initializing connection ...
Connected to target device with IDCODE 0x31050dd using USB-Blaster-2 (16777217) via JTAG at 16.00MHz.
Info : Active Harts Detected : 1
Info : Core[0] Hart[0] is in halted state
Info : [0] System architecture : RV32
Info : [0] Number of hardware breakpoints available : 1
Info : [0] Number of program buffers: 8
Info : [0] Number of data registers: 2
Info : [0] Memory access -> Program buffer
Info : [0] Memory access -> Abstract access memory
Info : [0] CSR & FP Register access -> Abstract commands

Waiting for debugger connection on port 40503 for core 0.
Press 'Q' to Quit.
Got a debugger connection from 127.0.0.1 on port 40503.
```

Alternatively, you can download the software ELF file using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to download the ELF file.

```
$ niosv-download <Working directory>/software/hal_app/debug/helloworld.elf -
g -r
```

3.4.3. Applying External Tool Configuration

External tool configuration is a generic feature where you can configure the Ashling RiscFree IDE for Altera FPGAs to include external tools. An example of such tools is the `juart-terminal` executable, which is used to display the Hello World message through the JTAG UART IP.

To perform external tool configuration for `juart-terminal`, follow these steps:

1. Go to **Run ➤ External Tools ➤ External Tools Configurations**.
2. Double click **Program** to open a **New_configuration** window.
3. Rename the configuration as `Nios V JTAG UART Output`.
4. In **Location**, click **Browse File system**.
5. Browse and select the `juart-terminal` file in the following paths:

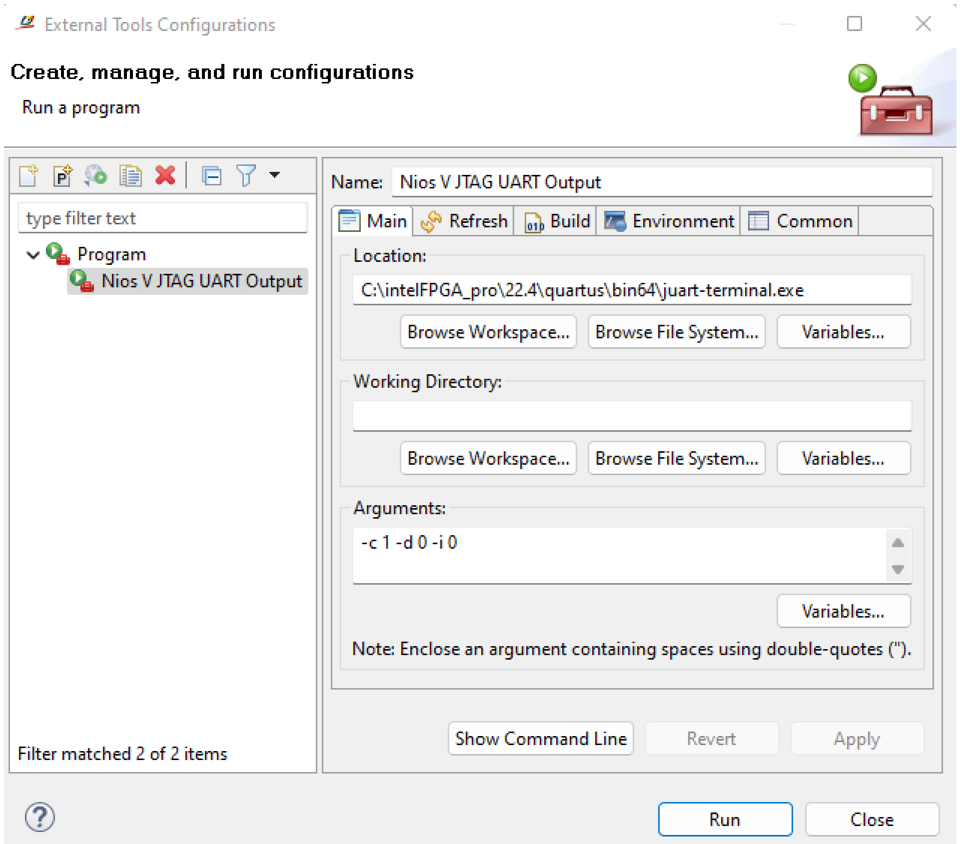
Table 10. Path

Operating System	Path
Windows	`<Quartus Prime installation directory>/quartus/bin64/juart-terminal.exe`
Linux	`<Quartus Prime installation directory>/quartus/linux64/juart-terminal`

6. Set the **Arguments** as "-c 1 -d 1 -i 0". This configures the JTAG UART connection is towards the JTAG UART IP at cable 1, device 1, and instance 0.
7. Click **Apply**.

Note: JTAG device chain index varies according to your setup. Run `jtagconfig --debug` to obtain the latest device chain index numbering.

Figure 141. External Tools Configuration

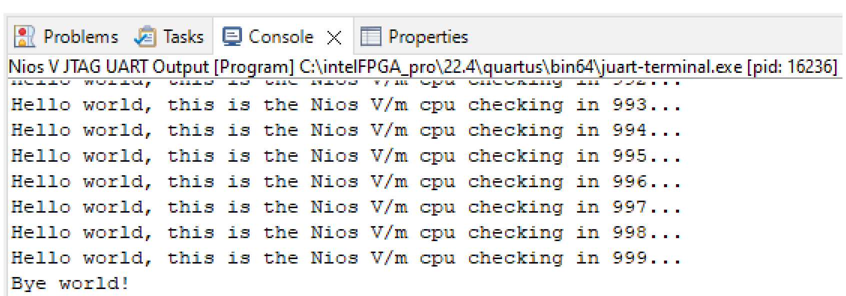


3.4.4. Run Hello World Application

With the **External Tool Configuration (Nios V JTAG UART Output)** defined successfully, you may proceed to use it to display the Hello World application.

1. Go to **Run ➤ External Tools ➤ External Tools Configurations**.
2. Select **Nios V JTAG UART Output**.
3. Click **Run**.
4. The Hello World message is printed on the **Console** tab.

Figure 142. Hello World Message (Console tab)



Alternatively, you can display the Hello World application using CLI.

1. Launch the Nios V Command Shell.

```
$ niosv-shell
```

2. Execute the command below to display the Hello World application.

```
$ juart-terminal -c 1 -d 0 -i 0
```

3.5. Debugging the System

Occasionally, the generated system might fail to run or display unexpected behaviors. You can debug the hardware and software system to pinpoint the root cause and ultimately return the system to normal. Optionally, you can also apply the debug tools to view a processor system's inner workings and running mechanisms while on-the-fly.

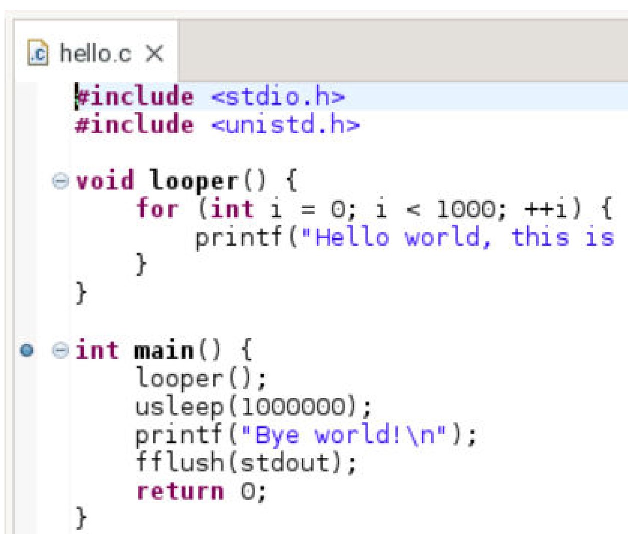
3.5.1. Software Debugging

3.5.1.1. Setting Initial Breakpoint

Before debugging any system, you need to have at least one breakpoint to suspend the execution inside the program.

1. Open the Ashling RiscFree IDE for Altera FPGAs.
2. Open the Nios V processor software project (hello.c).
3. Set a breakpoint at `main()` by double-clicking on the left-margin of the line containing `main()`.
4. Ensure that a blue circle have appeared beside `main()`.

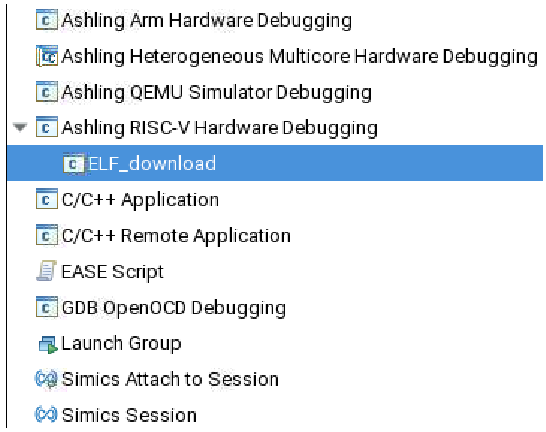
Figure 143. Initial Breakpoint at main()



3.5.1.2. Starting the Debugger

1. In the **Run** menu tab, select **Debug Configurations**.
2. Under **Ashling RISC-V Hardware Debugging**, find `ELF_download` (Created in Downloading the Software ELF file.
3. Select `ELF_download` to start the Debugger using the same settings.

Figure 144. Debug Configuration



4. In the **Main** tab, check the following settings
   a. **Project**: `hal_app`
   b. **C/C++ Application**: `<Working directory>/software/hal_app/build/helloworld.elf`
5. In the **Debugger** tab, check the following settings:
   a. For **Debug Probe Configuration**,
      i. **Debug probe**: development kit
      ii. **Transport type**: JTAG
      iii. **JTAG frequency**: 16 MHz
   b. **Target Configuration**: Click **Auto-detect Scan Chain** to list all possible cores. Select the appropriate **Device/TAP** and **Nios V Processor Core**.

Figure 145. Main Tab

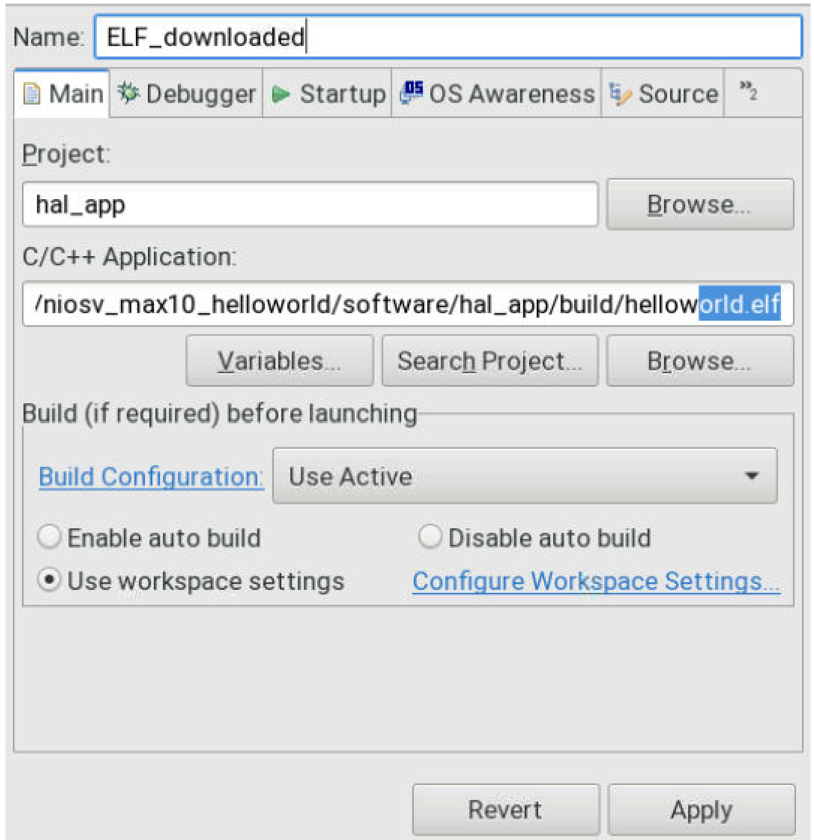
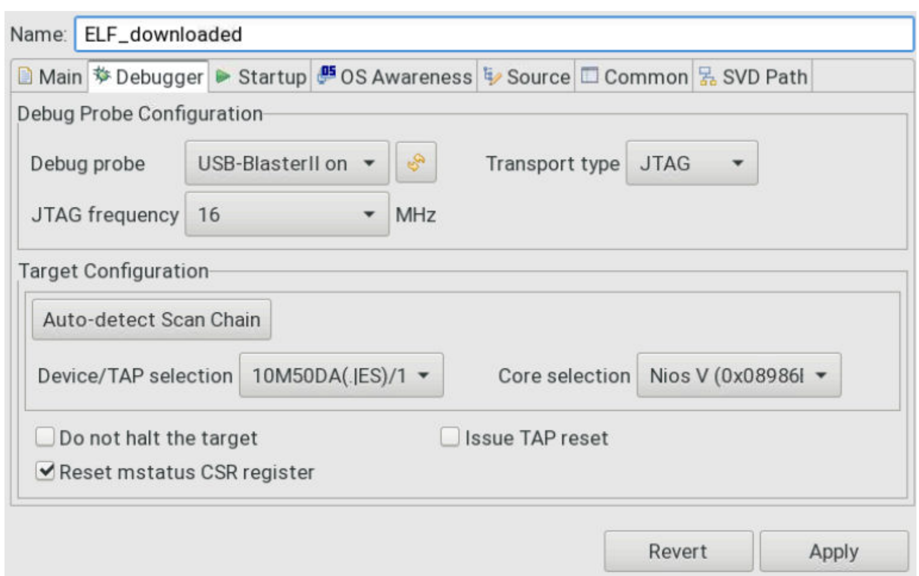
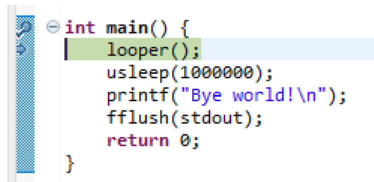


Figure 146. Debugger Tab



6. Click **Apply** and **Debug**.
7. The Ashling RiscFree IDE for Altera FPGAs switches to the **Debug Perspective**.
8. The program begins execution, and suspends at the initial breakpoint, `main()`.

Figure 147. Suspended at Initial Breakpoint



3.5.1.3. View Global Variables

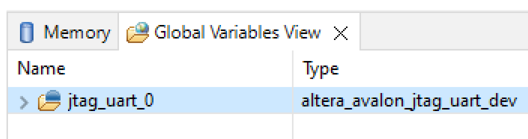
Ashling RiscFree IDE for Altera FPGAs has dedicated global variables.

To view the global variables in the application (`.elf`), follow these steps:

1. Go to **Window ➤ Show View ➤ Other... ➤ Debug ➤ Global Variables View**.
2. Right-click in the **Global Variables View** tab, and select the **Add Global Variables** icon.
3. Select the required variables to view.

Note: You can change the value of a variable. Right-click the variable and select **Change Value**.

Figure 148. Global Variables View



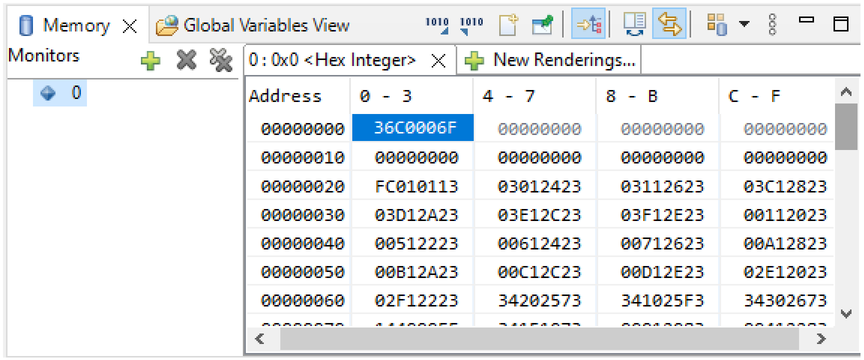
3.5.1.4. View Memory

Ashling RiscFree IDE for Altera FPGAs supports memory browser. You can view the content of On-Chip Memory (RAM) or other memory devices. This example targets the start address of On-Chip Memory (RAM), which the Hello World application begins.

To launch the memory browser, follow these steps:

1. Go to **Window ➤ Show View ➤ Memory Browser**.
2. Select **Add Memory Monitor**.
3. Provide the memory address 0x0, and click **OK**.

Figure 149. Memory Browser at Address 0x0



4. Go to `<Working directory>/software/app/build/Debug` folder.
5. Open the `hello.elf.objdump` file.
6. Search for `Disassembly of section .entry`.
7. The disassembly shows that the information at starting address 0 is 0x36c006f, which is exactly the same as in the **Memory Browser**.
8. You can continue to verify section `.exceptions`.

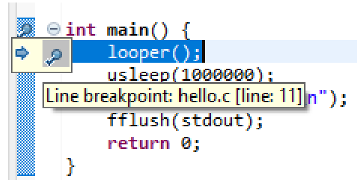
Figure 150. Disassembly of Hello World Application

```
Disassembly of section .entry:

00000000 <__reset>:
 * Jump to the _start entry point in the .text section if reset code
 * is allowed or if optimizing for RTL simulation.
 */

    /* Jump to the _start entry point in the .text section. */
    tail _start
   0:   36c0006f            j   36c <_start>
    ...
```

3.5.1.5. Setting a Second Breakpoint

1. Open the Nios V processor software project (`helloworld.c`).
2. Set a second breakpoint at `looper()` by double-clicking on the left-margin of the line containing `looper()`.
3. Ensure that a blue circle has appeared beside `looper()`.

Figure 151. Second Breakpoint at `looper()`



3.5.1.6. Control Debugger

Using the two breakpoints, you can control the debugging process using the Debug bar.

Figure 152. Execution Control (Debug bar)

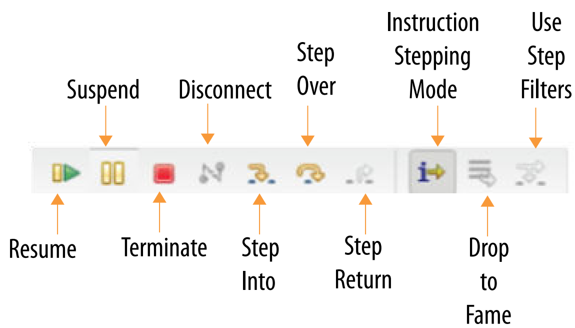


Table 11. Description of Execution Control

Function	Description
Resume	Continues the debugging process until the next breakpoint, or until the end of the debugging process.
Suspend	Halts the running code while debugging.
Terminate	Terminates the debug session. You need to relaunch the debug session after terminated.
Disconnect	Disconnects the debug tool connected to the RiscFree* IDE. Note: You are required to terminate the debug session before disconnecting the debug tool.
Step Into	Steps into the next method call (entering it) at the currently executing line of code.
Step Over	Steps over the next method call (without entering it) at the currently executing line of code. The method is still executed.
Step Return (Step Out)	Returns from a method which has been stepped into (exiting it). The remainder of the code that was skipped by returning is still executed.
Instruction Stepping Mode	Switches to instruction stepping mode.
Drop to Frame	Pops the current stack frame and puts control back out to the calling method, resetting any local variables.
Use Step Filters	Changes whether step filters should be used in the current Debug View.

4. Document Revision History for the AN 985: Nios V Processor Tutorial

Document Version	Changes
2025.08.28	Added a new section: *Hello World on MAX 10 FPGA Device*.
2024.07.24	• Updated *Hardware and Software Requirements*. • Added the following new sub-topics for *Building Hardware Design in Platform Designer:* — *Board-Aware Flow* — *Configurable Example Design*
2024.05.15	Updated the following topics: • *Building Hardware Design in Platform Designer* • *Adding Nios V/m Processor Intel FPGA IP* • *Connect Interfaces and Signals* • *Adding Nios V/m Processor Intel FPGA IP* • *Connect Interfaces and Signals* • *Creating an Application Project File* • *Importing Nios V Processor BSP Project* • *Importing Application Project*
2023.08.18	Initial release.